RTL Design Sherpa

DDR2 / LPDDR2 Family Controller Micro-Architecture Specification 0.1

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2 Document Information

2.1 DDR2 / LPDDR2 Family Controller Micro-Architecture Specification

Document Number: DDR2-LPDDR2-MAS-001 **Version:** 0.4 **Status:** Draft - reconciled with rearchitected RTL **Classification:** Open Source - Apache 2.0 License

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2.2 Document Purpose

This Micro-Architecture Specification (MAS) is the implementation-level companion to the DDR2 / LPDDR2 Hardware Architecture Specification (HAS). Where the HAS answers “what is the architecture and why,” this MAS answers “what is in the RTL and how does it work.”

The MAS is the document RTL designers, verification engineers, and integrators consult when writing or reviewing SystemVerilog code, building testbenches, and stitching the controller into an SoC.

Target Audience:

- RTL designers implementing the FUBs described in this document
- Verification engineers writing per-FUB cocotb tests
- Integrators stitching the controller into an SoC
- Software engineers writing low-level memory configuration code (driver authors, bring-up engineers)

Companion Documents:

- DDR2/LPDDR2 Family Controller HAS (`../pumice_has/`) — high-level architecture and design rationale

- DDR2/LPDDR2 DFI BFM documentation (in `RTLDesignSherpa-DV`) — verification-side reference

2.3 Document Scope

The MAS is the **micro-architecture** view: per-FUB inputs/outputs, internal state, FSM diagrams, datapath timing, register-level pipeline stages, and per-block timing budgets. It assumes the reader has read the HAS for context.

This document covers:

- Top-level integration wiring (`pumice_ctrl`)
- Each FUB's interface signal list, internal storage, datapath flow, FSM, and timing budget
- AXI4 and DFI v2.1 wire-level protocol details specific to this controller
- APB programming interface and the full CSR register map
- Programming sequences (init, refresh, power-down, multi-rank, error handling)
- Build-time and runtime configuration reference

This document does **not** cover:

- Higher-level architectural rationale (see HAS)
- Verification plans or coverage models (see `dv/README.md`)
- Floorplan or layout guidance (project-specific)
- Bit-level CSR YAML — that lives in `regs/` and is the source of truth for CSR generation

2.4 Revision History

Version	Date	Author	Notes
0.1	2026-06-14	RTL Design Sherpa	Initial skeleton — chapter outline, FUB list, port list scaffold
0.2	2026-07-07	RTL Design Sherpa	Narrow-device (x16) support: `DRAM_DEVICE_WIDTH` param; burst-length scaling to device-word units (§15); `addr_mapper` column granularity = device

Version	Date	Author	Notes
			word via `BYTE_OFFSET_WIDTH` (§3); `DFI_PHASE` CSR (rd_phase/wr_phase). Fixes the on-silicon DDR2 read failure (Nexys A7 x16).
0.3	2026-07-07	RTL Design Sherpa	§20 mode_register: document burst length as a fixed-per-instance init constant, decoded (not hardcoded) so the same RTL retargets DDR2 BL4 / DDR3-DDR4 BL8; BC4/burst-chop declared out of scope (always issue full BL).
0.4	2026-07-12	RTL Design Sherpa	Full reconciliation with the rearchitected RTL. ch02 block chapters rewritten to the live FUBs: retired txn_queue/bank_machine/xbank_timers/cmd_encoder/odt_ctrl/standalone page_predictor -> CAMs / FSM-free bank_timer / global_timers / dfi_cmd_formatter / arbiter inline open-page. `addr_mapper` = single `ADDR_MAP.bank_lsb` knob (scheme mux retired). LPDDR2 fully functional (bit-exact JESD209-2F CA + JEDEC MR init). `pumice_top_geared` host-width gearing. CSR map reconciled to the RDL. All block diagrams regenerated.

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3 Architecture and Datapath

This chapter is the implementation-level architectural orientation: the live three-layer hierarchy, the read/write datapath flow, and the clocking topology. Per-macro detail is in section 2; per-interface detail is in section 3.

3.1 Layer Hierarchy

The controller is a **three-layer core** wrapped by a top that adds the PeakRDL-generated register block. The top-of-tree is `pumice_top` (`rtl/top/pumice_top.sv`), with an optional host-width wrapper `pumice_top_geared` above it:

```
pumice_top_geared (rtl/top/pumice_top_geared.sv)   -- OPTIONAL host-
width wrapper
  |- axi4_dwidth_converter_wr/_rd (formal IP; only when
HOST_AXI_DATA_WIDTH != DW)
  |- pumice_top (rtl/top/pumice_top.sv)
       |- pumice_csr (regs/generated/rtl/, PeakRDL passthrough cpuif)
       |- pumice_core (rtl/top/pumice_core.sv)
            |- pumice_axi4_ifc          (rtl/macro/pumice_axi4_ifc.sv)
            |- pumice_mem_cmd_scheduler
(rtl/macro/pumice_mem_cmd_scheduler.sv)
            |- pumice_dfi_layer
(rtl/macro/pumice_dfi_layer.sv)
```

`pumice_core` wires the three macros, built bottom-up:

Macro (module)	Role	Chapter
`pumice_axi4_ifc`	Host AXI4 slave + burst splitters + write/read CAMs + snarf	2.1
`pumice_mem_cmd_scheduler`	Arbiter + per-bank/global timers + refresh + init + mode reg	2.2
`pumice_dfi_layer`	Single async-FIFO CDC + DFI-clock command/write/read datapath	2.4

The data path is **not** a standalone macro. Write and read burst buffers live in the CAMs inside `pumice_axi4_ifc` (SRAM-backed, de-FSM'd streaming readers); the DFI-clock serializer / aligner live in `pumice_dfi_layer`. Section 2.3 documents this split in place of the old `data_path_macro`.

Configuration (timings, DFI phases, page policy, address map) is delivered to `pumice_core` on ports, driven **by name** from the CSR `hwif_out.*` fields in `pumice_top`. There are no config ports on `pumice_top` other than the register cpuif – software programs every timing/phase/policy through the register bus.

3.2 Read / Write Datapath: AXI -> split -> intake -> CAM

The host AXI4 face is `pumice_axi4_ifc`. Each host burst is first split at DRAM-burst-byte boundaries by the shared `axi_master_wr_splitter` / `axi_master_rd_splitter` (one DRAM burst per split command), then handed to the dumb 1:1 intakes:

- `pumice_wr_intake` (`axi4_slave_wr` + AW-meta FIFO + wr-data FIFO + `addr_mapper`) pushes `(bank, row, col, id)` plus write data into `pumice_wr_data_cam`.
- `pumice_rd_intake` (`axi4_slave_rd` + `addr_mapper` + snarf probe) pushes `(bank, row, col, id)` into `pumice_rd_cmd_cam`, and probes the write CAM for read-your-write forwarding.

The salient property: there is exactly **one** stage where AXI-layer concepts (burst, ID, write strobe) cross into DRAM-layer concepts (rank, bank, row, column). That stage is `addr_mapper` inside each intake. Upstream everything is AXI; downstream of the CAMs everything is DRAM.

3.2.1 Why Two CAMs, Not One

The CAM split is between the write and read paths because the two CAMs hold **different metadata**, carry **different data**, and have **different lifetimes**:

CAM	Key	Data / storage	Retired on
`pumice_wr_data_cam`	(bank, row)	(col, id, age, slot) + write-data SRAM; fill/commit-drain/snarf movers	commit-done (B push)
`pumice_rd_cmd_cam`	(bank, row)	(col, id, age, slot) + read-return SRAM; return-fill/drain movers	last drained R beat

A single unified CAM would either carry both data sets (the write CAM needs a write-data SRAM the read CAM does not) or force an awkward "is_write" predicate on every scheduler lookup. Keeping them separate lets the read and write paths size independently.

Both CAMs are **de-FSM'd**: each stores burst data in an SRAM and exposes a streaming read engine that is FIFO-fed / oldest-pick beat-counter driven – no active/slot state latch. The `r_fdone` fill-complete flag gates schedulability (and, on the write side, snarf).

3.2.2 Snarf (read-your-write forwarding)

`pumice_rd_intake` probes the write CAM before scheduling a read. On a hit the read is streamed directly from the write CAM's SRAM via the **snarf mover** in `pumice_wr_data_cam` – there is no standalone `wr2rd_forward` block in the live path. Snarf is limited to unscheduled writes with the same id and same burst length.

3.3 Scheduler and Command Stream

`pumice_mem_cmd_scheduler` queries both CAMs in the (bank, row) dimension via `N_LU = NUM_BANKS` parallel lookup ports, picks one command with `pumice_cmd_arbiter` (the open-page decision is **inline** in the arbiter – there is no separate `page_predictor` in the issue path), and

emits a single abstract DRAM command {op, rank, bank, row, col, ap} into an output command FIFO. The scheduler is single-issue, PHY/nphases-agnostic, and runs entirely on aclk. Per-bank JEDEC readiness is stamped by pumice_bank_timers (wrapping the FSM-free bank_timer); cross-bank turnaround windows (tFAW/tRRD/tWTR/tRTW/tCCD) come from global_timers.

3.4 Clocking Topology

Two external clocks; exactly one CDC.

Clock	Domain members
aclk	Host AXI, burst splitters, both CAMs, and the whole command scheduler
dfi_clk	DFI command path, write serializer, read aligner, DFI 2.1 pin bus

The single CDC lives inside pumice_dfi_layer (pumice_dfi_cdc) and is built from **async gaxi FIFOs only** – the command, write-data, and read-data streams plus the init handshake cross there. No other clock crossing exists in the controller. If the SoC's AXI master is on a different clock than aclk, an external clock converter is required upstream. See section 1.3 for the reset topology.

3.5 DFI Phase / Rate Topology

The internal data unit is the **DFI word**: DW = DRAM_BEAT_WIDTH * DFI_RATE (128 by default). One AXI beat == one DFI word == DFI_RATE DRAM beats; one AXI burst (BL/DFI_RATE beats) == one DRAM burst (BL beats). The DFI command path in pumice_dfi_layer phase-packs the abstract command onto the DFI_RATE phases, placing the command and CS/ODT on wr_phase / rd_phase as programmed by the DFI_PHASE CSR; pumice_dfi_wr_serializer and pumice_dfi_rd_aligner handle the DFI-word to per-phase data split. The scheduler never sees the phase dimension – its issue rate is one command per aclk.

The old "AXI_DATA_WIDTH == DRAM_BEAT_WIDTH" coupling is gone. Host-width freedom is a separate concern handled by the optional pumice_top_geared wrapper, which inserts the formally-verified axi4_dwidth_converter_wr/_rd between a host-width AXI slave and the fixed-DW core (HOST == DW is a bit-identical generate bypass). See docs/AXI_DRAM_GEARING_SCOPE.md.

4 Top-Level Port List

This chapter is the **wire-level** port list of the `pumice_top` module (`rtl/top/pumice_top.sv`) – the exact signal names, directions, widths, and parameter dependencies that the integration script and SoC top-level use. The optional host-width wrapper `pumice_top_geared` re-exports the same set with a free `HOST_AXI_DATA_WIDTH` on the AXI slave face; see `docs/AXI_DRAM_GEARING_SCOPE.md`.

4.1 Parameters

`pumice_top` is fully parameterized – there is no `MEMTYPE`/`N_PHASES` string build any more; memory type is a runtime CSR field (`PHY_TIMING.memtype`) and the phase count is the integer `DFI_RATE`.

```
module pumice_top #(
    parameter int AXI_ID_WIDTH    = 8,
    parameter int AXI_ADDR_WIDTH  = 32,
    parameter int NUM_RANKS       = 1,
    parameter int NUM_BANKS       = 8,
    parameter int ROW_WIDTH       = 14,
    parameter int COL_WIDTH       = 10,
    parameter int DFI_RATE        = 2,
    parameter int DRAM_BEAT_WIDTH = 64,
    parameter int BL              = 8,   // DRAM beats per burst
    parameter int NUM_ENTRIES     = 8,   // CAM depth
    parameter int N_SRAM_SLOTS    = 8,   // per-CAM burst-data SRAM
slots
    parameter int CSR_ADDR_W      = 12
);
```

4.2 Parameter-Dependent Widths

Derived locally in the module (see `rtl/top/pumice_top.sv`):

- `DW` = `DRAM_BEAT_WIDTH * DFI_RATE` (host AXI data width == DFI word)
- `SW` = `DW/8`
- `IW` = `AXI_ID_WIDTH`
- `AW` = `AXI_ADDR_WIDTH`
- `PHW` = `(DFI_RATE > 1) ? $clog2(DFI_RATE) : 1`
- `DFI_DATA_WIDTH` = `DW`
- `DFI_STRB_WIDTH` = `DW/8`
- `DFI_EN_WIDTH` = `DFI_RATE`
- `DFI_VALID_WIDTH` = `DFI_RATE`

- `DFI_ADDR_BUS_W` = `ROW_WIDTH * DFI_RATE`
- `DFI_BANK_BUS_W` = `$clog2(NUM_BANKS) * DFI_RATE`
- `DFI_CTRL_BUS_W` = `1 * DFI_RATE`
- `DFI_CS_BUS_W` = `NUM_RANKS * DFI_RATE`

4.3 Clocks and Reset

Two clocks, two active-low resets. There is no APB CSR clock – the register block runs on `aclk` via a passthrough cpuif.

Port	Dir	Width	Domain	Notes
`aclk`	in	1	`aclk`	Host AXI + CAMs + scheduler
`aresetn`	in	1	`aclk`	Active-low async reset
`dfi_clk`	in	1	`dfi_clk`	DFI datapath + PHY pin bus
`dfi_rstn`	in	1	`dfi_clk`	Active-low async reset
`init_done_o`	out	1	`aclk`	Init sequencer done (STATUS)

4.4 Register cpuif (PeakRDL passthrough)

The register block `pumice_csr` is generated from `rtl/macro/pumice_csr.rdl` (via `bin/peakrdl_generate.py`). The top presents the raw PeakRDL passthrough cpuif – an SoC bridges its APB/AXI-Lite onto these signals.

Port	Dir	Width
`s_cpuif_req`	in	1
`s_cpuif_req_is_wr`	in	1
`s_cpuif_addr`	in	`CSR_ADDR_W`
`s_cpuif_wr_data`	in	32
`s_cpuif_wr_biten`	in	32
`s_cpuif_req_stall_wr`	out	1
`s_cpuif_req_stall_rd`	out	1
`s_cpuif_rd_ack`	out	1
`s_cpuif_rd_err`	out	1
`s_cpuif_rd_data`	out	32
`s_cpuif_wr_ack`	out	1

Port	Dir	Width
s_cpuif_wr_err	out	1

All config (timings, DFI phases, page policy, address map) is programmed through this bus by name via the generated dv/tbclasses/pumice_regmap.py; pumice_top fans hwif_out.* into pumice_core's config ports. See section 4 of this MAS (or the RDL) for the register map.

4.5 AXI4 Slave Port List

Standard AXI4 slave, single ID width IW, data width DW (= DFI word). All five channels are present (AW/W/B/AR/R) with the usual side-band (awcache/awprot/awqos/awregion/awlock/awuser and the AR equivalents). s_axi_awuser / s_axi_aruser / s_axi_wuser are single-bit at the top. Widths:

- Address: s_axi_awaddr / s_axi_araddr = AW
- Data: s_axi_wdata / s_axi_rdata = DW; s_axi_wstrb = SW
- ID: s_axi_awid / s_axi_bid / s_axi_arid / s_axi_rid = IW

The exact per-signal declaration is the AW/W/B/AR/R block in rtl/top/pumice_top.sv. AXI side-band awqos/arqos/awregion/arregion are carried through the burst splitters but do not drive scheduler priority.

4.6 DFI 2.1 Master Port List

The DFI pin bus is the wide (per-phase x DFI_RATE) form. Command control strobes are 1-bit-per-phase; address/bank are ROW_WIDTH/$clog2(NUM_BANKS) per phase.

Port	Dir	Width
dfi_address_o	out	DFI_ADDR_BUS_W
dfi_bank_o	out	DFI_BANK_BUS_W
dfi_cas_n_o	out	DFI_CTRL_BUS_W
dfi_ras_n_o	out	DFI_CTRL_BUS_W
dfi_we_n_o	out	DFI_CTRL_BUS_W
dfi_cs_n_o	out	DFI_CS_BUS_W
dfi_odt_o	out	DFI_CS_BUS_W
dfi_wrdata_o	out	DFI_DATA_WIDTH
dfi_wrdata_en_o	out	DFI_EN_WIDTH
dfi_wrdata_mask_o	out	DFI_STRB_WIDTH
dfi_rddata_en_o	out	DFI_EN_WIDTH

Port	Dir	Width
dfi_rddata_i	in	DFI_DATA_WIDTH
dfi_rddata_valid_i	in	DFI_VALID_WIDTH
dfi_init_start_o	out	1
dfi_init_complete_i	in	1

Notes:

- The DDR2 ras/cas/we strobes are present in all builds. For LPDDR2 (PHY_TIMING.memtype = 1) the command rides the CA bus packed onto dfi_address_o and these strobes idle; the memtype is a runtime field, so the same SV header serves both flavors.
- The DFI bus is driven in the dfi_clk domain. a7ddrphy (LiteDRAM's Xilinx 7-series PHY, the board target) handles the internal DFI-to-DRAM gearing; the controller does not drive per-phase read gearing beyond rd_phase.

4.7 SystemVerilog Header Snippet

The canonical source is rtl/top/pumice_top.sv – the module declaration there is authoritative. The parameter block above and the port groups (cpuif, AXI4, DFI 2.1) are a human-readable mirror of that file.

5 Clocks and Reset

5.1 Clock Domains

The controller has exactly two clock domains. Everything from the host AXI face through the command scheduler runs on aclk; the DFI datapath and PHY pin bus run on dfi_clk. The register block pumice_csr runs on aclk (there is no separate APB CSR clock).

Clock	Polarity	Domain members
aclk	Posedge	pumice_csr, pumice_axi4_ifc (AXI slave, burst splitters, both CAMs), pumice_mem_cmd_scheduler (arbiter, bank/global

Clock	Polarity	Domain members
		timers, refresh, init, mode register)
dfi_clk	Posedge	pumice_dfi_layer command path, write serializer, read aligner; the DFI 2.1 pin bus

aclk and dfi_clk are independent. The DFI-word datapath unit means one FIFO word equals one dfi_clk cycle, so the crossing is bubble-free at rate.

5.2 Reset Topology

Reset	Polarity	Type	Domain
aresetn	Active-low	Async assert	aclk
dfi_rstn	Active-low	Async assert	dfi_clk

Each domain has its own reset. pumice_csr takes rst = ~aresetn (the PeakRDL block uses active-high internally). All flops use the repo reset macros (reset_defs.svh); SRAMs inside the CAMs have no reset port. The SoC's PMU is expected to drive clean, de-glitched resets – there is no reset-glitch filter in the controller.

5.3 CDC

There is exactly **one** clock-domain crossing in the whole controller, and it lives in pumice_dfi_cdc inside pumice_dfi_layer. It is built from **async gaxi FIFOs only** (N_FLOP_CROSS = 2 by default). Four things cross it:

Stream	Direction	Payload
Command	aclk -> dfi_clk	{op, rank, bank, row, col, ap} (CMD_DW)
Write data	aclk -> dfi_clk	{last, strb, data} DFI-word (WD_DW)
Read data	dfi_clk -> aclk	{last, resp, data} DFI-word (RD_DW)
Init handshake	both	init_start out, init_complete back

There is **no** APB-to-MC CSR crossing, no quiet-point override-staging crossing, and no CDC in the AXI4 datapath – pumice_axi4_ifc runs entirely on aclk. If the SoC's AXI master is on a different clock, an external clock converter is required upstream. The register block also runs on aclk, so CSR writes take effect combinationally through hwif_out.* into the config ports of pumice_core (no staging register). Timing/phase/policy fields should be programmed while the controller is idle (before init_start, or at a quiet point) since they feed the timers and phase-packers directly.

5.4 Reset / Init Sequence

On power-on:

1. `aresetn` and `dfi_rstn` are asserted (both low). The PHY drives `dfi_init_complete_i = 0`.
2. The SoC deasserts both resets. `pumice_csr` becomes R/W-able on `aclk`; software programs the timing, DFI-phase, page-policy, and address-map fields by name (never by hardcoded offset – see `dv/tbclasses/pumice_regmap.py`). The controller idles with `init_done_o = 0`; AXI traffic is held off.
3. The `init_sequencer` (inside `pumice_mem_cmd_scheduler`) drives `dfi_init_start_o` and walks the per-memtype JEDEC MRS init sequence, emitting init commands into the same arbiter path as normal traffic and waiting on the programmed init timings (`t_init_wait`, `t_dll_wait`, `t_mrd_wait`, `t_rp_wait`, `t_rfc_wait`) plus `dfi_init_complete_i` from the PHY. The `mode_register` shadow captures the MRS writes and exposes CL/CWL/BL to the DFI layer.
4. On completion the sequencer asserts `init_done_o`; the controller begins honoring host AXI traffic and the refresh controller is enabled.

Because the CSR block and the scheduler share `aclk`, no inter-clock handshake gates step 2; software only needs both resets released and the register bus alive.

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6 Top-Level Integration (`pumice_core` / `pumice_top`)

Modules: `pumice_core.sv`, `pumice_top.sv` **Location:** `rtl/top/` **Category:** Integration (structural wiring) **Status:** Implemented

> **Note:** the early SWAG used a five-`*_macro` hierarchy (`pumice_core_macro`, `axi_frontend_macro`, `command_scheduler_macro`, `data_path_macro`, `dfi_v21_interface_macro`). That naming is retired. The live controller is a three-layer stack instantiated by `pumice_core`, wrapped by `pumice_top` (which adds the PeakRDL CSR block) and

optionally by `pumice_top_geared` (which adds a host-width AXI dwidth shim).

6.1 Purpose

The controller top is structural wiring only — every behavioral statement lives in a FUB or a layer macro. Two files make up the top:

- **`pumice_core`** — wires the three functional layers on their two clocks and exposes host AXI4 plus the DFI 2.1 pin bus. Config arrives on ports.
- **`pumice_top`** — instantiates `pumice_core` plus the PeakRDL-generated `pumice_csr` register block, and drives every core config port **by name** from the CSR `hwif_out.*` decode. It exposes the register cpuif, host AXI4, and the DFI pin bus.

An optional third file, `pumice_top_geared`, wraps `pumice_top` with a free `HOST_AXI_DATA_WIDTH` and inserts the repo's formally-verified `axi4_dwidth_converter_wr` / `axi4_dwidth_converter_rd` between a host-width AXI slave and the fixed-DW core. `HOST == DW` is a generate bypass (bit-identical). See `docs/AXI_DRAM_GEARING_SCOPE.md`.

6.2 `pumice_core` — the three layers

`pumice_core` instantiates exactly three layer modules and the nets between them. There is no FSM, no CSR decode, and no arithmetic beyond the packed-bus `assign` for the command word.

```
pumice_core
├── u_ifc   : pumice_axi4_ifc          (host AXI + wr/rd CAMs)
[aclk]
├── u_sched : pumice_mem_cmd_scheduler (arbiter + timers +
refresh/init)[aclk]
└── u_dfi   : pumice_dfi_layer         (single async CDC + DFI
datapath)[aclk→dfi_clk]
```

- Host AXI, the scheduler, and both CAMs run on **`aclk`** (`aresetn`).
- The DFI phase-packer and PHY interface run on **`dfi_clk`** (`dfi_rstn`).
- The **one** clock crossing in the whole controller is inside `pumice_dfi_layer` (async gaxi FIFOs only). `pumice_dfi_layer` is instantiated with `.ctl_clk(aclk)` / `.ctl_rstn(aresetn)` on the control side and `.dfi_clk` / `.dfi_rstn` on the PHY side.

The internal data unit is the **DFI word** (`DFI_DATA_WIDTH = DRAM_BEAT_WIDTH * DFI_RATE`, default 128). The host AXI data width equals the DFI word; a host that runs a different AXI width

uses the external `pumice_top_geared` shim (a separate edge concern — the core does not gear internally).

6.2.1 Layer instantiation inventory (`pumice_core`)

Instance	Module	Count	Clock	Role
`u_ifc`	`pumice_axi4_ifc`	1	`aclk`	Host AXI4 face; wr-data CAM + rd-cmd CAM
`u_sched`	`pumice_mem_cmd_scheduler`	1	`aclk`	Command arbiter + bank/global timers + refresh + init + mode-register shadow
`u_dfi`	`pumice_dfi_layer`	1	`aclk` → `dfi_clk`	Single async CDC + DFI cmd/wr/rd datapath

There are **no** `generate` fan-out blocks in `pumice_core`. Per-(rank, bank) fan-out (the bank timers) happens one level down, inside `pumice_mem_cmd_scheduler`.

6.2.2 Inter-layer nets

`pumice_core` declares and wires four net groups:

1. **Scheduler ↔ IFC CAM ports** — the per-bank scheduler lookup buses (`w_wr_lu_*`, `w_rd_lu_*`), the oldest-entry ports (`w_wr_old_*`, `w_rd_old_*`), the write commit handshake (`w_wr_commit_*`), and the read issue handshake (`w_rd_issue_*`). `N_LU = NUM_BANKS` (one lookup per bank).
2. **IFC commit-data → DFI wrdata** (`w_cm_*`: valid/ready/data/strb/last) and **DFI rddata → IFC rd-return** (`w_ret_*`: valid/ready/data/resp/last).
3. **Scheduler command stream → DFI** — the abstract command `{op, rank, bank, row, col, ap}`, flattened into `w_cmd_data` (width `CMD_DW = 4 + RKW + BKW + ROW_WIDTH + COL_WIDTH + 1`) by a single `assign`.
4. **Init handshake** — `w_init_start` / `w_init_complete` between the scheduler's init sequencer and the DFI layer's PHY-init side.

Note the parameter mapping at the IFC boundary: `pumice_core` passes `.BL(BL / DFI_RATE)` to `pumice_axi4_ifc` (the IFC/CAM view of burst length is in DFI words), while `pumice_dfi_layer` receives the full `.BL(BL)` (DRAM beats).

6.3 `pumice_top` — CSR + by-name config

`pumice_top` adds the register block and the config fan-out. It does two things:

1. Instantiates `pumice_csr` (PeakRDL passthrough cpuif) with the `hwif_in` / `hwif_out` struct pair. `hwif_in` is currently tied to `'{default:'0}` — status/observability readback is a follow-up; config-drive is wired first.
2. Instantiates `pumice_core` and drives **every** config port by name from the decoded `hwif_out.*` fields. Representative mappings:

Core port	CSR field (`hwif_out.*`)
`memtype_i`	`PHY_TIMING.memtype` (0 = DDR2, 1 = LPDDR2)
`page_policy_i`	`REFRESH_TUNING.page_policy_or`
`bank_lsb_i`	`ADDR_MAP.bank_lsb`
`hash_en_i`	`ADDR_MAP.hash_en`
`hash_seed_i`	`ADDR_MAP.hash_seed`
`t_rcd_i/t_rp_i/ t_ras_i/t_rc_i`	`TIMINGS_RC_RCD_RP_RAS.*`
`t_wr_i`	`TIMINGS_CL_CWL_WR.tWR`
`t_rtp_i/t_rtw_i`	`TIMINGS_RTP_RTW.*`
`t_faw_i/t_rrd_i/ t_wtr_i/t_ccd_i`	`TIMINGS_RRD_FAW_WTR_CCD.*`
`t_refi_i`	`TIMINGS_RFC_REFI.tREFI`
`refresh_burst_i`	`PHY_TIMING.refresh_burst`
`t_init_wait_i/ t_dll_wait_i`	`INIT_TIMING0.*`
`t_mrd_wait_i/ t_rp_wait_i/ t_rfc_wait_i`	`INIT_TIMING1.*`
`rd_phase_i/ wr_phase_i`	`DFI_PHASE.*` (sliced to `[PHW-1:0]`)
`t_phy_wrlat_i/ t_rddata_en_i`	`PHY_TIMING.*`

The CSR block is clocked on `aclk` with `.rst(~aresetn)` (PeakRDL uses an active-high reset; the top inverts `aresetn`). See `rtl/macro/pumice_csr.rdl` for the full field list.

6.4 What the top does not do

- No combinational logic beyond the single command-word `assign` in `pumice_core` and the by-name field selects in `pumice_top`.
- No CSR field expansion in `pumice_core` (that is `pumice_csr`'s job; the top only reads decoded `hwif_out` fields).
- No clock gating (a synthesis-script concern).

- No reset synchronization in the top — each layer takes its own domain reset (`aresetn` for `aclk` logic, `dfi_rstn` for the PHY domain); the CDC inside `pumice_dfi_layer` owns the cross-domain reset discipline.

This intentional poverty makes the top the obvious place for the structural-only sanity check: every line is an instantiation, a net declaration, or a port/field mapping.

---

7 `pumice_axi4_ifc` (AXI4 front-end macro)

Module: `pumice_axi4_ifc.sv` **Location:** `rtl/macro/` **Category:** Layer-1 macro (the AXI front-end of `pumice_core`) **FUBs bundled:** burst splitters + 2 intakes + 2 CAMs

7.1 Purpose

"Host AXI4 to CAM-cached DRAM requests." The first of the three core layers. It bolts the shared AXI burst splitters onto dumb 1:1 intakes, holds the write-data CAM (write buffer + snarf source) and the read-command CAM (read reorder buffer), presents the host AXI4 face, and exposes the scheduler lookup/oldest/commit/issue ports and the DFI wr-commit-data / rd-return ports outward to the other two layers.

```
host AXI4 -> [wr/rd splitter] -> pumice_wr_intake ->
pumice_wr_data_cam
                                -> pumice_rd_intake -> pumice_rd_cmd_cam
```

Runs entirely on `aclk`. This replaces the old `axi_frontend_macro` (`axi_intake` / `addr_mapper` / `wr_cmd_cam` / `rd_cmd_cam` / `wr2rd_forward`) – those names are retired.

7.2 FUBs

FUB	Role
`axi_master_wr_splitter` / `axi_master_rd_splitter`	Shared AMBA IP; split each host burst at DRAM-burst-byte boundaries (`ALIGN_MASK = BL*(DRAM_BEAT_WIDTH/8) - 1`), one DRAM burst per split command
`pumice_wr_intake`	`axi4_slave_wr` + AW-meta FIFO + wr-data FIFO + `addr_mapper`; emits `(bank,row,col,id)` push plus a wr-data stream; consumes commit-done for B response

FUB	Role
pumice_rd_intake	axi4_slave_rd + addr_mapper + snarf probe; emits (bank,row,col,id) push; probes the write CAM and, on a snarf hit, streams the R response from the write CAM SRAM; otherwise consumes the drained DFI return
pumice_wr_data_cam	Write CAM keyed on (bank,row) + write-data SRAM; fill / commit-drain / snarf movers; oldest + N_SCHED_LU lookup + commit ports; commit-data out to the DFI write path
pumice_rd_cmd_cam	Read CAM keyed on (bank,row) + read-return SRAM; return-fill / drain movers; oldest + N_SCHED_LU lookup + issue ports; DFI-return in, drain out to the rd intake

Address decode (addr_mapper) is driven by bank_lsb_i / hash_en_i / hash_seed_i (the ADDR_MAP CSR); there is no scheme selector. See section 4 of this MAS for the address-map field semantics.

7.3 External Boundaries

- **Upstream:** the host AXI4 slave port (SoC-facing).
- **To pumice_mem_cmd_scheduler:** for each CAM, the oldest_* snapshot, the N_SCHED_LU parallel sched_lu_* lookup ports (valid/bank/row in, hit/slot/col/id/age out), and the write commit / read issue handshakes. The scheduler drives these on aclk.
- **To pumice_dfi_layer:** write commit-data out (wr_cm_rd_*: valid/ready/data/strb/last) to the DFI write serializer; read DFI return in (rd_dfi_ret_*: valid/ready/data/resp/last) from the DFI read aligner.
- **Internal:** the read intake's snarf probe hits pumice_wr_data_cam directly (snarf_probe_* -> snarf_hit/snarf_rd_*); the write CAM's commit_done_* retires the AW-meta FIFO entry in pumice_wr_intake to emit the B response.

The busy_o output is the OR of the four sub-block busy flags.

7.4 De-FSM'd CAMs

Both CAMs store burst data in an SRAM and read it back with a streaming engine that is FIFO-fed / oldest-pick beat-counter driven – there is no active/slot state machine. The r_fdone fill-complete flag gates when a slot is schedulable (and, on the write side, snarfable). The write CAM's snarf mover is the sole read-your-write forwarding path; there is no separate forwarder.

7.5 Tests

FUB-level unit tests exist for each intake and each CAM (`dv/tests/fub/`), and a wrapper-level test drives `pumice_axi4_ifc` through the AXI4 BFMs (`AXI4MasterWrite`/`AXI4MasterRead` + `AXI4Sequence`). Drive AXI traffic through the BFMs – never hand-poke `s_axi_*`.

---

8 `pumice_mem_cmd_scheduler` (command scheduler macro)

Module: `pumice_mem_cmd_scheduler.sv` **Location:** `rtl/macro/` **Category:** Layer-2 macro (the decision layer of `pumice_core`) **FUBs bundled:** arbiter + bank/global timers + refresh + init + mode reg + cmd FIFO

8.1 Purpose

"What command should we issue this cycle?" – the decision-making layer. It wires the command arbiter to the per-bank safe timers, the global (bus/rank) turnaround timers, the refresh controller, the init sequencer plus mode-register shadow, and an output command FIFO. It emits a single abstract DRAM command stream `{op, rank, bank, row, col, ap}` for the DFI layer to phase-pack.

The scheduler is **single-issue**, **PHY / nphases-agnostic**, and runs on a single controller clock (`aclk`). The CAM sched-lookup / oldest / commit / issue ports are **external** – the CAMs live in `pumice_axi4_ifc`. This replaces the old `command_scheduler_macro` (`scheduler` / `xbank_timers` / `page_predictor` / `powerdown_ctrl` etc.); those names are retired.

8.2 FUBs

FUB	Role
`pumice_cmd_arbiter`	The single pick core. Picks one command per cycle from the CAM lookups + oldest snapshots under page policy; open-page decision is **inline** here (no separate predictor). Drives the write `commit` / read `issue` handshakes and the per-command `evt_*` strobes; emits the abstract command.
`pumice_bank_timers`	Stamps a `bank_timer` per (rank,bank) and fans the `evt_*` strobes; produces `bank_act/rdwr/pre_ready`, `bank_row_active`,

FUB	Role
	bank_open_row.
bank_timer	FSM-free per-bank JEDEC safe timers (tRCD/tRP/tRAS/tRC/tWR/tRTP) + a row-open register + a single auto-precharge bit. Combinational readiness off the countdown timers.
global_timers	Cross-bank/rank turnaround windows: tFAW + tRRD per rank, tWTR/tRTW/tCCD global. Produces _ok window flags.
refresh_ctrl	tREFI down-counter + 8-deep postpone; refresh_req/refresh_drain to the arbiter, refresh_grant back. Enabled after init.
init_sequencer	DDR2 + LPDDR2 JEDEC MRS init microprogram. Drives dfi_init_start_o, emits init commands into the arbiter, and writes the MR shadow. Gates traffic until init_done.
mode_register	MR shadow updated by the init MRS writes; decodes CL / CWL / BL (and AL / drive-strength / ODT). Supplies cl_o/cwl_o/bl_o to the DFI layer.
gaxi_fifo_sync	Output command FIFO (CMD_FIFO_DEPTH), packs {ap, col, row, bank, rank, op} (CMD_W bits) into the abstract command stream to the DFI layer.

8.3 FSM-Free Bank Timing

bank_timer is deliberately **stateless-in-the-control-sense**: preset / decrement countdown timers (rcd/ras/rc/rp/preblk), a row-open register, and one auto-precharge bit – no multi-state bank machine. The per-command safe_act/rd/wr/pre outputs are combinational off the timers with a single register stage, so the arbiter reads readiness with one cycle of latency. This replaced the old double-registered 3-state bank FSM (retired bank_machine), which caused refresh-vs-ACT and column-vs-PRE hazards. pumice_bank_timers stamps the timer per (rank,bank).

8.4 External Boundaries

- **CAM sched ports (to pumice_axi4_ifc):** for each of write and read, the N_LU = NUM_BANKS lookup buses (lu_valid/bank/row out; hit/slot/col/id/age in), the oldest_* snapshot in, and the write commit_* / read issue_* handshake out.
- **Command stream out (to pumice_dfi_layer):** cmd_valid/ready + {op, rank, bank, row, col, ap}.

- **Mode-register shadow out:** `cl_o` / `cwl_o` / `bl_o` to the DFI layer.
- **Init handshake:** `dfi_init_start_o` out and `dfi_init_complete_i` in (routed to/from the PHY via the DFI layer); `init_done_o` status out.
- **Config in:** the JEDEC timings (tRCD/tRP/tRAS/tRC/tWR/tRTP/tFAW/tRRD/tWTR/ tRTW/tCCD/tREFI), `refresh_burst`, the init waits (`t_init_wait/t_dll_wait/t_mrd_wait/t_rp_wait/t_rfc_wait`), `page_policy_i`, and `memtype_i` – all delivered from the CSR by name.

`busy_o` asserts while not init-done, on a pending refresh, while a command is buffered in the FIFO, or while any bank row is active.

8.5 Tests

The scheduler has a macro-level test that exercises the arbiter against the two real CAMs – combinational pickers must be macro-tested, since registered feedback latency can hide double-issue hazards that a FUB-only test misses. Each constituent FUB also has its own unit test in `dv/tests/fub/`.

---

9 Data Path (no standalone macro)

Category: Distributed across `pumice_axi4_ifc` and `pumice_dfi_layer` **Status:** the old `data_path_macro` (`wr_beat_sequencer` + `rd_cl_aligner`) is retired – there is no `data_path_macro.sv` in the live tree.

9.1 Purpose

"Move bytes between the host AXI buffers and the DFI data lanes." In the rearchitected core this is **not a separate macro**. The data path is split across two of the three core layers:

- The **burst buffers** live in the two CAMs inside `pumice_axi4_ifc` (section 2.1). Write data is buffered in `pumice_wr_data_cam`'s SRAM; read return data is buffered in `pumice_rd_cmd_cam`'s SRAM. Both use de-FSM'd streaming read engines rather than a beat-sequencer FSM.
- The **DFI-clock serialize / align** stages live in `pumice_dfi_layer` (section 2.4): `pumice_dfi_wr_serializer` presents write data on `dfi_wrdata` at

t_phy_wrlat, and pumice_dfi_rd_aligner captures dfi_rddata at t_rddata_en and reassembles DFI words.

The controller-to-PHY data crossing is carried as whole DFI words through the single CDC in pumice_dfi_layer (section 1.3), so there is no separate data-path clock boundary to describe.

9.2 Write Path (host -> DRAM)

1. pumice_wr_intake streams W beats (DFI-word granular {data, strb, last}) into pumice_wr_data_cam's SRAM; r_fdone marks the burst fully filled.
2. When the scheduler commits the write slot, the CAM's **commit-drain mover** streams the burst out on the wr_cm_rd_* port (data/strb/last) into the DFI layer's write FIFO.
3. pumice_dfi_wr_serializer (on dfi_clk) drives dfi_wrdata / dfi_wrdata_en / dfi_wrdata_mask at t_phy_wrlat after the WR fire strobe from the command path. AXI wstrb = 1 (write this byte) maps to the DFI mask as-carried by the CAM strobe payload.

The write CAM's SRAM is also the **snarf source** (read-your-write forwarding): pumice_rd_intake probes it and, on a hit, streams the R response straight from the write SRAM without a DRAM round-trip.

9.3 Read Path (DRAM -> host)

1. When the scheduler issues the read slot, pumice_dfi_rd_aligner (on dfi_clk) drives dfi_rddata_en at t_rddata_en after the RD fire strobe, captures dfi_rddata when dfi_rddata_valid asserts, and packs whole DFI words ({data, resp, last}) into the read FIFO.
2. The FIFO crosses to aclk (the single CDC) and lands as the dfi_ret_* stream into pumice_rd_cmd_cam's **return-fill mover**, which writes the read SRAM.
3. The CAM's **drain mover** streams the buffered burst out oldest-first, gated on data-ready, back through pumice_rd_intake to the AXI R channel.

9.4 Why No Beat-Sequencer FSM

The old wr_beat_sequencer / rd_cl_aligner FUBs carried an active/slot state latch and per-beat control FSM. The live CAMs replace that with FIFO-fed / oldest-pick beat counters and a r_fdone fill flag – fewer control-path hazards and a cleaner streaming shape. The only genuinely PHY-timed logic (t_phy_wrlat / t_rddata_en alignment) is the small serializer / aligner pair in the DFI layer.

9.5 Tests

Covered by the CAM FUB tests and the DFI-layer FUB tests (`pumice_dfi_wr_serializer`, `pumice_dfi_rd_aligner`) in `dv/tests/fub/`, plus the `pumice_axi4_ifc` wrapper test for the end-to-end buffered data flow.

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10 `pumice_dfi_layer` (DFI v2.1 interface macro)

Module: `pumice_dfi_layer.sv` **Location:** `rtl/macro/` **Category:** Layer-3 macro (the PHY-facing layer of `pumice_core`) **FUBs bundled:** single CDC + command path + write serializer + read aligner

10.1 Purpose

“Translate the internal command + data streams into DFI v2.1 wires, and hold the one clock crossing.” This layer owns the JEDEC command encoding for DDR2/LPDDR2, the multi-phase pack, and the **single** controller-to-PHY clock crossing. It presents the controller-clock command / wrdata / rddata streams on one side and the DFI 2.1 pin bus on the other.

Swap THIS macro when moving to DFI v3+ for newer DRAM generations – the other two core layers (AXI front-end, scheduler) are DFI-version-agnostic.

This replaces the old `dfi_v21_interface_macro` (`dfi_cmd_formatter` + `dfi_signal_pack`) and the retired `gear_dfi` block; the CDC that used to be implied elsewhere is now explicitly this layer’s `pumice_dfi_cdc`.

10.2 FUBs

FUB	Clock	Role
`pumice_dfi_cdc`	both	The **single** controller/PHY crossing – async gaxi FIFOs only. Carries cmd (`aclk->dfi_clk`), wrdata (`aclk->dfi_clk`), rddata (`dfi_clk->aclk`), plus the init handshake.
`pumice_dfi_cmd_path`	`dfi_clk`	Command FIFO -> DFI command bus. Uses `dfi_cmd_formatter` (+ `dfi_signal_pack`) to phase-place the JEDEC command on `wr_phase` / `rd_phase`; drives address/bank/ras/cas/we/cs/odt; emits `wr_fire` /

FUB	Clock	Role
		rd_fire strobes.
pumice_dfi_wr_serializer	dfi_clk	Write-data FIFO -> dfi_wrdata / dfi_wrdata_en / dfi_wrdata_mask at t_phy_wrlat after wr_fire.
pumice_dfi_rd_aligner	dfi_clk	Drives dfi_rddata_en at t_rddata_en after rd_fire; captures dfi_rddata on dfi_rddata_valid; packs DFI words into the read FIFO.

The internal datapath unit is the **DFI word** (dfi_wrdata width, all DFI_RATE phases). One FIFO word equals one DFI cycle, so the datapath is bubble-free at rate.

10.3 External Boundaries

- **Controller side (ctl_clk = aclk):** cmd_* in (from the scheduler, packed {op,rank,bank,row,col,ap}), wd_* in (write commit-data from pumice_wr_data_cam, {last,strb,data} DFI-word), rd_* out (read return to pumice_rd_cmd_cam, {last,resp,data}), and init_start_i / init_complete_o.
- **PHY side (dfi_clk):** the full DFI 2.1 pin bus – dfi_address / dfi_bank / dfi_cas_n / dfi_ras_n / dfi_we_n / dfi_cs_n / dfi_odt, dfi_wrdata / dfi_wrdata_en / dfi_wrdata_mask, dfi_rddata_en / dfi_rddata / dfi_rddata_valid, and dfi_init_start / dfi_init_complete.
- **Config in (dfi_clk domain):** memtype_i, rd_phase_i / wr_phase_i, t_phy_wrlat_i, t_rddata_en_i – delivered from the CSR by name.

10.4 Multi-Phase Pack Convention

For DFI_RATE = N, every DFI control bus is per-phase x N wide (for example, dfi_address_o = ROW_WIDTH * DFI_RATE, dfi_cs_n_o = NUM_RANKS * DFI_RATE). The command path places the active JEDEC command on the wr_phase / rd_phase slot as programmed by the DFI_PHASE CSR and drives NOP-equivalent idle on the other phases. The scheduler never sees the phase dimension; the command path absorbs all phase placement. dfi_rddata read gearing beyond rd_phase is left to the PHY (a7ddrphy handles its internal read gearing).

10.5 LPDDR2 Command Encoding

For memtype_i = LPDDR2, dfi_cmd_formatter takes the memtype branch and emits bit-exact JESD209-2F CA-bus commands (Table 60): the 10-bit CA bus over 2 edges, packed as a flat 20-bit word on dfi_address, with ras/cas/we idle. See rtl/LPDDR2_CA_ENCODING.md for the transcription. DDR2 uses the classic ras/cas/we/cs strobe encoding on the same pins.

10.6 Tests

Each FUB has its own unit test in `dv/tests/fub/` (`pumice_dfi_cdc`, `pumice_dfi_cmd_path`, `pumice_dfi_wr_serializer`, `pumice_dfi_rd_aligner`), plus the layer-level test that drives the full controller-to-PHY path against the DFI slave BFM.

---

11 AXI4 Slave + Intakes (`axi4_slave_wr` / `axi4_slave_rd`)

Modules: `axi4_slave_wr` / `axi4_slave_rd` (repo AMBA IP) inside `pumice_wr_intake` / `pumice_rd_intake` **Location:** intakes in `rtl/fub/`; slaves in the shared AMBA library **Category:** FUB (intakes) **Parent:** `pumice_axi4_ifc` **Status:** Implemented

> **Rearchitected:** the SWAG had a single `axi4_slave_fub` that also owned transaction state and an R-emit FSM. That block is retired. In the live design the AXI4 protocol engine is the repo's `axi4_slave_wr` / `axi4_slave_rd` skid IP, and each is wrapped by a thin **intake** FUB (`pumice_wr_intake`, `pumice_rd_intake`) that is FIFO-based and FSM-free. All in-flight transaction state lives downstream in the two CAMs (see ch02/04, ch02/05).

11.1 Purpose

The host-facing AXI4 protocol handling is split by direction and factored into two layers each:

- **`axi4_slave_wr` / `axi4_slave_rd`** — the repo's standard AXI4 slave skid buffers. They do the channel handshakes, provide `SKID_DEPTH_*` buffering, and present a clean post-skid "fub" face (`fub_axi_*`).
- **`pumice_wr_intake` / `pumice_rd_intake`** — the pumice-specific intake FUBs that sit on the post-skid face. They decode the burst address via `addr_mapper`, push a decoded command downstream to a CAM, stream write/read data through FIFOs, and return B/R responses.

Ahead of both intakes, `pumice_axi4_ifc` places the repo's `axi_master_wr_splitter` / `axi_master_rd_splitter` so that each burst delivered to an intake is **exactly one DRAM burst** (aligned to `DRAM_BURST_BYTES = BL * DRAM_BEAT_WIDTH/8`). The intakes therefore assume "one AXI burst == one DFI burst" and carry no burst-splitting logic.

11.2 `pumice_wr_intake` — structure

Exactly three things plus the address decoder, per the locked μarch spec:

1. **`axi4_slave_wr` (`u_slave_wr`)** — AXI protocol / skid buffering. Parameters `SKID_DEPTH_AW=2`, `SKID_DEPTH_W=4`, `SKID_DEPTH_B=2`.
2. **AW-meta FIFO (`u_aw_meta_fifo`)** — a `gaxi_fifo_sync`, depth `AW_FIFO_DEPTH` (4), width `AWM_W = 1 + IW + AW`, capturing `{err, awid, awaddr}` per burst.
3. **wr-data FIFO (`u_wr_data_fifo`)** — a `gaxi_fifo_sync`, depth `WDATA_FIFO_DEPTH` (16), width `WD_W = 1 + SW + DW`, capturing `{wlast, wstrb, wdata}` per AXI beat.
4. **`addr_mapper` (`u_addr_mapper`)** — combinational decode of the FIFO-head address into `{rank, bank, row, col}` for the downstream `aw_push`.

Plus a **B-response FIFO (`u_b_fifo`)**, depth `B_FIFO_DEPTH` (4), width `B_W = 2 + IW`.

11.2.1 Write data flow

1. **AW intake** — `fub_awvalid` writes `{err, awid, awaddr}` into the AW-meta FIFO. `fub_awready = w_awm_wr_ready` (accept while the FIFO has room).
2. **W intake** — `fub_wvalid` writes `{wlast, wstrb, wdata}` into the wr-data FIFO; `fub_wready = w_wd_wr_ready`. AW and W are decoupled — each has its own FIFO.
3. **Decode + push** — the AW-meta FIFO head address is decoded by `addr_mapper`; the decoded `{bank, row, col, id, err}` is presented on `aw_push_*` and valid whenever the AW-meta FIFO is non-empty. It pops on the downstream `aw_push_ready_i` handshake. (`aw_push_rank_o` is driven but left unconnected at the IFC — single-rank pick.)
4. **wr-data pop** — the wr-data FIFO head is exposed on `wdata_*` and drained by the wr-data CAM (`wdata_ready_i`) in commit order.
5. **B response** — a completed commit strobes `wr_done_valid_i` / `wr_done_id_i` from the CAM; that pushes `{resp, id}` into the B FIFO, which drives the AXI B channel.

11.2.2 Ragged-burst handling

The intake computes `w_aw_err = ((awlen+1)*GEAR != BL)` where `GEAR = AXI_DATA_WIDTH / DRAM_BEAT_WIDTH`. On a ragged burst the intake **self-generates** a SLVERR B response at the `aw_push` pop (err takes priority over a `wr_done`), and the downstream CAM drops the illegal command. A simulation `$error` fires when `RAGGED_ASSERT != 0` (a guardrail under `translate_off`); a companion assertion catches a B-push collision between the err path and a same-cycle `wr_done`.

11.3 `pumice_rd_intake` — structure

Mirror of the write intake, plus the read **snarf** path:

1. **`axi4_slave_rd` (`u_slave_rd`)** — AXI protocol / skid buffering (`SKID_DEPTH_AR=2`, `SKID_DEPTH_R=4`).
2. **`addr_mapper` (`u_addr_mapper`)** — decode `fub_araddr` → `{rank, bank, row, col}` at the AR inlet (combinational).
3. **Snarf probe** — the decoded `{bank, row, col, id, len}` is presented on the `snarf_probe_*` ports every cycle `fub_arvalid` is high; the wr-data CAM returns a combinational `snarf_hit_i`.
4. **Order FIFO (`u_order_fifo`)** — a `gaxi_fifo_sync`, depth `ORDER_FIFO_DEPTH` (8), width `ORD_W = 1 + IW`, storing `{source, id}` per admitted read to preserve AR order (`SRC_SNARF=1`, `SRC_DFI=0`).
5. **Source arbiter** — a combinational mux selecting the snarf stream or the DFI read-return stream by the order-FIFO head's `source` tag.
6. **rd-data FIFO (`u_rd_data_fifo`)** — a `gaxi_fifo_sync`, depth `RD_FIFO_DEPTH` (16), width `RD_W = 2 + 1 + IW + DW` = `{resp, last, id, data}`, feeding the AXI R channel.

11.3.1 Read admission and ordering

- An AR is admitted (`w_can_admit`) when the order FIFO has room **and**, for a MISS, `ar_push_ready_i` is asserted. `fub_arready = w_can_admit`.
- On a **HIT** (`snarf_hit_i`) the read is tagged `SRC_SNARF`, `snarf_accept_o` fires, and no `ar_push` is emitted (the data is streamed from the write CAM's SRAM — DRAM is stale for the in-flight write).
- On a **MISS** the read is tagged `SRC_DFI` and `ar_push_valid_o` fires with the decoded `{bank, row, col, id}` toward the scheduler / rd CAM.
- The source arbiter drains one burst at a time in AR order: the order-FIFO head's `source` selects which input stream is connected to the rd-data FIFO,

and the head is popped only when the last beat of that burst is accepted. So a snarf-ready read still waits behind an earlier DFI read at the head (in-order R per the AXI contract).

11.4 External interface

Both intakes present a full AXI4 write (AW/W/B) or read (AR/R) slave face plus:

Port group	Direction	Purpose
`bank_lsb_i` / `hash_en_i` / `hash_seed_i`	in	`ADDR_MAP` config forwarded to `addr_mapper`
`aw_push_*` / `ar_push_*`	out	Decoded command to the wr/rd CAM
`wdata_*` (wr)	out	Write-data pop to the wr-data CAM
`wr_done_*` (wr)	in	Commit-completion strobe from the CAM → B response
`snarf_probe_*` / `snarf_hit_i` / `snarf_accept_o` (rd)	in/out	Read-your-write probe into the wr CAM
`snarf_rd_*` (rd)	in	Snarf data stream from the wr CAM SRAM
`dfi_rd_*` (rd)	in	DFI read-return stream (MISS reads) from the rd CAM
`busy_o`	out	Any FIFO non-empty / slave busy

11.5 Notes

- There is **no** AXI ID side table, no `w_buf`/`b_fifo` in the SWAG sense, and no R-emit FSM. Per-ID metadata that must survive into the DRAM layer is carried in the CAM entries; everything else is dropped at the intake.
- Outstanding-count limits are structural: writes are bounded by the wr-data CAM depth (`NUM_ENTRIES`), reads by the rd-cmd CAM depth plus the order FIFO depth. There are no `MAX_PER_ID_*` parameters.
- The intakes are strict-FIFO and back-pressure cleanly: an intake stalls its AXI channel when the relevant FIFO or the downstream CAM is full.

12 Address Mapper (addr_mapper)

Module: addr_mapper.sv **Location:** rtl/fub/ **Category:** FUB (combinational; no clock) **Parent:** pumice_wr_intake / pumice_rd_intake **Status:** Implemented

> **Rearchitected:** the SWAG described a three-scheme decoder (ROW_MAJOR / BANK_INTERLEAVE / XOR_HASH) selected by a runtime scheme_active_i mux. That is retired. The live addr_mapper has **one** placement knob — bank_lsb_i — that slides the bank field through the (byte-stripped) column region, plus an optional bank XOR-hash. The classic schemes are now just settings of bank_lsb. There is no scheme selector, no SYNTH_* param, and no un-synthesized-scheme tie-off.

12.1 Purpose

addr_mapper decodes a flat AXI_ADDR_WIDTH-bit address into the DRAM-layer (rank, bank, row, col) tuple. It is **pure combinational** — no clock, no flop, no FSM. The output is valid the same cycle as the input. One instance sits at the head-address decode of each intake (pumice_wr_intake, pumice_rd_intake), so the CAMs store the decoded tuple rather than the raw AXI address.

The mapper is driven entirely from the ADDR_MAP CSR register (see rtl/macro/pumice_csr.rdl) via three runtime inputs; there are no per-scheme build parameters.

12.2 Parameters

Parameter	Default	Purpose
AXI_ADDR_WIDTH	32	Flat AXI byte-address width
NUM_RANKS	1	1, 2, or 4 (rank field width KW)
NUM_BANKS	8	4 or 8 (bank field width BW = $clog2(NUM_BANKS))
ROW_WIDTH	14	Row field width RW
COL_WIDTH	10	Column field width CW
BYTE_OFFSET_WIDTH	3	log2(beat byte size); low bits stripped to word

Parameter	Default	Purpose
		addr

12.3 Interface

Signal	Direction	Width	Description
axi_addr_i	input	AXI_ADDR_WIDTH	Flat AXI byte address
bank_lsb_i	input	5	Bank-field LSB in the word address (ADDR_MAP.bank_lsb)
hash_en_i	input	1	Enable the bank XOR-hash (ADDR_MAP.hash_en)
hash_seed_i	input	8	XOR-hash seed (ADDR_MAP.hash_seed)
rank_o	output	$clog2(max(NR,2))	Decoded rank
bank_o	output	$clog2(NUM_BANKS)	Decoded bank
row_o	output	ROW_WIDTH	Decoded row
col_o	output	COL_WIDTH	Decoded column

12.4 The single-knob layout

The address is first stripped of its byte offset to a **word address**:

```
w_word = axi_addr_i[AXI_ADDR_WIDTH-1 : BYTE_OFFSET_WIDTH]   (zero-
extended to 32b)
```

The bank field is BW bits wide and sits at bit position bank_lsb inside the word address. The column is split **around** the bank into a low part below it and a high part above it; the row and rank stack above the whole column region at **invariant** positions:

```
word address (low → high):
  [ col_lo(bank_lsb) | bank(BW) | col_hi(CW - bank_lsb) | row(RW) |
rank(KW) ]


  col = { col_hi, col_lo }
  row LSB is always at bit CW + BW      (only the bank slides; the
column
                                         auto-splits around it)
```

12.4.1 Field extraction (variable-base shifts/masks)

The RTL clamps `bank_lsb` to `[0, COL_WIDTH]` first (`w_blsb`), so `col_hi`'s width `CW - bank_lsb` never goes negative and the row/rank slices stay legal. Then, over 32-bit intermediates:

```
col_lo = w_word & ((1 << bank_lsb) - 1)
bank   = (w_word >> bank_lsb) & ((1 << BW) - 1)
col_hi = (w_word >> (bank_lsb + BW)) & ((1 << (CW - bank_lsb)) - 1)
row    = (w_word >> (CW + BW)) & ((1 << RW) - 1)
rank   = (w_word >> (CW + BW + RW)) & ((1 << KW) - 1)   // 0 when
NUM_RANKS == 1
col    = col_lo | (col_hi << bank_lsb)
```

12.5 The classic schemes are just `bank_lsb` settings

Legacy scheme	Equivalent setting
ROW_MAJOR	`bank_lsb == COL_WIDTH` — bank sits **above** the whole column
max BANK_INTERLEAVE	`bank_lsb == log2(cols/burst)` — minimal `col_lo`, so a burst's column walk stays inside one bank while consecutive lines round-robin banks
partial interleave	any `bank_lsb` in between
XOR_HASH	`hash_en == 1`, folded on top of **any** placement above

The default `ADDR_MAP.bank_lsb = 0x0A = COL_WIDTH` is therefore ROW_MAJOR. Software is expected to keep `log2(cols/burst) <= bank_lsb <= COL_WIDTH` so a DRAM burst's column walk never crosses a bank boundary; the RTL clamp enforces the upper edge.

12.6 Bank XOR-hash

When `hash_en_i` is set, each bank bit is XOR-folded with two row slices and the seed (a `genvar` loop of width `BW`):

```
bank_hashed[i] = bank_raw[i] ^ row[i] ^ row[MID] ^ hash_seed[i]
  where MID = (i + BW < ROW_WIDTH) ? (i + BW) : (ROW_WIDTH - 1)   //
clamped
w_bank = hash_en ? bank_hashed : bank_raw
```

This is the identical fold to the legacy XOR_HASH scheme and defeats power-of-two-stride hot-banking. `hash_seed_i` is runtime-modifiable, so a bring-up engineer can change the hash without rebuilding.

12.7 Mirror to the DV address model

This RTL is the bit-exact mirror of the DV-side address model; both must produce identical `(rank, bank, row, col)` for the same input and the same `bank_lsb`/`hash_en`/`hash_seed`. The mirror eliminates the class of bugs where the BFM decodes one way and the controller decodes another. It is exercised by the `addr_mapper` FUB test, which sweeps addresses and `bank_lsb` values and asserts bit-equality against the model.

12.8 Timing

Purely combinational. The longest path is the hash fold (a few XOR levels) plus the variable-base shift/mask extraction. At the board target this is comfortably within a cycle; if a future high-frequency target cannot meet timing, the mapper can be treated as a multi-cycle path (the placement knob only changes at quiet points).

12.9 Notes / flags

- `bank_lsb_i` is 5 bits wide, matching `ADDR_MAP.bank_lsb[4:0]`.
- `rank_o` width uses `$clog2(NUM_RANKS > 1 ? NUM_RANKS : 2)` so a single-rank build still has a legal 1-bit port; `rank_o` is tied to `'0` when `NUM_RANKS == 1`.
- There is **no** `bg_field_pos` / bank-group input (DDR2/LPDDR2 have no bank groups), no `scheme_active_i`, and no CSR `pslverr` path for unsupported schemes — those SWAG concepts are gone.

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13 Read Command CAM (`pumice_rd_cmd_cam`)

Module: `pumice_rd_cmd_cam.sv` **Location:** `rtl/fub/` **Category:** FUB **Parent:** `pumice_axi4_ifc` **Status:** Implemented (de-FSM'd streaming drain)

> **Rearchitected:** the SWAG CAM was keyed by AXI ID and drove a `match_pending` vector with per-slot beat counters and an issue FSM. The live `pumice_rd_cmd_cam` is keyed by `{bank, row}` with a free-running age, exposes `N_SCHED_LU` parallel scheduler lookups plus an oldest port, and is a **read reorder buffer**: DRAM read data returns in *issue* order into per-entry SRAM slots and drains to the intake in *AR (insert)* order.

The drain has **no** active/slot state latch — a burst beat-counter and the combinational oldest-valid pick are the only sequencing.

13.1 Purpose

`pumice_rd_cmd_cam` tracks every outstanding DRAM read (the MISS path — snarf hits never enter this CAM). It is the mirror of `pumice_wr_data_cam` on the issue side, but the data direction is opposite: data comes **in** from the DFI read return and drains **out** to `pumice_rd_intake`. There is no snarf port (that is a write-CAM concept).

It acts as a reorder buffer because DRAM read data returns in the order the scheduler *issued* commands (which is row-hit reordered), whereas the AXI R channel must see reads in AR order. Data is buffered per entry, then released oldest-first once complete.

13.2 Parameters

Parameter	Default	Purpose
`NUM_ENTRIES`	8	In-flight read slots (`PTRW = $clog2`)
`N_SCHED_LU`	4	Parallel scheduler lookup ports
`NUM_BANKS`	8	`BKW = $clog2(NUM_BANKS)`
`ROW_WIDTH`	14	
`COL_WIDTH`	10	
`AXI_ID_WIDTH`	8	`IW`
`AXI_DATA_WIDTH`	64	`DW` (the DFI word at the IFC)
`BL`	4	Beats per burst (`BCW = $clog2(BL)`)
`AGE_WIDTH`	16	Free-running age counter width
`N_SRAM_SLOTS`	`NUM_ENTRIES`	SRAM data slots (may be `< NUM_ENTRIES`)

13.3 Entry state (per slot)

Field	Description
`r_valid`	Slot occupied
`r_issued`	Scheduler has issued this read to DRAM
`r_ready`	Data complete (last DFI return beat seen)
`r_bank`	Decoded bank (key)
`r_row`	Decoded row (key)

Field	Description
r_col	Decoded column
r_id	AXI ID (echoed on drain)
r_resp	Captured DFI return response
r_age	Insert-time snapshot of r_age_ctr
r_ptr / r_pv	SRAM slot index (set on first return beat) + its valid flag

Relative age is w_rel[i] = r_age_ctr - r_age[i] (wrap-safe); larger rel = older. The burst data lives in a distributed-RAM array r_sram[N_SRAM_SLOTS*BL].

13.4 Interfaces

13.4.1 Insert (from pumice_rd_intake ar_push, AR order)

ins_valid_i / ins_ready_o with {bank, row, col, id}. ins_ready_o is w_have_free (any !r_valid[i]). On fire the slot is allocated, r_issued / r_ready cleared, key/id captured, and r_age <= r_age_ctr.

13.4.2 Scheduler lookups (N, keyed {bank, row})

For each of N_SCHED_LU ports the CAM returns the **oldest NOT-ISSUED** entry matching {bank, row} (max rel among valid && !issued), with its slot, col, id, and age. This lets the arbiter row-hit-schedule reads exactly like writes.

13.4.3 Oldest not-issued port

oldest_valid_o + {bank, row, col, id, slot} — the oldest valid && !issued entry, the arbiter's fallback ACT target.

13.4.4 Issue notify

issue_valid_i / issue_ready_o / issue_slot_i. The scheduler tells the CAM which slot it issued. On fire, r_issued[slot] <= 1 **and** the slot is pushed into an **issue-order FIFO** (u_issue_q, depth NUM_ENTRIES) so returns fill the right slot in issue order.

13.4.5 DFI read return (data in, issue order)

dfi_ret_valid_i / dfi_ret_ready_o with {data, resp, last}. The return-fill engine writes each beat into the SRAM slot of the **issue-FIFO head** (w_iq_rd_slot):

- On the first beat (r_ret_beat == 0) it pre-allocates a free SRAM slot (w_slot_free) and records it in r_ptr[head], marking r_sram_occ.

- `dfi_ret_ready_o` is gated on the issue FIFO being non-empty and (first-beat only) a free SRAM slot existing.
- On `dfi_ret_last_i` the entry is marked `r_ready`, `r_resp` captured, and the issue FIFO popped (`w_iq_rd_ready`).

13.4.6 Drain to `pumice_rd_intake` (AR order, oldest-first, ready-gated)

`drain_valid_o` / `drain_ready_i` with `{data, id, resp, last}`. The draining slot **is** the combinational oldest-valid pick `w_dro_slot` (max `rel` among all `valid`), which is stable across a burst: the entry stays valid until its own last-beat evict, and later inserts are always younger, so no other entry can become "more oldest" mid-burst. Drain is enabled when `w_dr_go = oldest-found && r_ready[oldest]` (data staged). Only a burst beat-counter `r_dr_beat` is registered; `drain_last_o = (r_dr_beat == BL-1)`.

13.5 De-FSM'd sequencing

There is **no** active/slot FSM on the return or drain path. The return path targets the issue-FIFO head; the drain path targets the oldest-valid pick. Both "active" slots are combinational functions of registered state and remain stable until their burst's last beat. The only registered burst state is the two beat-counters (`r_ret_beat`, `r_dr_beat`) plus the age counter and per-entry flags. This mirrors the write CAM and removes the SWAG's issue FSM entirely.

13.6 Eviction

On the drain last beat (`w_dr_fire && drain_last_o`): `r_valid[oldest] <= 0`, `r_pv` cleared, `r_dr_beat` reset, and the SRAM slot freed (`r_sram_occ[r_ptr[oldest]] <= 0`). AR-order release is a natural consequence of always draining the oldest entry.

13.7 Reset

`ALWAYS_FF_RST(aclk, aresetn, ...)`: clears `r_age_ctr`, both beat-counters, `r_sram_occ`, and per-entry `r_valid` / `r_issued` / `r_ready` / `r_pv`. `busy_o` is `oldest-found || issue-FIFO non-empty`.

13.8 Notes / flags

- Reads are keyed on `{bank, row}` only (not AXI ID); ID is carried for the R echo and drained with the data. There is no per-ID beat table.
- The `drain_id_o` port is driven but left unconnected at the IFC (the intake's order FIFO already carries the R-channel id); `oldest_id_o` and `sched_lu_id_o` are likewise informational — the arbiter's `unused` sink absorbs the id/slot buses it does not consume.
- No CSR/QoS priority in this CAM — selection is purely by wrap-safe age.

14 Write Data CAM (pumice_wr_data_cam)

Module: pumice_wr_data_cam.sv **Location:** rtl/fub/ **Category:** FUB **Parent:** pumice_axi4_ifc **Status:** Implemented (fill / commit-drain / snarf movers)

> **Rearchitected:** the SWAG wr_cmd_cam was an ID-keyed metadata CAM with w_buf_ptr / strb_ptr pointers into an external AXI write buffer and a b_pending / b_complete window driven by xbank_timers. The live pumice_wr_data_cam **owns** the write-data SRAM and is keyed on {bank, row, col} with a free-running age. It has three data movers over the SRAM — **fill** (on insert), **snarf-stream** (read-your-write), and **commit-drain** (to DFI write) — plus associative query ports. Commit completion drives B directly (no external write-window timer). All movers are FIFO-fed with beat-counters — no active/slot FSM.

14.1 Purpose

pumice_wr_data_cam holds every pending write: the decoded command key {bank, row, col} **and** the burst data in an SRAM slot. It presents the scheduler the same lookup/oldest interface as the read CAM so writes row-hit schedule identically, forwards data to in-flight reads (snarf), streams committed data to the DFI write path, and self-generates the B-completion strobe on evict.

14.2 Parameters

Parameter	Default	Purpose
NUM_ENTRIES	8	In-flight write slots (PTRW = $clog2)
N_SCHED_LU	4	Parallel scheduler lookup ports
NUM_BANKS	8	BKW = $clog2(NUM_BANKS)
ROW_WIDTH	14	
COL_WIDTH	10	
AXI_ID_WIDTH	8	IW
AXI_DATA_WIDT	64	DW (the DFI word at the IFC); SW = DW/8

Parameter	Default	Purpose
H		
BL	4	Beats per burst (BCW = $clog2(BL))
AGE_WIDTH	16	Free-running age counter width
N_SRAM_SLOTS	NUM_ENTRIES	SRAM data slots (may be < NUM_ENTRIES)

14.3 Entry state (per slot)

Field	Description
r_valid	Slot occupied
r_bank / r_row / r_col	Decoded command key
r_id	AXI ID (echoed on commit-done → B)
r_age	Insert-time snapshot of r_age_ctr
r_ptr / r_pv	SRAM slot index (set on first fill beat) + its valid flag
r_fdone	**Fill complete** — set on the last fill beat. Only then may the entry be scheduled or snarfed (otherwise commit-drain could outrun a gapped fill and read stale SRAM beats).
r_sched	Arbiter has committed this slot — excludes it from sched_lu / oldest the *next* cycle so the arbiter can pick another entry immediately (clean 1 cmd/clock).

Relative age w_rel[i] = r_age_ctr - r_age[i] (wrap-safe). The data and strobes live in distributed-RAM arrays r_sram / r_strb, each N_SRAM_SLOTS*BL deep.

14.4 Age selectors

Port	Pick rule	Rationale
Snarf lookup	**youngest** match (min rel)	Latest data for a read
Scheduler lookup	**oldest** match (max rel)	In-order commit per row
Oldest port	**oldest** valid (max rel)	Scheduler fallback

All three selectors are gated on r_valid && r_fdone && !r_sched.

14.5 Interfaces

14.5.1 Insert (from pumice_wr_intake aw_push)

ins_valid_i / ins_ready_o with {bank, row, col, id}. ins_ready_o is w_have_free && u_fill_q.wr_ready. On fire the slot is allocated (via the priority-encoded free slot), key/id/age captured, and r_fdone cleared. The slot index is pushed into the **fill FIFO** (u_fill_q, depth NUM_ENTRIES).

14.5.2 Fill (from pumice_wr_intake wdata pop)

wd_valid_i / wd_ready_o with {data, strb, last}, written to SRAM in insert order. The fill engine reads its slot from the fill-FIFO head:

- On the first beat (r_fill_beat == 0) a free SRAM slot is pre-allocated and recorded in r_ptr[head]; wd_ready_o requires a free slot on the first beat.
- Each beat writes r_sram / r_strb at r_ptr[head]*BL + r_fill_beat.
- On wd_last_i the entry's r_fdone is set and the fill FIFO popped.

14.5.3 Snarf lookup + stream (from pumice_rd_intake)

snarf_probe_* returns a combinational snarf_hit_o. Read-your-write forwarding is **limited to the safe case** — the hit requires all of:

1. !r_sched — a scheduled write is draining/evicting to DRAM, so its CAM data is racy.
2. r_id == snarf_probe_id_i — same-id W-before-R is the only AXI-ordered case where the read is *required* to see the write (cross-id has no ordering guarantee → that read takes the DRAM path).
3. snarf_probe_len_i == BL-1 — every admitted write is exactly BL beats, so a short/long read must not snarf a full-BL write.

On snarf_accept_i (the read was admitted as a hit) the youngest matching slot is pushed into a **snarf-request FIFO** (u_snarf_q). The snarf read engine streams snarf_rd_* from r_sram at the FIFO-head slot, beat-counter r_sn_beat, popping on the last beat.

14.5.4 Oldest port + scheduler lookups

Same shape as the read CAM: oldest_* presents the oldest schedulable entry; each of N_SCHED_LU ports returns the oldest {bank, row} match with its slot, col, id, and age.

14.5.5 Commit (from arbiter) + drain to DFI write

commit_valid_i / commit_ready_o / commit_slot_i. Commit **marks** the slot r_sched (immediate exclusion) and enqueues it into the **drain FIFO** (u_drain_q); commit_ready_o is the

drain FIFO's write-ready. The commit-drain read engine streams `cm_rd_*` (`{data, strb, last}`) from `r_sram` at the drain-FIFO-head slot, beat-counter `r_cm_beat`, to the DFI write path. Marking is decoupled from draining so the arbiter is never stalled by the DFI back-end.

14.5.6 Commit-done → B

On the commit-drain last beat, `commit_done_valid_o` + `commit_done_id_o` strobe the write intake's B-response path, and the entry is evicted: `r_valid`/`r_pv`/`r_fdone`/`r_sched` cleared and the SRAM slot freed. There is no separate `b_pending`/`b_complete` window and no `xbank_timers` completion driver — completion is simply "the committed burst finished streaming to DFI."

14.6 De-FSM'd sequencing

Each of the three movers (fill, snarf, commit-drain) is just a burst beat-counter. The "active" slot for each is its request-FIFO head, stable until popped on the last beat. There is no active/slot latch and no per-slot beats counter beyond the shared engine counters.

14.7 Difference from the read CAM

Aspect	Read CAM (`pumice_rd_cmd_cam`)	Write CAM (`pumice_wr_data_cam`)
Data direction	DFI return **in** → drain **out** to intake	Fill **in** from intake → commit **out** to DFI
Movers	return-fill + drain	fill + snarf-stream + commit-drain
Forwarding	none	snarf (read-your-write) — the wr2rd path
Extra flags	`r_issued`, `r_ready`	`r_fdone`, `r_sched`
Completion	AR-order drain to R channel	commit-done strobe → B channel

14.8 Reset / busy

`ALWAYS_FF_RST` clears the age counter, all three beat-counters, `r_sram_occ`, and per-entry `r_valid`/`r_pv`/`r_fdone`/`r_sched`. `busy_o` is oldest-found or any request FIFO non-empty.

14.9 Notes / flags

- The wr2rd (read-your-write) forwarding is **entirely** the snarf mover here; there is no standalone `wr2rd_forward` block in the live path (see ch02/21).
- `N_SRAM_SLOTS` may be smaller than `NUM_ENTRIES`: an entry can exist in the CAM (key + age) before its SRAM slot is allocated on the first fill beat, so more commands can be tracked than there are data slots.

15 Write-to-Read Forward — RETIRED as a standalone block

Status: RETIRED. There is no `wr2rd_forward.sv` in the live RTL. **Replaced by:** the **snarf mover inside `pumice_wr_data_cam`** (see §17, the Write Data Path chapter). **Probe driver:** `pumice_rd_intake` (the snarf probe on the AR path).

15.1 Why it was retired

The rearchitected AXI4 interface (`pumice_axi4_ifc`) folded write-to-read forwarding into the write CAM rather than keeping it as a separate FUB on the AR path. The old `wr2rd_forward` block sat between `addr_mapper` and `rd_cmd_cam` and compared each AR against a `wr_cmd_cam` snapshot bus; that snapshot bus, the standalone comparator, and the separate `w_buf` storage it read from no longer exist. Forwarding is now a native operation of `pumice_wr_data_cam`, which already holds the pending writes and their burst data in its SRAM.

15.2 Where forwarding lives now

Write-to-read forwarding ("snarf") is the **snarf mover** of `pumice_wr_data_cam`. The flow is:

1. `pumice_rd_intake` presents an incoming AR's decoded key `{bank, row, col}` plus its AXI id and `arlen` on the CAM's `snarf_probe_*` port.
2. `pumice_wr_data_cam` combinationally searches its entries and asserts `snarf_hit_o` for a **youngest** match, subject to the safe-case restrictions.
3. On `snarf_accept_i`, the matched slot is queued into the snarf request FIFO and the snarf mover streams the write's SRAM beats back on `snarf_rd_*` (non-destructively — the write still drains to DRAM normally). Otherwise the read takes the DRAM miss path through `pumice_rd_cmd_cam` (§18).

15.3 Restrictions (narrower than the old block)

The live snarf is limited to the safe case — a write is snarfable only if **all three** hold:

- **Unscheduled** (`!r_sched`): a scheduled write is draining/evicting to DRAM, so its CAM data is racy.

- **Same AXI id**: same-id write-before-read is the only AXI-ordered case where the read is *required* to see the write. Cross-id reads have no ordering guarantee and take the DRAM path.
- **Same burst length** (`arlen == BL-1`): every admitted write is exactly `BL` beats (ragged bursts are rejected in `pumice_wr_intake`), so a short or long read cannot snarf a full-BL write.

The match must also be fill-complete (`r_fdone`); among candidates the **youngest** wins (latest data). This differs from the old block's "last-write-wins by highest slot index, any matching id, conservative all-1 strb-coverage" policy — the live rule is youngest + same-id + same-BL + unscheduled.

15.4 Memory Ordering

Forwarding preserves AXI per-ID in-order semantics: the snarf path returns the same data a DRAM read would have, just earlier. Because it is restricted to same-id matches, it never forwards across an unordered cross-id pair.

15.5 Tests

Covered by the `pumice_wr_data_cam` FUB test and the `pumice_axi4_ifc` integration tests (snarf stream, snarf-vs-DRAM-path selection, and the same-id / same-BL / unscheduled exclusion scenarios). See §17 for the detailed test plan.

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16 Command Arbiter (`pumice_cmd_arbiter`)

Module: `pumice_cmd_arbiter.sv` **Location:** `rtl/fub/` **Category:** FUB (the pick core) **Parent:** `pumice_mem_cmd_scheduler` **Status:** Implemented (single-issue, single-rank v1)

> **Rearchitected:** the SWAG `scheduler.sv` was a multi-mask priority-encoder pipeline (Stages 1–4) over a wide `txn_queue` snapshot with a column-op FSM (S_NEED_PRE → S_NEED_ACT → S_NEED_RDWR → S_DONE) and a round-robin W/R toggle. That block is retired. The live pick core is `pumice_cmd_arbiter` — a **combinational, single-issue, FSM-free** priority picker that reads the CAM lookup/oldest ports and the per-bank / global readiness inputs and emits ONE abstract DRAM command

per cycle. It is instantiated inside `pumice_mem_cmd_scheduler` alongside the timers, refresh, init, and mode-register (see ch02/19, ch02/09, ch02/11, ch02/12).

16.1 Purpose

`pumice_cmd_arbiter` picks one abstract DRAM command each cycle and pushes it into the scheduler → DFI command interface. It is PHY/nphases-agnostic and single-issue; JEDEC timing is enforced by the per-bank (`pumice_bank_timers`) and global (`global_timers`) readiness inputs it consumes. Open-page vs auto-precharge is decided **inline** from `page_policy_i` — there is no standalone page predictor in the build (see ch02/08).

The wider `pumice_mem_cmd_scheduler` wrapper wires the arbiter to those timers, the refresh controller, the init sequencer + mode-register shadow, and an output command FIFO; the CAM sched-lookup / oldest / commit / issue ports pass through to `pumice_axi4_ifc`.

16.2 Priority (each cycle)

The pick is a single combinational `if/else` cascade producing an abstract command plus its side-effects (event strobes, CAM commit/issue, refresh grant), all gated on the command sink accepting the push:

1. **INIT** (`!init_done_i`) — forward the `init_sequencer` command verbatim (`init_cmd_op/bank/row`).
2. **REFRESH** (`refresh_req_i || refresh_drain_i`) — if any bank is active, precharge the lowest active, precharge-ready, un-guarded bank (one/cycle); once no bank is active, issue `OP_REF` and assert `refresh_grant_o`. The REF fires only under **`w_ref_safe`**: the registered view shows every row closed AND no row-affecting command is in flight or inside its 2-cycle guard window AND the previous REF's tRFC recovery has elapsed. The registered bank view alone is 2-3 cycles stale, which let a REFab collide with a just-issued ACT (no PRE) — silicon-confirmed as refresh-rate-correlated row corruption. **Mission-mode tRFC** (`TIMINGS_RFC_REFI.tRFC` -> `t_rfc_i`) is enforced by an arbiter down-counter loaded on each fired REF; while non-zero, ACT picks and further REFs are blocked (init-time refreshes use `INIT_TIMING1.t_rfc_wait` separately). The sequencing is audited by `dv/checkers/pumice_cmd_history_checker.sv` (bound in the macro TB, `$fatal` on violation).

3. **COLUMN row-hit** — an issuable RD/WR to an open row. **READ-PRIORITY**: a ready read wins over a ready write. Within each direction, **oldest** (max CAM relative age) wins the tie-break.
4. **FALLBACK** — no ready row-hit: `ACT` the oldest pending op's row on an idle, act-ready, un-guarded bank; or `PRE` a bank that is open on the wrong row.

16.2.1 Column issuability

For each bank `j` the arbiter drives the CAM lookups with `{bank j, that bank's open row}` (`bank_open_row_i`), gated valid by `bank_row_active_i`. A returned hit is issuable when:

- **RD**: `rd_lu_hit && bank_rdwr_ready && tccd_ok && twtr_ok`
- **WR**: `wr_lu_hit && bank_rdwr_ready && tccd_ok && trtw_ok`

The oldest issuable RD (max `rd_lu_age`) and oldest issuable WR are scanned in parallel across all `N_LU = NUM_BANKS` lookups; read-priority then selects RD over WR if both exist.

16.2.2 Auto-precharge (inline page policy)

```
w_ap = (page_policy_i == PAGE_POLICY_CLOSE);
```

A column op becomes `OP_RDA` / `OP_WRA` when `w_ap` is set, else `OP_RD` / `OP_WR`. `evt_ap_o = w_ap_out` tells the bank timers to model the auto-precharge. `PAGE_POLICY_OPEN` leaves rows open; `PAGE_POLICY_HAPPY_HYBRID` is treated as OPEN in v1 (the predictor hook is a TODO — see ch02/08).

16.3 Per-bank ACT/PRE re-issue guard

Because `pumice_bank_timers` register their readiness outputs (a 2-cycle latency from an `evt` to `act/pre_ready` dropping), a purely combinational arbiter would re-issue ACT/PRE to the same bank before the timers reflect it. The arbiter keeps a **2-cycle per-bank guard** (`r_guard0`, `r_guard1`; `w_guarded = guard0 | guard1`) set on any accepted **ACT, PRE or column**, and skips guarded banks in the refresh-PRE and fallback paths. Columns are included because a column fired <2 cycles ago has not yet dropped the bank's registered `pre_ready` (tRTP/tWR load), so an unguarded PRE — normal or refresh-drain — could precharge on stale readiness.

Column ops still need no guard **against each other**: both CAMs exclude a just-committed/issued slot from their lookup/oldest ports the next cycle (`r_sched` on the write side, `r_issued` on the read side), so the arbiter never re-issues the same slot. The shared-DQ-bus occupancy (a BL burst owns the DQ bus for `BL/DFI_RATE` dfi cycles) is a `dfi_clk`-domain constraint enforced **downstream** in `pumice_dfi_cmd_path` (`COL_BURST_CYC`), not here — the CDC decouples `aclk` command issue from `dfi_clk` DQ timing.

16.4 Interface (arbiter)

Group	Direction	Notes
page_policy_i	in	OPEN / CLOSE / HAPPY_HYBRID (HAPPY == OPEN in v1)
init passthrough	in	init_done_i, init_cmd_{valid,op,bank,row}_i
refresh	in/out	refresh_req_i, refresh_drain_i, refresh_grant_o
per-bank readiness	in	bank_{act,rdwr,pre}_ready_i, bank_row_active_i, bank_open_row_i ([NUM_RANKS][NUM_BANKS])
global readiness	in	tfaw_ok_i, trrd_ok_i (per-rank), twtr_ok_i, trtw_ok_i, tccd_ok_i
wr CAM lookup + oldest + commit	in/out	wr_lu_* (drive + read), wr_oldest_*, wr_commit_{valid,slot}_o
rd CAM lookup + oldest + issue	in/out	rd_lu_*, rd_oldest_*, rd_issue_{valid,slot}_o
event strobes	out	evt_{act,rd,wr,pre,ap}_o, evt_{rank,bank,row}_o (to timers)
command push	out/in	cmd_{valid,op,rank,bank,row,col,ap}_o, cmd_ready_i

16.5 Side-effects (all gated on w_fire = w_valid && cmd_ready_i)

- evt_act/rd/wr/pre_o — strobe the bank + global timers on an accepted issue.
- wr_commit_valid_o / wr_commit_slot_o — on an accepted WR/WRA, mark the wr CAM slot scheduled (enqueue to its drain FIFO).
- rd_issue_valid_o / rd_issue_slot_o — on an accepted RD/RDA, mark the rd CAM slot issued (enqueue to its issue FIFO).
- refresh_grant_o — on an accepted OP_REF.

cmd_valid_o is asserted whenever a command is picked; cmd_rank_o is tied to rank 0 (RK0) — v1 is a single-rank pick.

16.6 v1 scope / TODO

- **Single-rank** pick (`RK0 = 0`); multi-rank arbitration is a TODO.
- **HAPPY_HYBRID** == **OPEN**; the per-bank page predictor is not in the build.
- Write-drain watermark and powerdown are TODO.
- Unused CAM buses (`wr_lu_id_i`, `rd_lu_id_i`, `*_oldest_slot_i`) are absorbed by an explicit `unused` sink.

16.7 `pumice_mem_cmd_scheduler` wrapper

The macro instantiates, on a single `aclk`:

Instance	Module	Role
`u_init`	`init_sequencer`	JEDEC MRS init; gates traffic until `init_done`
`u_mode_reg`	`mode_register`	MRS-updated CL/CWL/BL shadow → `cl_o/cwl_o/bl_o`
`u_refresh`	`refresh_ctrl`	tREFI + postpone; enabled after init
`u_bank_timers`	`pumice_bank_timers`	per-(rank,bank) safe timers + open-row
`u_global_timers`	`global_timers`	tFAW/tRRD (per-rank), tWTR/tRTW/tCCD (global)
`u_arbiter`	`pumice_cmd_arbiter`	the pick core (above)
`u_cmd_fifo`	`gaxi_fifo_sync`	output command FIFO, packs `{op,rank,bank,row,col,ap}` (`CMD_W = 4 + RKW + BKW + ROW_WIDTH + COL_WIDTH + 1`, depth `CMD_FIFO_DEPTH=8`)

The arbiter pushes into `u_cmd_fifo`; the FIFO read side is the macro's `cmd_*_o` command stream to the DFI layer. `busy_o` is `!init_done || refresh_req || cmd-fifo-non-empty || any-bank-active`.

Issue rate is **one command per `aclk`**. The arbiter never sees the DFI multi-phase dimension — the DFI layer packs phases downstream. Selection is oldest-first by wrap-safe CAM age; there is no QoS, no lookahead window, and no `SCHEDULER_MODE` OOO/INORDER synthesis switch.

17 Page Policy (inline in `pumice_cmd_arbiter`)

Live location: `rtl/fub/pumice_cmd_arbiter.sv` (inline decision) **Standalone module:** `rtl/fub/page_predictor.sv` — present but **not in the default build Category:** FUB behaviour (arbiter-embedded) **Parent:** `pumice_mem_cmd_scheduler` **Status:** Inline OPEN/CLOSE decision implemented; HAPPY_HYBRID hook is a TODO

> **Rearchitected:** the SWAG planned a per-(rank, bank) HAPPY hybrid predictor as a distinct scheduler FUB feeding a `predict_open` / `predict_hit` hint into the scheduler's auto-precharge stage. In the live build there is **no standalone predictor instantiated** — the open-page vs auto-precharge decision is made **inline** in `pumice_cmd_arbiter` from the `page_policy_i` CSR field. A `page_predictor.sv` file still exists in the repo (a 2-bit saturating counter, optional/reference) but is not wired into `pumice_mem_cmd_scheduler` or `pumice_core`; confirm against the filelists in `rtl/filelists/`.

17.1 The live decision

The arbiter computes a single auto-precharge bit directly from the page policy:

```
// pumice_cmd_arbiter.sv
logic w_ap;
assign w_ap = (page_policy_i == PAGE_POLICY_CLOSE);
```

and applies it to every column op it picks:

`page_policy_i`	Column op emitted	Effect
`PAGE_POLICY_OPEN`	`OP_RD` / `OP_WR`	Rows stay open; the arbiter row-hit-schedules subsequent column ops to the open row
`PAGE_POLICY_CLOSE`	`OP_RDA` / `OP_WRA`	Every column op auto-precharges (`w_ap_out = 1`)
`PAGE_POLICY_HAPPY_HYBRID`	(treated as OPEN in v1)	Predictor hook is a TODO; behaves as OPEN

`evt_ap_o` is driven from `w_ap_out`, which tells `pumice_bank_timers` to model the auto-precharge (transition the bank toward precharged after the column op) rather than leaving the row active. There is no per-bank hint, no saturating counter, and no `HAPPY_HYBRID`-specific RTL in the live path — HAPPY currently falls through to the OPEN branch.

`page_policy_i` originates at `pumice_top` from `hwif_out.REFRESH_TUNING.page_policy_or` (see ch02/01), so page policy is a **runtime CSR** setting, not a build parameter.

17.2 Why inline

Open-page management in this controller is not a prediction problem: the arbiter already scans, every cycle, which banks have an open row (`bank_row_active_i` / `bank_open_row_i`) and which pending CAM entries are a row hit to that open row (the per-bank CAM lookups). Given that live row-hit visibility, the only remaining choice is the static "leave open vs auto-precharge" policy bit, which is exactly what the inline `w_ap` expresses. A separate predictor structure — a table, a warmup counter, per-entry hysteresis, an outcome-training broadcast — would add significant state for a marginal gain over the CSR-selected OPEN/CLOSE policy, and none of that is in the shipping build.

17.3 The retained `page_predictor.sv` file

For reference, the file that remains in the repo implements a per-(rank, bank) **2-bit saturating "predict open" hint**:

Counter	Meaning
`00`	strongly closed
`01`	weakly closed
`10`	weakly open
`11`	strongly open

It updates on the `evt_act_i` strobe: the first ACT on a bank records the row with no update; a subsequent ACT to the same row saturates the counter up, a different row saturates it down. `predict_open_o[r][b]` is the counter MSB, all strict-flopped. This matches the module header and the two `ALWAYS_FF_RST` blocks in `page_predictor.sv`.

If HAPPY_HYBRID is promoted from TODO to a real path, this module is the intended building block: the arbiter would consult `predict_open_o[rank][bank]` to choose `OP_RD/WR` (predict open) vs `OP_RDA/WRA` (predict closed) when `page_policy_i == PAGE_POLICY_HAPPY_HYBRID`, replacing the current OPEN fall-through. Until then it is not instantiated.

17.4 Notes / flags

- The elaborate SWAG predictor (4 K-entry hashed table, BRAM storage, warmup counter, per-entry hysteresis, outcome-training broadcast,

accuracy telemetry, PAGE_PREDICTOR_TABLE_BITS / HYSTERESIS_BITS params) does **not** exist in any live RTL — that was SWAG only.
- The page_predictor.sv that *does* exist is the simpler per-bank 2-bit counter documented above; treat it as optional/reference, gated by the filelists.
- powerdown_ctrl.sv is likewise present-but-optional and not in the default top build.

---

18 Global Timers (global_timers)

Module: global_timers.sv **Location:** rtl/fub/ **Category:** FUB **Parent macro:** pumice_mem_cmd_scheduler **Status:** Implemented (per-rank tFAW/tRRD; global tWTR/tRTW/tCCD; strict-flop outputs)

> **Replaces the retired cross-bank timer block.** The original xbank_timers FUB owned a mix of per-bank AND cross-bank counters. That mix has been split:
>
> - **Per-bank** timing (tRCD, tRP, tRAS, tRC, tWR/tRTP) moved into bank_timer, stamped per (rank, bank) by pumice_bank_timers.
> - **Cross-bank / channel-wide** turnaround (tFAW, tRRD, tWTR, tRTW, tCCD) lives here in global_timers.
>
> The retired block's cross-rank extras (tRTRS, tCS, the last_rd_rank / last_cmd_rank chains, and the wide blocks_*[NR][NB] broadcast) are **not** in the live RTL. global_timers emits compact per-rank / global *_window_ok readiness bits that the arbiter ANDs with bank_timer's safe_* outputs.

18.1 Purpose

global_timers holds the controller-wide DRAM constraints that span banks and therefore cannot live inside a single bank's timer:

- **tFAW** — no more than four ACT commands within a rolling `t_faw_i` window. Tracked **per rank** (each rank has its own 4-deep sliding window) so a multi-rank build enforces the device-local four-activate limit independently.
- **tRRD** — minimum cycles between any two ACTs, **per rank**.
- **tWTR** — cycles since the last WR (global; the DQ bus is shared).
- **tRTW** — cycles since the last RD (global; shared DQ bus).
- **tCCD** — cycles since the last column command (global; paces back-to-back RD/WR across banks on the shared DQ bus).

The arbiter reads five readiness outputs — `tfaw_window_ok_o[r]`, `trrd_window_ok_o[r]` (per rank), and `twtr_global_ok_o`, `trtw_window_ok_o`, `tccd_window_ok_o` (global) — and ANDs the relevant subset into its eligibility decision alongside the per-bank `safe_*` bits.

18.2 Constraint Scope

Constraint	Scope	Why
`tFAW`	per rank	Four-Activate-Window — device-local thermal/power limit
`tRRD`	per rank	Minimum ACT-to-ACT stagger inside one DRAM device
`tCCD`	global (channel)	Column-to-column pacing on the shared DQ bus
`tWTR`	global (channel)	Write → read turnaround on the shared DQ bus
`tRTW`	global (channel)	Read → write turnaround on the shared DQ bus

There is **no cross-rank (tRTRS / tCS) tracking** in the live module. `NUM_RANKS` only sizes the per-rank tFAW/tRRD arrays; for `NUM_RANKS == 1` those arrays are a single element and the per-rank outputs collapse to a single bit. The DQ-bus turnaround counters (tWTR/tRTW/tCCD) are single global counters shared across all ranks because the DQ bus is physically shared.

18.3 Synthesis Parameters

Parameter	Default	Effect
NUM_RANKS	1	Number of per-rank tFAW windows and tRRD counters
NUM_BANKS	8	Present for interface symmetry; not used to size counters
RKW	clog2(NUM_RANKS)	Width of evt_act_rank_i (min 1)
BKW	clog2(NUM_BANKS)	Bank-select width (interface symmetry)

All timing reload values are runtime CSR-backed 8-bit inputs (t_faw_i, t_rrd_i, t_wtr_global_i, t_rtw_i, t_ccd_i) — not parameters.

18.4 Per-Rank Trackers

18.4.1 tFAW window — r_faw_slots[NUM_RANKS][4]

Each rank owns a 4-deep pool of 8-bit down-counters (logic [NUM_RANKS-1:0][3:0][7:0]). Every cycle each non-zero slot decrements. On an ACT to a rank, the slot with the **smallest remaining count** is reloaded with t_faw_i:

```
if (evt_act_i) begin
    // pick the slot with the smallest remaining count for this rank
    automatic int unsigned slot_pick = 0;
    automatic logic [7:0] slot_min = 8'hFF;
    for (int unsigned i = 0; i < 4; i++) begin
        if (r_faw_slots[evt_act_rank_i][i] < slot_min) begin
            slot_min  = r_faw_slots[evt_act_rank_i][i];
            slot_pick = i;
        end
    end
    r_faw_slots[evt_act_rank_i][slot_pick] <= t_faw_i;
    r_trrd_cnt [evt_act_rank_i]            <= t_rrd_i;
end
```

The window is “OK” (an ACT may issue) when **at least one** of the rank’s four slots is at 0 — i.e. fewer than four of the most-recent ACTs are still inside the tFAW window:

```
w_tfaw_ok[k] = 1'b0;
for (int unsigned i = 0; i < 4; i++)
    if (r_faw_slots[k][i] == 8'd0) w_tfaw_ok[k] = 1'b1;
```

The “pick smallest count” load policy is equivalent to “replace the oldest entry,” which is the correct rolling-window behavior. It avoids a timestamp FIFO (which would need a subtractor at check time); the check is a 4-way compare-to-zero, one LUT layer.

18.4.2 tRRD — `r_trrd_cnt[NUM_RANKS]`

One 8-bit down-counter per rank, reloaded with `t_rrd_i` on each ACT to that rank (see the ACT block above). `trrd_window_ok_o[r]` is `(r_trrd_cnt[r] == 0)`. An ACT on rank 0 does not affect `r_trrd_cnt[1]`.

18.5 Global Trackers (channel-wide)

Three single 8-bit counters shared across all ranks:

Counter	Reload trigger	Reload value	Drives
`r_twtr_cnt`	`evt_wr_i` (WR event)	`t_wtr_global_i`	`twtr_global_ok_o`
`r_trtw_cnt`	`evt_rd_i` (RD event)	`t_rtw_i`	`trtw_window_ok_o`
`r_tccd_cnt`	`evt_wr_i` OR `evt_rd_i` (any column command)	`t_ccd_i`	`tccd_window_ok_o`

```
if (evt_wr_i) begin r_twtr_cnt <= t_wtr_global_i; r_tccd_cnt <= t_ccd_i; end
if (evt_rd_i) begin r_trtw_cnt <= t_rtw_i;        r_tccd_cnt <= t_ccd_i; end
```

Each `*_ok` output is asserted when its counter is 0. tCCD reloads on **either** a read or a write, so it paces every column-to-column gap on the shared bus. tWTR loads on writes and gates reads; tRTW loads on reads and gates writes.

At DDR2-800 tRTW is typically 0 cycles, but the counter is still synthesized for parameter consistency and forward-compatibility with DDR3+ where tRTW matters.

> **Note on the tWTR trigger.** In the live RTL, tWTR reloads on the WR *command* event (`evt_wr_i`), not on a separate last-write-data-beat strobe. The CSR-supplied `t_wtr_global_i` value therefore carries whatever additional WL+BL/2 offset the timing programming folds in; there is no data-path `wr_last_beat` input to this module.

18.6 Readiness Outputs (strict-flop)

The `*_window_ok_o` outputs are **registered** (strict-flop house style). Each cycle the module computes the next-cycle window-ok values combinationally, then flops them:

```
// next-cycle tFAW-ok (combinational, per rank)
tfaw_window_ok_o <= w_tfaw_ok;
for (int unsigned k = 0; k < NUM_RANKS; k++)
    trrd_window_ok_o[k] <= (r_trrd_cnt[k] == 8'd0);
twtr_global_ok_o <= (r_twtr_cnt == 8'd0);
trtw_window_ok_o <= (r_trtw_cnt == 8'd0);
tccd_window_ok_o <= (r_tccd_cnt == 8'd0);
```

Out of reset, every `*_ok` output is driven to its permissive value (`'1` / `1'b1`) so the arbiter is not spuriously blocked before any command has issued.

Because the outputs are flopped, the arbiter sees a one-cycle-registered view of the counters — the same single-stage discipline as `bank_timer`'s combinational `safe_*`, one flop deeper. The arbiter accounts for this uniform one-cycle readiness latency across both timer FUBs.

18.7 Observability

All observability outputs are combinational "counter non-zero" flags:

Signal	Width	Meaning
`obs_faw_nz_o`	NR	Rank's tFAW window currently full (`! w_tfaw_ok[r]`)
`obs_trrd_nz_o`	NR	`r_trrd_cnt[r] != 0`
`obs_twtr_nz_o`	1	`r_twtr_cnt != 0`
`obs_trtw_nz_o`	1	`r_trtw_cnt != 0`
`obs_tccd_nz_o`	1	`r_tccd_cnt != 0`

18.8 Interface

18.8.1 Clocks / Reset

Signal	Direction	Description
`mc_clk`	input	Controller clock
`mc_rst_n`	input	Active-low async reset

18.8.2 Timing config (CSR-backed)

Signal	Width	Constraint
t_faw_i	8	tFAW
t_rrd_i	8	tRRD
t_wtr_global_i	8	tWTR
t_rtw_i	8	tRTW
t_ccd_i	8	tCCD (CAS-to-CAS)

18.8.3 Events from the arbiter

Signal	Width	Description
evt_act_i	1	ACT issued this cycle
evt_act_rank_i	RKW	Rank of the ACT (selects tFAW/tRRD)
evt_rd_i	1	RD issued this cycle
evt_wr_i	1	WR issued this cycle

18.8.4 Readiness to the arbiter

Signal	Width	Description
tfaw_window_ok_o	NR	Per-rank: rank may issue another ACT
trrd_window_ok_o	NR	Per-rank: tRRD elapsed
twtr_global_ok_o	1	Global: tWTR elapsed (RD may follow WR)
trtw_window_ok_o	1	Global: tRTW elapsed (WR may follow RD)
tccd_window_ok_o	1	Global: tCCD elapsed (next column cmd OK)

18.8.5 Observability

obs_faw_nz_o[NR], obs_trrd_nz_o[NR], obs_twtr_nz_o, obs_trtw_nz_o, obs_tccd_nz_o.

18.9 Timing Budget

Event → readiness is single-cycle plus the output flop:

```
cycle T:   arbiter asserts evt_act_i, evt_act_rank_i = r
cycle T:   r_trrd_cnt[r] <- t_rrd_i; picked tFAW slot <- t_faw_i
cycle T+1: trrd_window_ok_o[r] = 0 (registered); another ACT to rank r
blocked
```

The window-ok comparators are one LUT layer (compare-to-zero) feeding the output flop, so the path is short. The arbiter combines these with the per-bank `safe_*` bits; the aggregate eligibility AND is the arbiter's own critical path (§2.7), not this module's.

18.10 Verification Notes (cocotb test plan)

Scenario	What it proves
Two ACTs to same rank, gap < tRRD	Second blocked; `trrd_window_ok_o[r]` low
Two ACTs to same rank, gap == tRRD	Second issues exactly at tRRD
Five ACTs to one rank inside tFAW window	Fifth blocked; all four slots non-zero
Five ACTs spanning the tFAW boundary (4 in + 1 aged out)	Fifth issues; oldest slot returned to 0
Multi-rank: ACTs on rank 0 do not block ACTs on rank 1	Per-rank tRRD / tFAW isolation
WR then RD before tWTR elapses	RD blocked; `twtr_global_ok_o` low
RD then WR before tRTW elapses	WR blocked; `trtw_window_ok_o` low
Back-to-back column commands before tCCD elapses	Second column blocked; `tccd_global_ok_o` low
Reset → all `*_ok` outputs assert permissively	Reset default correctness
tFAW "pick smallest" evicts the oldest slot	Rolling-window load policy

18.11 Open Questions / Future Work

- **Cross-rank turnaround (tRTRS / tCS).** The live module has no rank-switch DQ-handoff or chip-select-setup tracking. On the DDR2/LPDDR2 board targets (single rank, or same-rank streaming) these are met by IOB margin. A true multi-rank DDR3/DDR4 build would add a `last_rd_rank` / `last_cmd_rank` pair and cross-rank gate here — reserved but not implemented.
- **tWTR from data-path event.** tWTR currently reloads on the WR command event, relying on the CSR value to fold in WL+BL/2. A more precise

implementation would load from the last-write-data-beat strobe; add a `wr_last_beat_i` input if characterization shows the command-relative approximation costs bandwidth.

- **Bank-group tCCD (DDR4).** DDR4 splits tCCD into same-group (tCCD_L) vs different-group (tCCD_S). DDR2/LPDDR2 have no bank groups, so the single global tCCD is exact here; the DDR4 family controller adds per-group tracking.

---

19 Global Timers (`global_timers`)

Module: `global_timers.sv` **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_mem_cmd_scheduler` **Status:** implemented (v2 H)

19.1 Purpose

Controller-wide constraint trackers that span banks. Where `pumice_bank_timers` holds per-(rank, bank) JEDEC "safe" countdowns, `global_timers` holds the **windows that apply across banks** — some per-rank (device-local limits), some truly global (shared DQ bus):

Constraint	Meaning	Tracked as
tFAW	At most 4 ACT commands within `t_faw_i` cycles	Per-rank: a 4-deep sliding window of countdown timers
tRRD	Minimum cycles between any two ACT commands	Per-rank: single countdown, reloaded on each ACT
tWTR	Cycles after a WR before any RD	Global: single countdown, reloaded on each WR
tRTW	Cycles after a RD before any WR	Global: single countdown, reloaded on each RD
tCCD	Cycles between back-to-back column (RD/WR) commands	Global: single countdown, reloaded on each RD *and* WR

tFAW and tRRD are **per-rank** because they enforce device-local thermal/power limits — each rank has its own 4-deep tFAW slot array and its own tRRD countdown, so multi-rank silicon

meters each device independently. tWTR, tRTW, and tCCD are **global** because they gate the shared DQ bus, which every rank contends for.

The FUB exposes `_window_ok` flags to the scheduler:

```
tfaw_window_ok_o [NUM_RANKS]   // per-rank: 1 = at least one tFAW slot
is at 0
trrd_window_ok_o [NUM_RANKS]   // per-rank: 1 = tRRD has elapsed
twtr_global_ok_o               // global : 1 = tWTR has elapsed
trtw_window_ok_o               // global : 1 = tRTW has elapsed
tccd_window_ok_o               // global : 1 = tCCD has elapsed
```

All five `_window_ok` outputs are strict-flop registered. Default after reset is `'1` / `1'b1` (all windows open), which is correct given no ACTs/WRs/RDs have been issued.

19.2 Timer mechanics

All timers are 8-bit down-counters that saturate at 0 (they decrement each `mc_clk` while non-zero). Reload values come in on the `t_*_i` ports:

- **`evt_act_i`** (with `evt_act_rank_i`): installs `t_faw_i` into the targeted rank's tFAW slot with the smallest remaining count (the freed slot), and reloads that rank's tRRD with `t_rrd_i`.
- **`evt_wr_i`**: reloads the global tWTR (`t_wtr_global_i`) *and* tCCD (`t_ccd_i`).
- **`evt_rd_i`**: reloads the global tRTW (`t_rtw_i`) *and* tCCD (`t_ccd_i`).

`tfaw_window_ok_o[r]` is high when rank `r` has at least one tFAW slot at 0 (i.e. fewer than 4 ACTs are still inside the window).

19.3 Observability

Combinational `obs_*` "non-zero" flags mirror each counter for CSR/debug observation:

```
obs_faw_nz_o  [NUM_RANKS]   obs_trrd_nz_o [NUM_RANKS]
obs_twtr_nz_o               obs_trtw_nz_o               obs_tccd_nz_o
```

19.4 Parameters

Parameter	Default	Purpose
`NUM_RANKS`	1	Per-rank tFAW/tRRD array depth
`NUM_BANKS`	8	(`BKW` derivation)

19.5 Scope

- **Per-rank tFAW / tRRD** are implemented (v2 H). Multi-rank silicon may still need the per-rank windows tightened against rank-level scheduler coordination.
- **tCCD is tracked here** now (it was previously baked into burst-length pacing in the scheduler).

19.6 Tests

Verified by `dv/tests/fub/test_global_timers.py`: `smoke`, `tfaw_4_acts`, `trrd_spacing`, `twtr_after_wr`, `trtw_after_rd` (plus tCCD coverage as the suite is extended for the v2 H additions).

---

20 Refresh Controller (`refresh_ctrl`)

Module: `refresh_ctrl.sv` **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_mem_cmd_scheduler` **Status:** Implemented — FSM-free tREFI/pending/drain accumulator with a REFab/REFpb selector and REFpb bank rotor.

> Refresh **recovery** (tRFC) is not this block's job: the arbiter (`pumice_cmd_arbiter`) loads a down-counter from `TIMINGS_RFC_REFI.tRFC` on each fired REF and blocks ACT/REF picks until it expires — see ch02/07. `refresh_ctrl` only meters *when* refreshes are owed.
>
> Scope note: this module is `refresh_ctrl`. It is a compact, FSM-free accumulator design — three registered counter blocks plus a bank rotor, all strict-flopped at the outputs. It tracks the tREFI interval, keeps a JEDEC-capped count of postponed refreshes, meters how many REF commands the scheduler should burst back-to-back, and (for LPDDR2 REFpb) rotates a target bank across grants.
>
> Features that earlier drafts of this chapter described as current architecture are **not implemented** and are not part of this module: DARP per-bank age selection, periodic ZQCS piggyback, per-rank PASR mask

propagation, self-refresh coordination, LPDDR2 temperature (MR4) scaling, per-rank REFab round-robin, a multi-state FSM, and per-(rank, bank) `refresh_req`/`refresh_gnt` handshake arrays. Refresh does not use a bank-machine handshake: it raises a single `refresh_req_o` and wins against other scheduler commands by priority inside `pumice_cmd_arbiter`. Those retired features are collected under Open Questions / Future Work.

20.1 Purpose

`refresh_ctrl` owns DRAM refresh scheduling. It produces:

- `refresh_req_o` — a single request line that elevates refresh to highest priority in the scheduler. It stays high while any refresh is pending.
- `pending_refreshes_o` — the count of postponed refreshes (JEDEC max 8).
- `refresh_drain_active_o` — a hint that the scheduler should keep granting REF back-to-back to work down a burst quota.
- `refresh_kind_o` / `refresh_bank_o` — the REFab-vs-REFpb selector and, in REFpb mode, the target bank produced by the bank rotor.
- `obs_*` — observability taps on internal state for future CSR readout.

This is the heart of the controller's correctness story: get refresh wrong and DRAM forgets data. The design is deliberately simple — three registered counter blocks and a bank rotor, with no control FSM.

20.2 Synthesis Parameters

Parameter	Default	Effect
`NUM_BANKS`	`8`	Number of banks the REFpb rotor cycles through (`0..NUM_BANKS-1`)
`BA_W`	`$clog2(NUM_BANKS)`	Width of `refresh_bank_o` / the bank rotor (derived)

The module imports `pumice_pkg::*` and uses the standard reset macros (`reset_defs.svh`). Clock is `mc_clk`; reset is `mc_rst_n` (active-low).

20.3 Behavioral Blocks

There is no FSM. The module is four registered blocks feeding a strict-flop output stage.

20.3.1 1. tREFI Counter (`r_refi_cnt`)

A 16-bit down-counter that measures the refresh interval in MC cycles.

- Only ticks when `enable_i` is high (`enable_i` comes from `init_sequencer` — refresh is gated off until init completes). While `enable_i` is low, the counter is held reloaded at `t_refi_i`.
- When it reaches 0 (`w_refi_expired`), it reloads `t_refi_i` and that expiry event drives one increment of the pending accumulator.

20.3.2 2. Pending Accumulator (`r_pending`)

A 4-bit count of postponed refreshes, capped at `MAX_PENDING = 8` (the JEDEC ceiling of postponed refreshes).

- A tREFI expiry (while enabled) adds 1, saturating at 8. At saturation the RTL comment flags a looming DRAM data-retention violation — the workload has blocked refresh longer than JEDEC allows.
- A grant (`w_grant_accept = refresh_grant_i && r_pending > 0`) subtracts 1.
- A simultaneous expiry and grant net to zero change.
- `refresh_req_o` is `(r_pending > 0)`, registered.

20.3.3 3. Drain Quota (`r_burst_remaining`)

A 4-bit counter that meters how many REFs the scheduler should issue back-to-back once refresh wins arbitration.

- When the previous burst is fully drained (`r_burst_remaining == 0`) and there is pending work (`r_pending > 0`), it (re)loads `min(refresh_burst_i, r_pending)`. If that clamp evaluates to 0 it loads 1 instead, so a burst always makes forward progress.
- Each accepted grant decrements it.
- `w_drain_active = (r_burst_remaining > 0) && (r_pending > 0)` is exported as `refresh_drain_active_o`. While it is high, the scheduler should keep granting REF back-to-back rather than yielding to reads/writes.

`refresh_burst_i` is a 4-bit input (1..8) that sets the desired drain count per request cycle.

20.3.4.4. REFpb Bank Rotor (`r_bank_rotor`)

Selects the target bank for LPDDR2 per-bank refresh.

- On each accepted grant, if `refpb_mode_i` is set, the rotor increments and wraps `0..NUM_BANKS-1`.
- In REFab mode (`refpb_mode_i == 0`) it is held at 0.
- `refresh_bank_o` is the registered rotor value; it is only meaningful in REFpb mode.
- `r_grants_total` (16-bit) counts total accepted grants for observability.

`refresh_kind_o` is simply the registered copy of `refpb_mode_i` (0 = REFab, 1 = REFpb), forwarded to the scheduler / `dfi_cmd_formatter`.

All final outputs are strict-flopped (registered) in a single output stage.

20.4 Interface

20.4.1 Parameters and Clocking

Signal	Direction	Width	Description
`mc_clk`	input	1	Memory-controller clock
`mc_rst_n`	input	1	Active-low asynchronous reset

20.4.2 Inputs

Signal	Width	Description
`t_refi_i`	16	Refresh interval in MC cycles (tREFI). Reload value for `r_refi_cnt`.
`refresh_burst_i`	4	Desired drain count per request cycle (1..8). Loads the burst quota.
`refpb_mode_i`	1	0 = REFab, 1 = REFpb (LPDDR2). Selects `refresh_kind_o` and enables the rotor.
`enable_i`	1	From `init_sequencer` (init done). Gates the tREFI counter and accumulation.
`refresh_grant_i`	1	Pulsed by the scheduler when it issues a REF on the DFI bus.

20.4.3 Outputs

Signal	Width	Description
refresh_req_o	1	Refresh pending: (r_pending > 0), registered.
pending_refreshes_o	4	Current postponed-refresh count (0..8).
refresh_drain_active_o	1	Burst quota still owed and pending nonzero — keep granting REF.
refresh_kind_o	1	Registered refpb_mode_i (0 = REFab, 1 = REFpb).
refresh_bank_o	BA_W	REFpb target bank from the rotor (valid in REFpb mode).

20.4.4 Observability (obs_*, future CSR readout)

Signal	Width	Source
obs_refi_cnt_o	16	Current r_refi_cnt value
obs_drain_remaining_o	4	Current r_burst_remaining value
obs_bank_rotor_o	BA_W	Current r_bank_rotor value
obs_grants_total_o	16	Total accepted grants (r_grants_total)

These are wired out for future CSR/telemetry hookup; no CSR block consumes them yet.

20.5 Timing / Behavior

- **Reset.** All registers clear to 0. refresh_req_o, refresh_drain_active_o, refresh_kind_o, refresh_bank_o, and every obs_* output reset to 0.
- **Init gating.** With enable_i low, r_refi_cnt is held at t_refi_i and no pending refreshes accumulate — refresh is inert until init_sequencer releases enable_i.
- **Single-cycle latency.** Because there is no FSM, request/drain/bank outputs simply track the counter state one flop behind. When tREFI expires, r_pending rises the next cycle and refresh_req_o follows the cycle after (output flop stage).
- **Grant handshake.** The module never blocks on a grant. It exports its request and drain hint; the scheduler decides when to pulse refresh_grant_i, and the module bookkeeps against those pulses.

- **Priority, not handshake.** Refresh wins against other scheduler commands by priority in `pumice_cmd_arbiter`; there is no per-bank `refresh_req`/`refresh_gnt` array and no bank-machine handshake.

20.6 Verification Notes (cocotb test plan)

Scenario	What it proves
`enable_i` low: `t_refi_i` loaded, no ticks, `r_pending` stays 0, `refresh_req_o` stays low	Init gating
`enable_i` high: `r_refi_cnt` counts down and reloads on expiry	tREFI countdown + reload
One tREFI expiry with no grant: `r_pending == 1`, `refresh_req_o` asserts	Pending accumulate + request
Withhold grants across 8+ expiries: `r_pending` saturates at 8, does not overflow	JEDEC max-8 postpone cap
Expiry and grant in the same cycle: `r_pending` unchanged	Simultaneous tick+grant net-zero
Grants with pending > 0: `r_pending` decrements; `refresh_req_o` drops at 0	Grant accounting
`refresh_burst_i = N` with `r_pending >= N`: quota loads N, `refresh_drain_active_o` high	Drain quota load + active hint
`refresh_burst_i > r_pending`: quota clamps to `r_pending`	Burst clamp (no overcount)
`refresh_burst_i` clamp resolves to 0 but pending > 0: quota loads 1	Forward-progress floor
Grants during drain: `r_burst_remaining` counts down; `refresh_drain_active_o` deasserts at 0	Drain metering
REFab mode (`refpb_mode_i = 0`): `refresh_bank_o` stays 0, `refresh_kind_o == 0`	REFab selector + rotor held
REFpb mode (`refpb_mode_i = 1`): `refresh_bank_o` rotates `0..NUM_BANKS-1` across grants	REFpb bank rotor wrap
`obs_*` taps track internal counters	Observability wiring

20.7 Open Questions / Future Work

The following were described in earlier SWAG/HAS drafts but are **not implemented** in `refresh_ctrl`. They are recorded here as possible future scope, not as current behavior:

- **DARP per-bank age selection.** REFpb currently uses a plain round-robin bank rotor. An age-aware selector (refresh the oldest idle non-masked bank first, falling back to oldest-first) would need bank-state and last-refresh-age inputs the module does not have today.
- **Periodic ZQCS piggyback.** No ZQCS interval counter or ZQCS sequencing exists. If added, it could piggyback on the refresh window (all banks idle) rather than triggering a separate bus-blocking event.
- **Per-rank PASR mask propagation.** No PASR bank/segment masks, no MR16/MR17 propagation. LPDDR2 partial-array self-refresh is unsupported.
- **Self-refresh coordination.** No `sr_entry_req`/`sr_entry_gnt` handshake with a power-state block; the tREFI counter is not paused for self-refresh.
- **LPDDR2 temperature scaling.** No MR4-driven tREFI derating. `t_refi_i` is the single reload value regardless of temperature class.
- **Per-rank REFab round-robin.** Single-rank only; there is no rank pointer or per-rank REF dispatch. `NUM_BANKS` (banks), not ranks, is the only structural parameter.
- **Multi-state FSM / per-bank handshake.** The design is FSM-free and uses a single priority-based request rather than per-(rank, bank) `refresh_req`/`refresh_gnt` arrays. Any of the above features would likely reintroduce sequencing state.

---

21 Init Sequencer (`init_sequencer`)

Module: `init_sequencer.sv` **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_mem_cmd_scheduler` **Status:** Implemented — DDR2 and LPDDR2 sequences both complete (DDR2 validated on the Nexys A7 board; LPDDR2 passes the full sim suite for family reuse)

Renamed / re-scoped: an early sketch called this an `init_engine_fub` built on a microprogram step-record ROM (opcodes, a step interpreter FSM, `SIM_INIT_SCALE`, `WAIT_FOR_BIT` polling, ZQ retry loops, an `init_error` path, IRQ outputs, and a CSR MR-override path). **None of that ROM machinery was built.** The implemented module is a small hard-coded FSM that issues a fixed JEDEC bring-up sequence. It drives MR-write strobes into `mode_register` (which owns the MR shadow + live CL/CWL/BL decode), and it issues the real DRAM commands (PRECHARGE, REFRESH, MRS/MRW) through the command path while `init_busy_o` is high.

21.1 Purpose

`init_sequencer` walks the JEDEC cold-boot DRAM initialization sequence — the deterministic recipe of DFI init handshake, PRECHARGE-ALL, mode-register loads, and REFRESH settling that must complete before any normal traffic can issue. It is memtype-specific: `memtype_i` selects between the DDR2 sequence (mirrors LiteDRAM's proven Nexys A7 recipe) and the LPDDR2 MRW sequence (JEDEC JESD209-2F).

Two things happen for each mode-register step:

1. A real DRAM command is issued. The sequencer drives a single-cycle command request (`init_cmd_valid_o` / `init_cmd_op_o` / `init_cmd_bank_o` / `init_cmd_row_o`) that the parent scheduler forwards to `dfi_cmd_formatter`. Because the scheduler stays idle during init (it owns nothing else while `init_busy_o` is high) and the formatter is always `cmd_ready`, a single-cycle pulse issues exactly one command — no grant handshake is needed.
2. The `mode_register` shadow is updated in lockstep (`mr_seq_we_o`) so the controller's live CL/CWL/BL decode tracks what was actually programmed.

The init sequencer sits above the scheduler in the command-priority hierarchy. `init_done_o` also drives `refresh_ctrl.enable_i` — periodic refresh does not start until init reports done.

History (why the sequence is "full"). An earlier version was a "simplified" 4-MR shadow-only walk that never issued MRS/PRECHARGE/REFRESH to the DRAM. On the DFI-loopback sim the memory model stores/returns data regardless of init state, so it passed. On real DDR2 the read DLL never locked (no proper reset + refresh

sequence) and no IDELAY tap found a read eye. The full sequence documented here fixed on-board bring-up.

21.2 Synthesis Parameters

Parameter	Default	Effect
ROW_WIDTH	14	Width of the init_cmd_row_o request field (carries MR data; must hold DDR2 MR0=0x533 which needs bit 10, so a narrower COL-based path would truncate).
NUM_BANKS	8	Number of DRAM banks.
BKW	derived	$clog2(NUM_BANKS) — width of init_cmd_bank_o (bank / MR index).

There is no MEMTYPE synthesis parameter — the sequence is selected at run-time by the memtype_i input, so one elaboration supports both families.

21.3 Interface

21.3.1 Clock / reset

Signal	Direction	Description
mc_clk	input	Memory-controller clock.
mc_rst_n	input	Active-low async reset (house macros).

21.3.2 Configuration

Signal	Direction	Width	Description
memtype_i	input	enum	MEMTYPE_DDR2 selects the DDR2 sequence; anything else selects the LPDDR2 MRW chain.

21.3.3 JEDEC init-sequence waits (CSR-backed, MC cycles)

All countdowns run internally as 16-bit values; the 8-bit inputs are zero-extended. Defaults live in the CSR (mentioned as 512 / 256 / 8 / 8 / 16), so these were promoted from hardcoded constants to `INIT_TIMING`-programmable.

Signal	Width	Internal name	JEDEC parameter
`t_init_wait_i`	16	`W_INIT`	CKE / tINIT settle (also LPDDR2 tINIT4).
`t_dll_wait_i`	16	`W_DLL`	DLL lock tDLLK (also LPDDR2 tZQINIT).
`t_mrd_wait_i`	8	`W_MRD`	tMRD, post mode-register-set.
`t_rp_wait_i`	8	`W_RP`	tRP, post precharge.
`t_rfc_wait_i`	8	`W_RFC`	tRFC, post auto-refresh.

21.3.4 DFI status

Signal	Direction	Description
`dfi_init_start_o`	output	Asserted (registered) once `r_state != S_RESET`. Tells the PHY to begin its own init.
`dfi_init_complete_i`	input	PHY reports its DLL-lock / IO training complete. Gates the exit from `S_DFI_INIT`.

21.3.5 MR-shadow write port (muxed with CSR by `pumice_mem_cmd_scheduler`)

Signal	Direction	Width	Description
`mr_seq_we_o`	output	1	Write-enable strobe into `mode_register`.
`mr_seq_index_o`	output	5	MR index (0..3) being shadowed.
`mr_seq_data_o`	output	16	MR data being shadowed.

21.3.6 DRAM command request into the scheduler (issued while init_busy)

Signal	Direction	Width	Description
`init_cmd_valid_o`	output	1	Single-cycle command pulse.
`init_cmd_op_o`	output	`dram_op_e`	`OP_PREA`, `OP_REF`, or `OP_MRS`.
`init_cmd_ban`	output	`BKW`	MR index for MRS (DDR2) / bank field.

Signal	Direction	Width	Description
k_o			
init_cmd_row_o	output	ROW_WIDTH	MR data for MRS on the wide row path (see MR tables).

21.3.7 Legacy ZQCL handshake (DDR3+; unused for DDR2/LPDDR2)

Signal	Direction	Description
zqcl_req_o	output	Tied to 0 — DDR2 has no ZQCL command.
zqcl_grant_i	input	Unused (tied off via a _unused sink).

21.3.8 Status

Signal	Direction	Description
init_busy_o	output	Registered; high whenever r_state != S_DONE. Reset value is 1.
init_done_o	output	Registered; high when r_state == S_DONE. Drives refresh_ctrl.enable_i.

21.4 FSM

A single 5-bit state_e enum drives everything. Each command state is occupied for exactly one cycle (an unconditional transition to S_WAIT), so the combinational command decode produces a single-cycle pulse per command. S_WAIT holds for the programmed inter-command delay: it counts r_wait down and, when it reaches zero, jumps to r_next (the resume state latched by the command state).

State	Value	Role	Wait after (into S_WAIT)
S_RESET	0	Reset entry; unconditionally advances.	— (→ S_DFI_INIT)
S_DFI_INIT	1	Wait for dfi_init_complete_i, then arm W_INIT.	W_INIT
S_PREA1	2	DDR2: Precharge All (pre-EMR).	W_RP → S_EMR2
S_EMR3	3	DDR2: MRS EMR(3).	W_MRD → S_EMR1
S_EMR2	4	DDR2: MRS EMR(2).	W_MRD → S_EMR3
S_EMR1	5	DDR2: MRS EMR(1).	W_MRD →

State	Value	Role	Wait after (into S_WAIT)
			S_MR0_DLL
S_MR0_DLL	6	DDR2: MRS MR0 + DLL reset (MR0.VAL	0x100; reset 0x533).
S_PREA2	7	DDR2: Precharge All (pre-refresh).	W_RP → S_REF1
S_REF1	8	DDR2: Auto Refresh #1.	W_RFC → S_REF2
S_REF2	9	DDR2: Auto Refresh #2.	W_RFC → S_MR0
S_MR0	10	DDR2: MRS MR0, DLL-reset bit cleared (MR0.VAL; reset 0x433).	W_MRD → S_OCD_DEF
S_OCD_DEF	11	DDR2: MRS EMR(1) + OCD default (0x380).	W_MRD → S_OCD_EXIT
S_OCD_EXIT	12	DDR2: MRS EMR(1) + OCD exit (0x000).	W_MRD → S_DONE
S_WAIT	13	Inter-command countdown; r_wait==0 → r_next.	—
S_DONE	14	Terminal; init_done_o high, self-loop.	—
S_L_RESET	15	LPDDR2: MRW(MR63) Reset.	W_INIT → S_L_ZQ
S_L_ZQ	16	LPDDR2: MRW(MR10) ZQ Init Calibration.	W_DLL → S_L_MR1
S_L_MR1	17	LPDDR2: MRW(MR1) BL8 / nWR3.	W_MRD → S_L_MR2
S_L_MR2	18	LPDDR2: MRW(MR2) RL3 / WL1.	W_MRD → S_L_MR3
S_L_MR3	19	LPDDR2: MRW(MR3) drive strength 40ohm.	W_MRD → S_DONE

memtype_i only branches once: on exit from S_DFI_INIT, r_next is set to S_PREA1 for DDR2 or S_L_RESET otherwise. From there each path is a fixed chain of command states separated by S_WAIT.

S_DONE is a hard self-loop — there is no restart, no init_error, and no software-triggered re-run in the RTL. A fresh init requires mc_rst_n.

21.5 DDR2 Sequence

Mirrors LiteDRAM's get_ddr2_phy_init_sequence (litedram/init.py), the reference proven on the Nexys A7:

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i` (PHY runs its own DLL-lock / IO training). Then wait `W_INIT` (tINIT / CKE settle).
2. Precharge All (`S_PREA1`), wait `W_RP`.
3. Load the extended mode registers in JEDEC JESD79-2 order **EMR(2), EMR(3), EMR(1)** — the state chain is `S_EMR2 → S_EMR3 → S_EMR1` — each followed by `W_MRD`. (Note the enum numbering: `S_EMR2` = 4, `S_EMR3` = 3, so the values are out of numeric order but the executed order is 2 → 3 → 1.)
4. Load MR0 + DLL reset (`S_MR0_DLL`, MR0.VAL | 0x100; reset 0x533: BL8 / CL3 / tWR3, DLL_RESET); wait `W_DLL` for the DLL to lock (~200 DRAM clocks).
5. Precharge All (`S_PREA2`), wait `W_RP`.
6. Auto Refresh x2 (`S_REF1`, `S_REF2`), each followed by `W_RFC`.
7. Load MR0 with the DLL-reset bit cleared (`S_MR0`, MR0.VAL; reset 0x433); wait `W_MRD`.
8. EMR(1) + OCD Default (`S_OCD_DEF`, 0x380) → EMR(1) + OCD Exit (`S_OCD_EXIT`, 0x000).
9. `S_DONE`: `init_done_o = 1`.

MR data is carried on `init_cmd_row_o` (ROW_WIDTH wide, MR index on `init_cmd_bank_o`), because MR0 = 0x533 sets bit 10 and a 10-bit column-based path would truncate it.

21.5.1 DDR2 MR values

Localparams, transcribed exactly:

Symbol	Value	State	Meaning	Shadowed?
`DDR2_MR0_DLL`	`MR0.VAL | 0x100` (reset 0x0533)	`S_MR0_DLL`	MR0: BL8 / CL3 / tWR3, DLL_RESET ORed in.	Yes (MR0)
`DDR2_MR0`	`MR0.VAL` (reset 0x0433)	`S_MR0`	MR0 with DLL_RESET cleared.	Yes (MR0)
`DDR2_MR1`	`0x0000`	`S_EMR1`, `S_OCD_EXIT`	EMR(1): Rtt disabled, ODS full.	Yes (MR1)

Symbol	Value	State	Meaning	Shadowed?
DDR2_MR1_OCD	0x0380	S_OCD_DEF	EMR(1) + OCD calibration default.	No — transient calibration; shadow left at final EMR value.
DDR2_MR2	0x0000	S_EMR2	EMR(2).	Yes (MR2)
DDR2_MR3	0x0000	S_EMR3	EMR(3).	Yes (MR3)

DDR2_MR0_DLL decomposes as log2(BL=4)=2 | (CL=3<<4)=0x30 | (tWR=3<<9)=0x400 = 0x433, plus reset_dll = 1<<8 = 0x100, giving 0x533. OCD default = EMR | (7<<7) = 0x380; OCD exit = EMR = 0.

21.6 LPDDR2 Sequence

JEDEC JESD209-2F §3.4.1 power-up + §3.5 mode registers:

1. Assert dfi_init_start_o; wait dfi_init_complete_i; wait W_INIT (tINIT settle).
2. MRW(MR63) = Reset (S_L_RESET, OP don't-care); wait W_INIT (tINIT4).
3. MRW(MR10) = 0xFF ZQ Init Calibration (S_L_ZQ); wait W_DLL (tZQINIT).
4. Configure the device: MRW(MR1) = BL8 / nWR3 (S_L_MR1, 0x23), MRW(MR2) = RL3 / WL1 (S_L_MR2, 0x01), MRW(MR3) = DS 40ohm (S_L_MR3, 0x02) — each followed by W_MRD (tMRW).
5. S_DONE.

The CSR waits are reused: W_INIT covers tINIT4, W_DLL covers tZQINIT, W_MRD covers post-MRW settling.

LPDDR2 mode-register writes must reach indices up to MR63, which a 3-bit bank port cannot express. So the index (MA) and data (OP) are packed together into the wide row request by the

`mrw_row()` function as `{MA[5:0], OP[7:0]}` (`init_cmd_row_o[13:8]` = MA index, `init_cmd_row_o[7:0]` = OP data); `dfi_cmd_formatter` unpacks this for the LPDDR2 CA MRW word.

21.6.1 LPDDR2 MR values

Symbol	OP value	State	Meaning	Shadowed?
`LPDDR2_MR63_OP`	`0x00`	`S_L_RESET`	MRW(63) Reset (OP don't-care).	No — issued, not shadowed (MR63 exceeds the 5-bit index).
`LPDDR2_MR10_OP`	`0xFF`	`S_L_ZQ`	MR10 ZQ Init Calibration.	No — issued, not shadowed.
`LPDDR2_MR1_OP`	`0x23`	`S_L_MR1`	MR1: nWR3 | BL8.	Yes (MR1).
`LPDDR2_MR2_OP`	`0x01`	`S_L_MR2`	MR2: RL3 / WL1.	Yes (MR2).
`LPDDR2_MR3_OP`	`0x02`	`S_L_MR3`	MR3: DS 40ohm.	Yes (MR3).

Only MR1 / MR2 / MR3 update the `mode_register` shadow (`mr_seq_we_o`), because those are the registers that feed the CL/CWL/BL decode. MR63 and MR10 are issued to the DRAM but not shadowed.

21.7 Init-Busy Gating

`init_busy_o` is high from reset (its reset value is 1) through every state except `S_DONE`. While busy:

- The parent `pumice_mem_cmd_scheduler` forwards the sequencer's command request to `dfi_cmd_formatter` and issues nothing of its own — the scheduler parks so exactly one command reaches DFI per `init_cmd_valid_o` pulse.
- `refresh_ctrl.enable_i` is low (driven by `init_done_o`), so periodic refresh does not start until init completes.

When the FSM reaches `S_DONE`, `init_busy_o` drops and `init_done_o` rises; the scheduler and refresh controller take over normal operation.

21.8 Verification Notes (cocotb test plan)

Scenario	What it proves
DDR2 init from cold reset to `init_done_o`	Full DDR2 state chain executes in order.
Observe the DDR2 command stream: PREA, MRS×3 (EMR2/EMR3/EMR1), MRS MR0_DLL, PREA, REF×2, MRS MR0, MRS OCD_DEF, MRS OCD_EXIT	Exact JEDEC order + op/index/data on each `init_cmd_*` pulse.
DDR2 MR values on `init_cmd_row_o` match 0x533 / 0x433 / 0x380 / 0x000 (MR0.VAL resets)	Wide row path carries bit 10 without truncation.
LPDDR2 init from cold reset to `init_done_o`	Full LPDDR2 MRW chain executes.
LPDDR2 `mrw_row()` packing: MR63/MR10 reach the DRAM; only MR1/MR2/MR3 set `mr_seq_we_o`	Index/data packing + selective shadowing.
`dfi_init_complete_i` held low → FSM stalls in `S_DFI_INIT`, `init_busy_o` stays high	DFI handshake gating.
Each command state produces exactly one `init_cmd_valid_o` cycle, then `S_WAIT`	Single-cycle pulse behavior.
Program short `t_*_wait_i` values → inter-command `S_WAIT` countdown matches	CSR-backed wait timing.
`mr_seq_*` shadow tracks the MRS writes (MR0/MR1/MR2/MR3 for DDR2)	Shadow lockstep with issued commands; OCD_DEF not shadowed.
After `init_done_o`, `refresh_ctrl.enable_i` asserts and refresh begins	Init → refresh handoff.

21.9 Open Questions / Future Work

- **No restart / error path.** `S_DONE` is terminal and there is no `init_error`, timeout, or software-triggered re-run — a fresh init requires `mc_rst_n`. If a DFI handshake never completes, the FSM simply parks in `S_DFI_INIT` forever. A future variant could add a timeout on `dfi_init_complete_i` and a status/restart CSR, but that machinery is not in the current RTL.

- **MR4 readback (LPDDR2).** LPDDR2 supports reading MR4 for device temperature class. The current sequencer only writes mode registers; it does no MRR readback. Folding a temperature read into or after init is possible future work but is not implemented.
- **ZQCL / ZQCS.** The `zqcl_req_o` / `zqcl_grant_i` ports exist only as a legacy DDR3+ handshake and are tied off; DDR2 has no ZQCL and LPDDR2 does its ZQ init via the MR10 MRW. If the family is extended to DDR3/DDR4, this port becomes a real ZQ-calibration handshake and the FSM would gain a ZQ state.
- **Per-rank MRS.** The sequencer issues a single stream of commands with no rank iteration; multi-rank support would require a rank field and a per-rank MRS loop. Out of scope for the current single-rank Nexys A7 target.

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22 Power-Down Controller (`powerdown_ctrl`)

Module: `powerdown_ctrl.sv` **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_mem_cmd_scheduler` **Status:** Live but optional — a single-channel idle-detect power-down requester. Not instantiated in the default top build. Deep Power Down, per-rank control, and refresh/init interlocks are TODO (see Open Questions).

> **Renamed / rescoped.** An earlier SWAG called this `power_state_fub` and described a per-rank FSM array with a separate DEEP_POWER_DOWN state, a quiet-point detector, self-refresh coordination handshake with the refresh manager, split APD/SRF idleness counters, an init-time CKE handoff mux, and a large CSR/IRQ interface. **None of that is implemented.** The actual RTL is a single channel-wide 4-state FSM (`S_AWAKE / S_ARMING / S_REQ / S_ASLEEP`) that detects controller idleness and requests either Precharge Power Down (PDE) or Self Refresh (SR). CKE is dropped for all ranks together. This chapter describes what the RTL actually does.

22.1 Purpose

`powerdown_ctrl` is a small idle-detector that asks the scheduler for permission to put the DRAM into a low-power state, then drops the DFI clock-enable (`CKE`) when granted. It supports two low-power flavors, selected by two enable inputs:

1. **PDE (Precharge Power Down)** — CKE low, banks may stay precharged; wakes on any new controller activity.
2. **SR (Self Refresh)** — CKE low while the DRAM self-refreshes internally. SR takes priority over PDE when both are enabled.

The FSM only drops and raises `CKE`. It does **not** issue the SREFE/SREFX DRAM commands, does **not** run any exit-timing counters (tXP / tXSDLL / tXSR), and does **not** coordinate a handshake with the refresh controller. All command issue, precondition enforcement (all banks precharged before SR), and exit timing are the responsibility of the parent scheduler (`pumice_mem_cmd_scheduler`) — this FUB is purely a request/grant + CKE-drive block. It trusts the scheduler not to issue a grant before DRAM init completes.

22.2 Synthesis Parameters

Parameter	Default	Effect
`NUM_RANKS`	`1`	Number of ranks. `CKE` is driven identically to all ranks (see below).
`CS_WIDTH`	`NUM_RANKS`	Width of the `dfi_cke_o` output vector.

There is no `MEMTYPE` parameter and no build-time DPD/PDE/SR selection — the enable inputs select behavior at run time.

22.3 FSM

The block is a **single** channel-wide FSM (`state_e`, 3-bit encoded), not a per-rank array. There is no separate Deep Power Down state.

State	Encoding	Meaning	`dfi_cke_o`
`S_AWAKE`	`3'd0`	CKE high, normal operation; idle counter held at 0	`'1`

State	Encoding	Meaning	dfi_cke_o
S_ARMING	3'd1	Controller idle; counting cycles toward entry	'1
S_REQ	3'd2	pdn_req_o high, awaiting pdn_grant_i	'1
S_ASLEEP	3'd3	Granted; CKE low, in PDE or SR per latched r_kind	'0

Reset state is S_AWAKE with dfi_cke_o = '1 (CKE high out of reset).

22.3.1 PDE-vs-SR Selection

The request is enabled when either mode is enabled: w_request_enabled = enable_pde_i || enable_sref_i.

- enable_sref_i high → on reaching the idle threshold, request **SR** (pdn_kind_o = 1). SR has priority over PDE.
- enable_sref_i low, enable_pde_i high → request **PDE** (pdn_kind_o = 0).
- Neither enabled → never request; the FSM stays in S_AWAKE.

The kind is latched into r_kind at the S_ARMING → S_REQ transition (r_kind <= enable_sref_i) and is held through S_ASLEEP.

22.3.2 Transitions

```
S_AWAKE:
  r_idle_cnt <- 0
  if (w_request_enabled && controller_idle_i) -> S_ARMING

S_ARMING:
  if (!controller_idle_i || !w_request_enabled) -> S_AWAKE, r_idle_cnt
<- 0
  else if (r_idle_cnt >= idle_threshold_i)      -> S_REQ, r_kind <-
enable_sref_i
  else                                          r_idle_cnt <-
r_idle_cnt + 1

S_REQ:
  if (!controller_idle_i || !w_request_enabled) -> S_AWAKE, r_idle_cnt
<- 0
  else if (pdn_grant_i)                         -> S_ASLEEP
                                                   (r_grants_sr++ if
r_kind else r_grants_pde++)

S_ASLEEP:
  if (!controller_idle_i)                       -> S_AWAKE, r_idle_cnt
<- 0
```

```
  // wakes on ANY new activity; SR exit timing (tXSDLL/tXSR) is
enforced
  // by the scheduler, not here
```

Any loss of `controller_idle_i` (or de-assertion of both enables) at `S_ARMING` or `S_REQ` backs the FSM off to `S_AWAKE`. From `S_ASLEEP`, the FSM wakes on any new controller activity and returns to `S_AWAKE`; the scheduler is responsible for honoring exit latency before issuing commands to the DRAM.

The grant counters `r_grants_pde` / `r_grants_sr` (16-bit) increment on each successful grant and are exposed as observability outputs.

22.3.3 CKE Behavior

`dfi_cke_o` is a strict-flop output: `'1` out of reset and whenever the FSM is not in `S_ASLEEP`; `'0` in `S_ASLEEP`. `CKE` is `CS_WIDTH`-wide but every bit is driven identically — the block powers down **all ranks together**. Per-rank CKE control is a documented TODO.

All outputs are registered (strict-flop), so `pdn_req_o`, `pdn_kind_o`, `sref_active_o`, `dfi_cke_o`, and the `obs_*` signals lag the internal state by one cycle.

22.4 Interface

Clock `mc_clk`, active-low reset `mc_rst_n`.

22.4.1 Control / Configuration Inputs

Signal	Direction	Width	Description
`idle_threshold_i`	input	16	Idle cycles to count in `S_ARMING` before requesting power-down
`enable_pde_i`	input	1	Enable Precharge Power Down requests
`enable_sref_i`	input	1	Enable Self Refresh requests; takes priority over PDE
`controller_idle_i`	input	1	Controller idle indication from the scheduler

22.4.2 Request / Grant + CKE

Signal	Direction	Width	Description
`pdn_req_o`	output	1	High while in `S_REQ` (requesting power-down)
`pdn_kind_o`	output	1	Requested kind: `0` = PDE, `1` = SR

Signal	Direction	Width	Description
pdn_grant_i	input	1	Scheduler grant; S_REQ → S_ASLEEP on assertion
sref_active_o	output	1	High while asleep in SR (S_ASLEEP && r_kind == 1)
dfi_cke_o	output	CS_WIDTH	DFI clock-enable; '0 in S_ASLEEP, else '1

22.4.3 Observability (future CSR readout)

These outputs are provided for future CSR readback; there is no CSR/APB interface wired into this FUB today.

Signal	Direction	Width	Description
obs_state_o	output	3	Current FSM state (state_e encoding)
obs_idle_cnt_o	output	16	Current idle counter value
obs_grants_pde_o	output	16	Count of PDE grants taken
obs_grants_sr_o	output	16	Count of SR grants taken

22.5 Verification Notes (cocotb test plan)

The block is small and self-contained; a directed FSM testbench is sufficient.

Scenario	What it proves
Reset → S_AWAKE, dfi_cke_o = '1, all obs_* cleared	Reset behavior and CKE-high default
Neither enable set; hold idle → FSM never leaves S_AWAKE	No request when disabled
enable_pde_i only; idle for idle_threshold_i cycles → pdn_req_o, pdn_kind_o = 0	PDE arming/threshold and request
enable_sref_i set (with or without PDE) → pdn_kind_o = 1	SR priority over PDE
S_REQ then pdn_grant_i → S_ASLEEP, dfi_cke_o = '0, sref_active_o per kind	Grant, sleep, CKE drop, SR-active flag

Scenario	What it proves
Activity (`!controller_idle_i`) during `S_ARMING` → back to `S_AWAKE`, counter cleared	Arming back-off
Activity during `S_REQ` (before grant) → back to `S_AWAKE`	Request withdrawal on new activity
Wake from `S_ASLEEP` on `!controller_idle_i` → `S_AWAKE`, `dfi_cke_o = '1`	Wake path
Both enables dropped mid-arming / mid-request → back-off to `S_AWAKE`	Enable de-assertion handling
Multiple sleep cycles → `obs_grants_pde_o` / `obs_grants_sr_o` increment correctly	Grant counters
`idle_threshold_i = 0` → request on first idle cycle in `S_ARMING`	Threshold edge case

22.6 Open Questions / Future Work

All of the following are called out in the RTL header comments and are **not** implemented today:

- **Deep Power Down (DPD).** LPDDR2-only. Would need an `enable_dpd_i` input and `dfi_dram_clk_disable_o` cooperation from `dfi_signal_pack`. There is no DPD state in the current FSM.
- **Per-rank power-down.** The block currently powers down **all ranks together** (identical CKE on every `dfi_cke_o` bit). A per-rank FSM array with independent CKE routing is future work.
- **`dfi_init_complete` interlock.** The FSM currently trusts the scheduler not to issue `pdn_grant_i` before DRAM init completes. A hard interlock input is future work.
- **Self-refresh command / handshake ownership.** SREFE/SREFX command issue, the all-banks-precharged precondition, and exit-timing enforcement (tXP / tXSDLL / tXSR) live in the scheduler, not here. There is no `refresh_ctrl` coordination handshake in this FUB.
- **CSR / observability wiring.** The `obs_*` outputs are intended for future CSR readback but are not connected to any APB register block yet.

- **Not in the default top build.** The module is live and verifiable standalone, but is not instantiated in the default `pumice` top today.

---

23 Mode Register (`mode_register`)

Module: `mode_register.sv` **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_mem_cmd_scheduler` **Status:** implemented (DDR2 + LPDDR2 decode)

23.1 Purpose

Per-rank Mode Register shadow + live decode of MR-derived timing values for use by the rest of the controller.

On `mr_we_i`, write `mr_data_i` into shadow MR[`mr_index_i`] for `mr_rank_i` (indices < `MAX_MR_IDX` only). `init_sequencer` drives this during DRAM bring-up; a CSR/APB hot-update path drives it later.

The `memtype_i` input (`memtype_e`: `MEMTYPE_DDR2` / `MEMTYPE_LPDDR2`) selects the decode branch for every output. The live decode reads from rank 0 (multi-rank designs must program matching MR values across ranks; mixed-per-rank MRs are a TODO).

23.2 Live Decoded Outputs

Output	Source (DDR2)	Source (LPDDR2)
`cl_o`	MR0[6:4]	RL, from the MR2[3:0] RL&WL enum
`cwl_o`	CL − 1	WL, from the MR2[3:0] RL&WL enum
`bl_o`	MR0[2:0]	MR1[2:0] (4/8; BL16 clips to 8)
`al_o`	MR1[5:3]	0 (not used)
`drv_strength_o`	MR1[1]	0 (informational)
`odt_o`	MR1[6,2]	0

For LPDDR2, CL and CWL are **not** independent MR fields — they are both derived from a single MR2[3:0] “RL & WL” enum per JESD209-2F:

MR2[3:0]	RL (`cl_o`)	WL (`cwl_o`)
`0001`	3	1

MR2[3:0]	RL (cl_o)	WL (cwl_o)
0010	4	2
0011	5	2
0100	6	3
0101	7	4
0110	8	4

(default / illegal codes fall back to RL3/WL1.)

For DDR2, `cl_o` is the raw MR0[6:4] field and `cwl_o` is simply `CL − 1` (saturating at 0). `bl_o` decodes MR0[2:0] = `010` → BL4, `011` → BL8.

All outputs are strict-flop registered. Consumed by:

- `pumice_cmd_arbiter` / `pumice_mem_cmd_scheduler` — use `cl_o`, `cwl_o`, `al_o` to time RD/WR latencies.
- write data path (`pumice_dfi_wr_serializer` via `t_phy_wrlat`) — sized against `cwl_o` for the WR-to-wrdata window.
- read data path (`pumice_dfi_rd_aligner` via `t_rddata_en`) — sized against `cl_o` for the RD-to-rddata window.
- `dfi_cmd_formatter` — uses `bl_o` to size column commands.

23.3 Burst Length: Fixed per Instance, Parameterized for Family Reuse

A memory controller supports exactly **one** DRAM burst length, chosen at init and fixed for the life of the instance — the datapath (beat sequencing, prefetch depth, column-stride math, tWR/tRTP-derived timing windows) is sized around that single value. pumice does **not** switch BL per transaction; the host/AXI burst is arbitrary and `axi_intake` splits it into fixed-BL DRAM commands.

`bl_o` is therefore a **build/init constant**, not a per-command control. It is decoded from the mode register rather than hardcoded so the **same RTL retargets across the DDR family** — the only thing that changes is the programmed value and the device/beat widths:

Family	Burst length	Source	pumice-beat count (via `bl_dram_beats`)
DDR2	BL4 (fixed)	MR0[2:0]=`010`	`bl_o >> log2(DRAM_BEAT_WIDTH/DRAM_DEVICE_WIDTH)`
DDR	BL8	MR0[1:0]	same expression, `bl_o=8`

Family	Burst length	Source	pumice-beat count (via `bl_dram_beats`)
3	(fixed)		
DDR4	BL8 (fixed)	MR0[1:0]	same expression, `bl_o=8`

Everything downstream keys off the decoded `bl_o` and the device/beat widths (see §15 “Narrow-Device (x16) Support”), so no burst-length constants are baked into the sequencers or the address decode. `BYTE_OFFSET_WIDTH` keys off **device width, not BL**, so the column granularity is already BL-agnostic.

BC4 / burst-chop (DDR3/DDR4): out of scope — always issue the full fixed BL (BL8 at the RTL reset; the Nexys A7 board runs BL4). DDR3 and DDR4 allow BC4 and BL8-on-the-fly (A12). That is the *only* case where one instance would see two effective burst lengths per transaction, which would break the fixed-BL datapath invariant. The design decision is to **not** support BC4: always transfer the full fixed BL8. Revisit only if a workload demonstrably needs the partial-burst bandwidth (it usually does not).

23.4 Scope

- `MAX_MR_IDX=17` (indices 0..16) covers both DDR2 (MR0..MR3) and LPDDR2 (up to MR16). The shadow array is `[NUM_RANKS][MAX_MR_IDX]`.
- **LPDDR2 BL16 clips to BL8** because `bl_o` is 4-bit. A follow-up widens `bl_o` to `[4:0]` and updates the 3 downstream macros that consume it.
- `mr_req_o` is tied 0 — no hot MR updates are issued via the scheduler. Init does the MR loads directly through this FUB’s CSR write port. The `mr_req_*` channel is still flopped (all-zero) for port consistency and lands when the APB CSR slave provides a write-during-traffic path with a quiet-point handshake.

23.5 Tests

Verified by `dv/tests/fub/test_mode_register.py`: `smoke_ddr2`, `ddr2_cl_sweep`, `reset_values`, `ddr2_bl_sweep`, `ddr2_al_sweep`, `multi_rank` (plus LPDDR2 RL/WL-enum coverage as the suite is extended for the `memtype_i` LPDDR2 branch).

24 Write Data Path (pumice_wr_data_cam + pumice_dfi_wr_serializer)

Modules: pumice_wr_data_cam.sv, pumice_dfi_wr_serializer.sv **Location:** rtl/fub/ **Category:** FUB **Parents:** pumice_axi4_ifc (CAM), pumice_dfi_layer (serializer) **Status:** implemented

The old single-block wr_beat_sequencer / wr_data_path_fub no longer exists. In the rearchitected controller the write data path is split across two clock domains and two FUBs:

- pumice_wr_data_cam (MC domain, inside pumice_axi4_ifc) — a write-command CAM plus a wr-data SRAM with three de-FSM'd movers.
- pumice_dfi_wr_serializer (DFI domain, inside pumice_dfi_layer) — a purely mechanical DFI-word streamer that drives dfi_wrdata.

The single clock-domain crossing between them is pumice_dfi_cdc (async gaxi FIFOs). This chapter documents both halves.

Architectural context: HAS §3.7 and _SWEEP_GROUND_TRUTH.md §8. Both halves follow the repository "streaming pipeline + flags" model — no enumerated datapath FSM in the CAM movers (the DFI serializer has a tiny 3-state pacing FSM only, described below). Sequence is enforced by data-flow handshakes and burst beat-counters, not by slot state latches.

24.1 Purpose

The write data path stages the AXI write burst that arrives on the AW/W channels, holds it while the scheduler picks a moment to commit it to DRAM, and — when the DFI layer is ready — streams it onto the DFI write bus one DFI word per cycle.

pumice_wr_data_cam is the MC-domain staging structure. It stores each pending write's key {bank, row, col}, its AXI id, a free-running age, and the burst's beats in an SRAM slot. It exposes associative query ports so the pumice_cmd_arbiter can find row-hit writes, an oldest-entry port as a scheduler fallback, and a snarf (read-your-write) forwarding port for pumice_rd_intake. On commit it drains the SRAM slot into the DFI layer.

pumice_dfi_wr_serializer is the DFI-domain endpoint. On a WR command strobe (wr_fire_i) it waits t_phy_wrlat DFI cycles, then pops one DFI word per cycle from its write-data FIFO onto dfi_wrdata / dfi_wrdata_en / dfi_wrdata_mask until the burst's last word.

24.2 pumice_wr_data_cam — three movers over one SRAM

The CAM never runs a datapath state machine. It has one SRAM (r_sram[N_SRAM_SLOTS*BL], plus a companion strobe array r_strb) and three movers, each of which is just a burst beat-counter reading the head of a request FIFO:

Mover	Trigger source	Direction	Beat counter
fill	u_fill_q (insert-order slot FIFO)	wd_data_i → SRAM	r_fill_beat
snarf	u_snarf_q (accept-order slot FIFO)	SRAM → snarf_rd	r_sn_beat
commit	u_drain_q (scheduled-slot FIFO)	SRAM → cm_rd	r_cm_beat

Each mover streams straight off its request-FIFO head, which is stable until popped. The head slot IS the “active” slot; “active” simply means the FIFO is non-empty; the FIFO is popped on the burst’s last beat. This is the de-FSM’d streaming reader pattern — no active/slot latch.

24.2.1 Entry state

Per entry (NUM_ENTRIES, default 8): r_valid, r_bank, r_row, r_col, r_id, r_age, plus:

- r_ptr — SRAM slot index, set on the entry’s first fill beat.
- r_pv — pointer-valid (first beat has been staged).
- r_fdone — **fill COMPLETE** (wd_last has been seen). An entry may be scheduled or snarfed only once r_fdone is set; otherwise the commit-drain mover could outrun a gapped fill and read stale SRAM beats.
- r_sched — the scheduler has committed this slot. It is set the cycle commit_valid_i fires and immediately excludes the entry from the sched_lu / oldest ports, so the arbiter can pick the next entry the very next cycle (clean one-command-per-clock). It clears on evict.

r_age_ctr is a free-running counter; relative age is w_rel[i] = r_age_ctr - r_age[i], which is wrap-safe.

24.2.2 SRAM slot pre-allocation

N_SRAM_SLOTS may be less than NUM_ENTRIES. r_sram_occ is a per-SRAM-slot occupancy bitmap. On the first fill beat of an entry a free slot is chosen (w_slot_free, highest-index-first scan), recorded into r_ptr, and marked occupied. The slot is freed on the entry’s commit evict. wd_ready_o deasserts on the first beat if no SRAM slot is free.

24.2.3 Age-based selectors

Three selectors run combinationally over the entry array (all skip already-scheduled and not-fill-complete entries where applicable):

- **oldest port** — oldest valid, fill-complete, unscheduled entry (max relative age). Scheduler fallback.
- **scheduler lookups** (`N_SCHED_LU`, default 4) — for each `{bank, row}` query, the oldest matching fill-complete unscheduled entry. This gives the arbiter in-order commit per row.
- **snarf lookup** — the *youngest* match (min relative age), so a read sees the latest write data.

24.3 Snarf: write-to-read forwarding (was the standalone `wr2rd_forward`)

There is no standalone `wr2rd_forward` FUB anymore. Write-to-read forwarding is the **snarf mover inside `pumice_wr_data_cam`**. When `pumice_rd_intake` presents an AR's decoded key on the snarf probe port, the CAM combinationally searches for a matching in-flight write. The match is deliberately narrow — a write is snarfable only when all three hold:

1. **Not yet scheduled** (`!r_sched`). A scheduled write is draining/evicting to DRAM, so its CAM data is racy.
2. **Same AXI id** (`r_id[i] == snarf_probe_id_i`). Same-id write-before-read is the only AXI-ordered case where the read is *required* to observe the write; cross-id has no ordering guarantee, so a cross-id read takes the DRAM path instead.
3. **Same burst length** — `snarf_probe_len_i == BL-1`. Every admitted write is exactly `BL` beats (ragged bursts are rejected upstream in `pumice_wr_intake`), so this reduces to `arlen == BL-1`. A short or long read must not snarf a full-BL write.

The match must also be fill-complete (`r_fdone`). Among candidates the youngest is chosen. `snarf_hit_o` asserts when a valid probe finds a match with matching length. On `snarf_accept_i`, the matched slot is pushed into `u_snarf_q`; the snarf mover then streams `snarf_rd_data_o` beats (non-destructively — snarf does not evict the write) in accept order.

> The old chapter's "last-write-wins highest-slot-index" rule and the "any matching in-flight write regardless of id" behavior are gone. The live policy is youngest-match, same-id, same-BL, unscheduled only.

24.4 Commit drain and B response

The arbiter's commit does two things atomically: it sets `r_sched` on the slot (immediate exclusion from the pick ports) and enqueues the slot into `u_drain_q`. The commit mover works through scheduled slots at its own pace, streaming `cm_rd_data_o` / `cm_rd_strb_o` to the DFI write path and popping `u_drain_q` on the last beat. On that last beat the entry is evicted (`r_valid`, `r_pv`, `r_fdone`, `r_sched` cleared; SRAM slot freed) and `commit_done_valid_o` / `commit_done_id_o` strobe, which drives the `pumice_wr_intake` B response. Decoupling the scheduled-mark from the drain is what allows one-command-per-clock scheduling.

24.5 `pumice_dfi_wr_serializer` — the DFI-domain endpoint

This FUB is purely mechanical and lives in the DFI clock domain. Its internal datapath unit is the DFI word (`DFI_DATA_WIDTH`, default 128), which already carries all `DFI_RATE` phases; `dfi_signal_pack` splits it to the pins, so there is **no per-beat packing here**.

Interface (abridged):

Signal	Dir	Description
`t_phy_wrlat_i`	in	WR-cmd → wrdata_en delay (DFI cycles)
`wr_fire_i`	in	WR command strobe (from `pumice_dfi_cmd_path`)
`wd_valid/ready/data/strb/last`	in/out	Pre-staged write-data FIFO (DFI-word granular)
`dfi_wrdata_o`	out	`= wd_data_i` while driving
`dfi_wrdata_en_o`	out	all-ones while driving, else 0
`dfi_wrdata_mask_o`	out	`~wd_strb_i` while driving (AXI wstrb=1 write → DFI mask=1 means "mask", i.e. mask = ~strb)

It has a small 3-state pacing FSM (`S_IDLE`, `S_WAIT`, `S_DRIVE`) — the only FSM in the whole write path, and it exists to time `t_phy_wrlat`, not to route data:

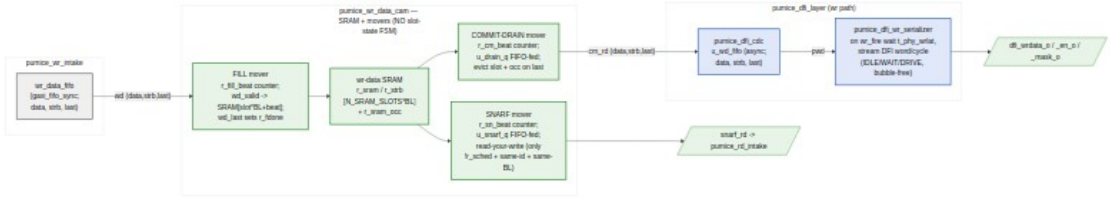
- `S_IDLE`: on an available fire, if `t_phy_wrlat==0` drive word 0 combinationally this cycle (via `w_start_now`) and go straight to `S_DRIVE` (or stay idle for a single-word burst); if `==1` go to `S_DRIVE`; else load `r_wait = t_phy_wrlat-1` and go `S_WAIT`.
- `S_WAIT`: count down, enter `S_DRIVE` at fire+wrlat.

- `S_DRIVE`: pop one DFI word per cycle. On the last word, if another fire is pending continue **seamlessly** into the next burst (zero bubbles); else return to `S_IDLE`.

A `r_pending` counter (3-bit) absorbs `wr_fire` strobes that arrive while a burst is in flight. This matters: the DFI cmd path paces column commands exactly `BL_WORDS` DFI cycles apart (the burst's DQ-bus occupancy), so the next burst's first word is due the cycle right after the current burst's last word. Because the data is pre-staged in the FIFO (the CAM drains it when the command is scheduled), at `wr_fire` the burst is already waiting, so the drive never stalls and bursts run back-to-back with zero bubbles.

> Fixed bug (see project notes): a fire arriving mid-burst used to be dropped, stranding the write in the PHY. It is now counted in `r_pending`.

24.6 Block Pipeline View



Write Data Path — streaming CAM movers + DFI-word serializer, no datapath FSM

Source: 15_wr_data_path_pipeline.mmd

24.7 `pumice_wr_data_cam` interface

24.7.1 Insert (from `pumice_wr_intake` aw_push)

Signal	Dir	Width	Description
`ins_valid_i`	in	1	Allocate an entry
`ins_ready_o`	out	1	Free entry AND fill-FIFO room
`ins_bank_i`	in	`BKW`	Decoded bank
`ins_row_i`	in	`ROW_WIDTH`	Decoded row
`ins_col_i`	in	`COL_WIDTH`	Decoded column
`ins_id_i`	in	`IW`	AXI id

24.7.2 Fill data (from pumice_wr_intake wdata pop)

Signal	Dir	Width	Description
wd_valid_i	in	1	Write-data beat valid
wd_ready_o	out	1	Fill-FIFO head present AND SRAM slot free
wd_data_i	in	DW	Beat data
wd_strb_i	in	SW	Byte strobes
wd_last_i	in	1	Last beat (sets r_fdone, pops fill FIFO)

24.7.3 Snarf lookup + stream (to/from pumice_rd_intake)

Signal	Dir	Description
snarf_probe_*	in	Probe key {bank,row,col}, id, len
snarf_hit_o	out	Valid probe matched (youngest, same-id, same-BL)
snarf_accept_i	in	Read admitted as a snarf; push slot to snarf FIFO
snarf_rd_valid/ ready/data/last	out /in	Snarf data stream (accept order)

24.7.4 Oldest port + scheduler lookups (to pumice_cmd_arbiter)

Signal	Dir	Description
oldest_valid_o + oldest_{bank,row, col,id,slot}_o	out	Oldest fill-complete unscheduled entry
sched_lu_valid_i[N] + sched_lu_{bank,ro w}_i	in	Per-port {bank,row} queries
sched_lu_hit_o[N] + sched_lu_{slot,co l,id,age}_o	out	Oldest matching entry per port

24.7.5 Commit / drain (to pumice_dfi_layer write path)

Signal	Dir	Description
commit_valid_i	in	Arbiter commits a slot
commit_ready_o	out	Room in the drain FIFO
commit_slot_i	in	Slot to schedule
cm_rd_valid/	out	Commit-drain data stream

Signal	Dir	Description
ready/data/strb/last	/in	
commit_done_valid_o + commit_done_id_o	out	Evict strobe → drives B response
busy_o	out	Any pending fill / snarf / drain / oldest

24.8 pumice_dfi_wr_serializer parameters

Parameter	Default	Effect
DFI_DATA_WIDTH	128	DFI-word width (= DRAM_BEAT_WIDTH*DFI_RATE)
DFI_RATE	2	Phases per DFI word (= DFI_EN_WIDTH)
DFI_STRB_WIDTH	DW/8	Byte-mask width
WRLAT_W	8	Width of t_phy_wrlat_i

24.9 Verification Notes (cocotb test plan)

Scenario	What it proves
Insert + fill a BL burst; r_fdone sets only on wd_last	Fill mover + fill-complete gate
Commit an entry; SRAM drains in order; evict + commit_done on last	Commit mover + B-response strobe
Snarf hit: same-id, same-BL, unscheduled write returns its data	Snarf youngest-match forwarding
Snarf miss: cross-id read, or scheduled write, or mismatched len	Snarf policy exclusions
Oldest / sched-lookup exclude scheduled + not-fill-complete entries	Pick-port correctness (1 cmd/clock)
SRAM slot exhaustion: wd_ready_o / ins_ready_o deassert	Pre-allocation backpressure
DFI serializer: t_phy_wrlat sweep (0, 1, N); first word at fire+wrlat	Write-latency alignment
Back-to-back WR fires paced BL_WORDS apart: zero-bubble drive	r_pending seamless continuation

Scenario	What it proves
`wr_fire` mid-burst is counted, not dropped	`r_pending` regression
`dfi_wrdata_mask_o == ~wd_strb_i` while driving	Strobe → mask inversion

24.10 Open Questions / Future Work

- **Cross-burst commit interleave.** The commit mover drains one scheduled slot at a time. Interleaving beats from multiple scheduled slots would need a per-slot beat-counter shadow. Punt to a later performance mode.
- **Multi-outstanding DFI reads/writes.** The serializer's `r_pending` counter is 3-bit; deeper pipelining is bounded by the DFI cmd path's pacing, not by this FUB.
- **Strobe-less writes.** Hosts that always write full beats could elide the strobe SRAM and mask path at elaboration. Minor area win; punt.

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25 Read Data Path (`pumice_rd_cmd_cam` + `pumice_dfi_rd_aligner`)

Modules: `pumice_rd_cmd_cam.sv`, `pumice_dfi_rd_aligner.sv` **Location:** `rtl/fub/` **Category:** FUB **Parents:** `pumice_axi4_ifc` (CAM), `pumice_dfi_layer` (aligner) **Status:** implemented

The old single-block `rd_cl_aligner` / `rd_data_path_fub` no longer exists. In the rearchitected controller the read data path is split across two clock domains and two FUBs:

- `pumice_dfi_rd_aligner` (DFI domain, inside `pumice_dfi_layer`) — drives `dfi_rddata_en` at the right cycle for an issued READ and captures `dfi_rddata` words into the read return FIFO.
- `pumice_rd_cmd_cam` (MC domain, inside `pumice_axi4_ifc`) — an outstanding-read reorder buffer: DRAM data returns in *issue* order and buffers per entry; it drains to `pumice_rd_intake` in *AR* order.

The single clock-domain crossing between them is `pumice_dfi_cdc` (async gaxi FIFOs). This chapter documents both halves.

Architectural context: HAS §3.7 and `_SWEEP_GROUND_TRUTH.md` §8. The CAM has no datapath FSM — the movers are burst beat-counters over the SRAM. The DFI aligner has a small pacing FSM for the `dfi_rddata_en` window only. Sequence is enforced by data-flow handshakes, not by slot state.

25.1 Purpose

The read data path is the miss path: reads that cannot be satisfied by a snarf from the write CAM (see §17) go to DRAM. `pumice_rd_cmd_cam` is the mirror of `pumice_wr_data_cam` — entries keyed `{bank, row, col}` with a free-running age, an oldest port, and `N_SCHED_LU` scheduler lookups, so reads row-hit-schedule exactly like writes. It also acts as a **reorder buffer**: it accepts AR inserts in AR order, lets the scheduler issue them in an order it chooses (row-hit optimized), receives DRAM data in *issue* order, and drains it back to the host in *AR/oldest* order.

`pumice_dfi_rd_aligner` is the DFI-domain endpoint. On a RD command strobe (`rd_fire_i`) it drives the `dfi_rddata_en` capture window starting `t_rddata_en` DFI cycles later, and pushes one DFI word per `dfi_rddata_valid` cycle into the read return FIFO as `{data, resp, last}`.

Compared to the write CAM there is **no snarf port** here — snarf is a write-CAM concept (§17). Data flows *in* from the DFI return and *out* to `pumice_rd_intake`.

25.2 `pumice_rd_cmd_cam` — reorder buffer, two movers over one SRAM

The CAM stores each burst's return data in an SRAM (`r_sram[N_SRAM_SLOTS*BL]`) and runs two de-FSM'd movers, each a burst beat-counter:

Mover	Trigger source	Direction	Beat counter
return-fill	`u_issue_q` (issue-order slot FIFO)	dfi_ret → SRAM	`r_ret_beat`
drain	oldest-valid pick, data-ready gated	SRAM → drain	`r_dr_beat`

25.2.1 Entry state and age

Per entry (NUM_ENTRIES, default 8): r_valid, r_issued (scheduler has issued it to DRAM), r_ready (return data complete), r_bank, r_row, r_col, r_id, r_resp, r_age, plus r_ptr (SRAM slot, set on the first return beat) and r_pv (pointer valid). r_age_ctr is free-running; relative age w_rel[i] = r_age_ctr - r_age[i] is wrap-safe.

SRAM slot pre-allocation is identical to the write CAM: r_sram_occ bitmap, highest-index-first free scan, allocate on first return beat, free on drain evict.

25.2.2 The three age selectors

- **issue-side oldest** — oldest valid **not-yet-issued** entry (max rel). Scheduler fallback; drives the oldest_* port.
- **scheduler lookups** (N_SCHED_LU, default 4) — per {bank, row} query, the oldest not-issued match. Row-hit read scheduling.
- **drain-side oldest** — oldest valid entry over **all** valid entries (max rel), regardless of issued/ready. This is w_dro_slot.

25.2.3 Ordering guarantee

Inserts happen in AR order and each new insert is *younger*, so the drain-side oldest pick w_dro_slot is stable across a burst: the entry stays valid until its own last-beat evict, and later inserts can never become "more oldest". The drain mover therefore needs no active latch — the draining slot IS the oldest-valid pick. It only fires when that oldest entry's data is staged: w_dr_go = w_dro_found && r_ready[w_dro_slot]. This is what enforces AR-order release even though DRAM returns arrive in issue order.

25.2.4 Issue-order return fill

When the scheduler issues a read it notifies the CAM (issue_valid_i / issue_slot_i), which sets r_issued[slot] and pushes the slot into u_issue_q. DRAM returns arrive in the same order the reads were issued, so the return-fill mover writes each return burst into the issue-FIFO head slot's SRAM region. On the burst's dfi_ret_last, it sets r_ready, captures r_resp, and pops u_issue_q.

25.3 pumice_dfi_rd_aligner — the DFI-domain endpoint

This FUB mirrors pumice_dfi_wr_serializer. Its internal unit is the DFI word (= dfi_rddata width); BL_WORDS = BL/DFI_RATE words per burst.

It has two independent concerns:

1. dfi_rddata_en window (small pacing FSM). A 3-state FSM (S_IDLE, S_WAIT, S_EN) plus a pending-fire counter r_epend:

- `S_IDLE`: on an available fire, if `t_rddata_en==0` drive word 0's enable combinationally (`w_en_now`) and seed the remaining-cycle counter for `BL_WORDS-1` more en cycles; if `==1` go straight to `S_EN`; else load `r_ewait = t_rddata_en-1` and go `S_WAIT`.
- `S_WAIT`: count down, enter `S_EN` at fire+`t_rddata_en`.
- `S_EN`: assert `dfi_rddata_en_o` for the window. On the last en-cycle, if another fire is pending reseed for a contiguous back-to-back window (zero bubbles); else return to `S_IDLE`.

Like the write serializer, the DFI cmd path paces column commands `BL_WORDS` apart, so windows abut with zero bubbles. A `rd_fire` arriving mid-window is counted in `r_epend` (previously it was dropped, leaving the second read with no capture window and stranding it in the PHY).

2. Capture (flag-and-counter only). On each `dfi_rddata_valid` (`w_word_valid = | dfi_rddata_valid_i`) a word is pushed to the read FIFO: `rd_data_o = dfi_rddata_i`, `rd_resp_o = OKAY`, and `rd_last_o` asserts on word `BL_WORDS-1`. A single word counter `r_rcnt` tracks progress; it wraps on the last word.

> v1 note: `rd_resp_o` is hardwired `RESP_OKAY` — the DFI rddata-error signal is not yet wired. The aligner supports one outstanding read window (single-issue, tRTW/tCCD spaced by the scheduler / `global_timers`).

25.4 Block Pipeline View

Read Data Path — DFI en-window aligner + issue-order fill + AR-order drain reorder buffer

Source: 16_rd_data_path_pipeline.mmd

25.5 `pumice_rd_cmd_cam` interface

25.5.1 Insert (from `pumice_rd_intake` ar_push, AR order)

Signal	Dir	Width	Description
`ins_valid_i`	in	1	Allocate an entry

Signal	Dir	Width	Description
ins_ready_o	out	1	Free entry available
ins_bank/row/col/id_i	in	key + id	Decoded key + AXI id

25.5.2 Scheduler ports (to pumice_cmd_arbiter)

Signal	Dir	Description
sched_lu_valid_i[N] + sched_lu_{bank,row}_i	in	Per-port {bank,row} queries
sched_lu_hit_o[N] + sched_lu_{slot,col,id,age}_o	out	Oldest not-issued match per port
oldest_valid_o + oldest_{bank,row,col,id,slot}_o	out	Oldest not-issued entry

25.5.3 Issue notify (from scheduler)

Signal	Dir	Description
issue_valid_i	in	Scheduler issued this slot to DRAM
issue_ready_o	out	Room in the issue-order FIFO
issue_slot_i	in	Slot issued (records issue order for return fill)

25.5.4 DFI return (from pumice_dfi_layer read path, issue order)

Signal	Dir	Description
dfi_ret_valid_i	in	Return beat valid
dfi_ret_ready_o	out	Issue-FIFO head present AND SRAM slot free
dfi_ret_data_i	in	Return beat data
dfi_ret_resp_i	in	Return response (captured on last)
dfi_ret_last_i	in	Last beat (sets r_ready, pops issue FIFO)

25.5.5 Drain (to pumice_rd_intake R source, AR order)

Signal	Dir	Description
drain_valid_o	out	Oldest entry with data ready

Signal	Dir	Description
drain_ready_i	in	Downstream accepts
drain_data_o	out	Beat data
drain_id_o	out	AXI id (rid echo)
drain_resp_o	out	rresp
drain_last_o	out	Last beat (evicts entry, frees SRAM slot)
busy_o	out	Oldest-valid present OR issue FIFO non-empty

25.6 pumice_dfi_rd_aligner parameters

Parameter	Default	Effect
DFI_DATA_WIDTH	128	DFI-word width
DFI_RATE	2	Phases per DFI word (= DFI_EN_WIDTH, DFI_VALID_WIDTH)
BL_WORDS	4	DFI words per read burst (= BL/DFI_RATE)
RDEN_W	8	Width of t_rddata_en_i

25.7 Out-of-Order Completion (across AXI IDs)

The CAM preserves AXI ordering the way the standard requires: per-ID reads complete in issue order (a single in-flight read returns all its beats in burst order), while cross-ID reads may complete out of order because the scheduler is free to issue them in row-hit order. The reorder buffer's AR-order drain (oldest-first, data-ready gated) makes the release order deterministic; the drain_id_o echo tags each beat with its original AXI id.

25.8 Verification Notes (cocotb test plan)

Scenario	What it proves
Insert AR, issue, DFI return, drain: BL burst round-trips	Full miss-path smoke
Returns in issue order buffered into correct SRAM slots	Issue-order return fill
Out-of-issue-order scheduling but AR-order drain release	Reorder-buffer ordering guarantee

Scenario	What it proves
Drain gated on `r_ready`: oldest entry with data not yet complete stalls	Data-ready gate
SRAM slot exhaustion deasserts `dfi_ret_ready_o`	Return-side pre-allocation backpressure
DFI aligner `t_rddata_en` sweep (0, 1, N); en-window at fire+t_rddata_en	Read-latency window alignment
Back-to-back RD fires paced `BL_WORDS` apart: contiguous en windows	`r_epend` seamless continuation
`rd_fire` mid-window counted, not dropped	`r_epend` regression
`rd_last_o` on DFI word `BL_WORDS-1`	Word counter / burst boundary

25.9 Open Questions / Future Work

- **DFI rddata error propagation.** `rd_resp_o` is fixed OKAY in v1; wire the PHY rddata-error to `rd_resp_o` (and thence `r_resp` → AXI rresp) during bring-up.
- **Multi-outstanding read windows.** The aligner is single-issue in v1; the `r_epend` counter already tolerates paced back-to-back fires, but overlapping windows would need a deeper capture-side counter.
- **Read-path parallelism for reorder / col-major.** Deferred perf item — the drain mover releases one oldest burst at a time.

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26 DFI Command Formatter (`dfi_cmd_formatter` + `dfi_signal_pack`)

Module: `dfi_cmd_formatter.sv` (command decode) / `dfi_signal_pack.sv` (final DFI pipe stage) **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_dfi_layer` **Status:** Implemented (DDR2 truth table + bit-exact LPDDR2 CA-bus encoding; both memtypes functional)

Replaces the retired command encoder. The original `cmd_encoder_fub` (with its `ddr2_cmd_encoder` / `lpddr2_cmd_encoder` generate-branch sub-modules and a separate `odt_ctrl_fub`) is gone. The live design is `dfi_cmd_formatter.sv` — a single module with a runtime `memtype_i` branch (not an elaboration-time generate) — followed by `dfi_signal_pack.sv`, the final registered DFI-bus pipeline stage.

LPDDR2 is now fully implemented. The retired chapter listed LPDDR2 as "deferred to v2." The live formatter builds the bit-exact JESD209-2F Table 60 CA-bus word for every command; the transcription lives in `rtl/LPDDR2_CA_ENCODING.md`, and the DV BFM (`lpddr_ca.py`) encodes/decodes against the identical layout.

ODT is absorbed here. There is no standalone ODT block (see the retired ODT chapter). `dfi_cmd_formatter` drives `dfi_odt_o`; the DDR2 ODT rule follows from the same truth table. In v1 `dfi_odt_o` is driven to 0 (the decode leaves `w_p0_odt` at its NOP default for every op); the JEDEC ODT-timing hooks live in `mode_register`'s `odt_o` decode.

26.1 Purpose

`dfi_cmd_formatter` translates the scheduler's abstract command — `(cmd_op, rank, bank, row, col, len)` plus an implicit auto-precharge encoded in the op (RDA/WRA vs RD/WR) — into wire-level DFI v2.1 control-bus signals, packed across `DFI_RATE` phases. For DDR2 it drives the classic `{ras_n, cas_n, we_n}` strobes plus `dfi_address` / `dfi_bank`; for LPDDR2 it packs the multiplexed 10-bit CA bus (two edges = a flat 20-bit word) onto `dfi_address`.

`dfi_signal_pack` is the final pipeline register on the DFI v2.1 bus. It latches the formatter's command bus together with the write/read data-enable signals and drives reset-safe NOP values during reset. It owns `dfi_dram_clk_disable` (currently held 0).

Both modules run in the DFI clock domain inside `pumice_dfi_layer` (see the DFI layer / gearing chapter); the formatter is instantiated via `pumice_dfi_cmd_path`.

26.2 Synthesis Parameters (`dfi_cmd_formatter`)

Parameter	Default	Effect
`NUM_RANKS`	1	Width of the per-rank `dfi_cs_n` / `dfi_odt` output

Parameter	Default	Effect
NUM_BANKS	8	Bank-field width (BKW)
ROW_WIDTH	14	Row operand width (also carries DDR2 MR data and LPDDR2 MRW field)
COL_WIDTH	10	Column operand width
BURST_LEN_WIDTH	8	cmd_len_i width (currently unused — tied into unused_v1)
DFI_RATE	2	Phases per DFI-layer word; sets the multi-phase bus widths
DFI_ADDR_WIDTH	14	Per-phase address width; the LPDDR2 CA word needs DFI_ADDR_BUS_W >= 20
DFI_BANK_WIDTH	3	Per-phase bank width
DFI_CTRL_WIDTH	1	Per-phase width of each ras/cas/we strobe
DFI_CS_WIDTH	NUM_RANKS	Per-phase CS/ODT width

The multi-phase bus widths (DFI_*_BUS_W = DFI_*_WIDTH * DFI_RATE) are derived. memtype_i is a **runtime input** (MEMTYPE_DDR2 / MEMTYPE_LPDDR2), not a parameter — both encoding paths are synthesized and the branch is selected live, so one bitstream can serve either family.

26.3 Command Interface and Runtime Phase Placement

The formatter takes a valid/ready command handshake from the DFI-layer command FIFO:

Signal	Width	Description
cmd_valid_i	1	Command present
cmd_ready_o	1	Always 1 (registered) — formatter never stalls
cmd_op_i	dram_op_e	The chosen operation
cmd_rank_i	RKW	Target rank (selects the active CS_n)
cmd_bank_i	BKW	Target bank (MR index for MRS)
cmd_row_i	ROW_WIDTH	Row (ACT); DDR2 MR data; LPDDR2 packed

Signal	Width	Description
		MRW field
cmd_col_i	COL_WIDTH	Column (RD/WR)
cmd_len_i	BURST_LEN_WIDTH	Burst length (reserved; unused in v1)
rd_phase_i	PHW	DFI sub-phase carrying the READ command
wr_phase_i	PHW	DFI sub-phase carrying the WRITE command

rd_phase_i / wr_phase_i are CSR-driven (DFI_PHASE CSR). The decoded command is placed on wr_phase for WR/WRA, rd_phase for RD/RDA, and phase 0 for everything else. This matches a PHY that consumes a per-command rdphase/wrphase off the DFI bus. Defaults are 0, which reproduces the legacy "everything on phase 0" behavior — notably the LiteDRAM a7ddrphy takes the command on phase 0 and applies its rdphase internally, so on the board target these stay 0. (See the gearing chapter §"DFI_PHASE CSR".)

26.4 DDR2 Encoding Branch

For memtype_i == MEMTYPE_DDR2, a combinational block builds the phase-0 command fields (w_p0_*). The default is an all-deselected NOP; when a command is valid, w_p0_cs_n is set to the active-rank mask (w_active_rank_mask, bit r low for the target rank) and the strobes/address are driven per the JEDEC JESD79-2 truth table (transcribed verbatim from the RTL):

Op	cs_n	ras_n	cas_n	we_n	bank	address
OP_NOP	active mask	1	1	1	0	0
OP_ACT	active mask	0	1	1	bank	row
OP_RD	active mask	1	0	1	bank	col (A10 = 0)
OP_RDA	active mask	1	0	1	bank	col with bit 10 set (auto-PRE)
OP_WR	active mask	1	0	0	bank	col (A10 = 0)
OP_WRA	active mask	1	0	0	bank	col with bit 10 set (auto-PRE)
OP_PRE	active	0	1	0	bank	0 (A10 = 0, single-bank)

Op	cs_n	ras_n	cas_n	we_n	bank	address
	mask					
OP_PREA	active mask	0	1	0	0	bit 10 set (all-bank)
OP_REF	active mask	0	0	1	0	0
OP_MRS	active mask	0	0	0	MR idx	MR data from cmd_row_i

cs_n is per-rank: bit r is 0 for the selected rank, 1 elsewhere; all-1 is an all-deselected NOP. The A10 auto-precharge bit is set by OR-ing 1 << 10 into the address for RDA/WRA, and doubles as the all-bank flag for PREA.

MRS carries data on the ROW field, not the column. The RTL comment is explicit: MR0 = 0x533 needs bit 10 (tWR[1]), which a COL_WIDTH (10-bit) field would truncate — so the formatter reads MR data from cmd_row_i (ROW_WIDTH) and the MR index from cmd_bank_i. This matches init_sequencer, which drives MR data on init_cmd_row_o.

Unhandled DDR2-irrelevant ops (OP_REFPB, OP_ZQCS/ZQCL, self-refresh entry/ exit) fall through the default arm and emit NOP — they are driven via CKE / a separate sequencer, not the strobe truth table.

26.5 LPDDR2 Encoding Branch (bit-exact JESD209-2F Table 60)

LPDDR2 has no ras_n/cas_n/we_n. The command AND a scrambled address are multiplexed onto a 10-bit CA bus over two clock edges (rising r, falling f), carried on the DFI bus as a flat 20-bit word. The formatter builds two 10-bit half-words w_ca_r / w_ca_f and concatenates them:

```
assign w_lpddr2_ca = {w_ca_f, w_ca_r};   // [19:10] = falling, [9:0] = rising
```

This is the exact layout locked in rtl/LPDDR2_CA_ENCODING.md and enforced by a round-trip conformance test against the DV decoder (lpddr_ca.py). Per JEDEC, any other bit ordering is prohibited, so bit-exactness is a spec requirement.

Command decode is by the rising-edge opcode bits {CA0r, CA1r, CA2r, CA3r}, transcribed from the RTL unique case:

Op	Rising opcode bits set	Payload placement
OP_ACT	CA1r = 1	bank → CA7r..CA9r; row hi R8..R12 →

Op	Rising opcode bits set	Payload placement
		CA2r..CA6r; row lo R0..R7 → CA0f..CA7f; R13 → CA8f; R14 → CA9f
OP_RD / OP_RDA	CA0r = 1, CA2r = 1 (read)	bank → CA7r..CA9r; C1,C2 → CA5r,CA6r; AP → CA0f; C3..C11 → CA1f..CA9f
OP_WR / OP_WRA	CA0r = 1, CA2r = 0 (write)	same column/bank/AP placement as read
OP_PRE	CA0r=1, CA1r=1, CA3r=1 (AB=0)	bank → CA7r..CA9r
OP_PREA	CA0r=1, CA1r=1, CA3r=1, CA4r=1 (AB=1)	bank don’t-care
OP_REF	CA2r=1, CA3r=1 (all-bank)	—
OP_REFPB	CA2r=1 (per-bank; CA3r=0)	bank implied by device counter
OP_MRS (MRW)	all opcode bits 0	MA0..MA5 → CA4r..CA9r; MA6,MA7 → CA0f,CA1f; OP0..OP7 → CA2f..CA9f
default (NOP)	CA0r=CA1r=CA2r=CA3r=1	—

Auto-precharge is `w_ca_ap = (op == OP_RDA) || (op == OP_WRA)`, placed on CA0f. C0 is implied 0 and never transmitted (only C1..C11 ride the bus). Row/column/ bank are zero-extended (`w_row15`, `w_col12`) so wider geometries (R14, C10, C11) light up their reserved pins cleanly.

MRW field packing. LPDDR2 mode-register addresses reach MR63 (MA[5:0]), which a 3-bit bank port cannot express. The scheduler therefore carries the MRW fields in the ROW request as `{MA[5:0], OP[7:0]}` — `w_mr_ma = {2'b0, cmd_row_i[13:8]}`, `w_mr_op = cmd_row_i[7:0]` — matching `init_sequencer`’s `mrw_row()` packing. MA[7:6] are 0 (MR <= 63).

26.6 Multi-Phase Packing and CS_n

After the decode, a combinational stage packs the command into the multi-phase buses (`w_dfi_*`). Every phase starts at NOP (cs_n = 1, ras/cas/we = 1); then:

- **DDR2:** the decoded `w_p0_*` fields are placed on the target phase `w_cmd_phase` (rd_phase for reads, wr_phase for writes, phase 0 otherwise); the other phases stay NOP.
- **LPDDR2:** the whole 20-bit CA word is written to `w_dfi_address[19:0]` (low bits), ras/cas/we stay idle, and `dfi_cs_n` is asserted for the target rank on phase 0 when the op is not NOP. The two CA edges are already inside the word, so there is no per-DFI-phase command placement for LPDDR2.

The active-rank CS_n mask is built once (`w_active_rank_mask`): bit `r` = 0 when `r == cmd_rank_i`, else 1. For `NUM_RANKS == 1` only bit 0 exists.

26.7 `dfi_signal_pack` — Final Registered DFI Stage

`dfi_signal_pack` is a pure one-cycle registered pipeline. It latches the formatter's command bus plus the data-enable/mask signals from the write/read serializers and drives them to the PHY the next DFI cycle. Its value is the reset-safe defaults it guarantees during reset / before first issue:

Output	Reset value	Meaning
`dfi_cs_n_o`	all-1	all-deselected
`dfi_ras_n_o/cas_n_o/we_n_o`	all-1	NOP
`dfi_cke_o`	0	DRAM held CKE-low until init
`dfi_odt_o`	0	ODT off
`dfi_wrdata_en_o` / `dfi_rddata_en_o`	0	no data movement
`dfi_wrdata_mask_o`	all-1	mask all bytes (write nothing)
`dfi_dram_clk_disable_o`	0	clock enabled (power-state TODO)

It owns `dfi_dram_clk_disable` for a future power-state machine (held 0 today). Phase staggering and CKE power-down driving are v2 TODOs noted in the RTL header; today it is a transparent width-preserving flop.

26.8 Interface (`dfi_cmd_formatter` outputs)

Signal	Width	Description
`dfi_address_o`	DFI_ADDR_BUS_W	DDR2 row/col operand per phase; LPDDR2 flat CA word
`dfi_bank_o`	DFI_BANK_BUS_W	Per-phase bank (DDR2)
`dfi_cas_n_o`	DFI_CTRL_BUS_W	Per-phase CAS strobe
`dfi_ras_n_o`	DFI_CTRL_BUS_W	Per-phase RAS strobe
`dfi_we_n_o`	DFI_CTRL_BUS_W	Per-phase WE strobe
`dfi_cs_n_o`	DFI_CS_BUS_W	Per-phase per-rank chip-select
`dfi_odt_o`	DFI_CS_BUS_W	Per-phase per-rank ODT (0 in v1)

All outputs are strict-flopped; `dfi_signal_pack` widens/relatches them onto the PHY pins unchanged.

26.9 Timing Budget

The formatter is combinational decode into a single output flop; `dfi_signal_pack` adds one more flop. The worst path is the LPDDR2 ACT with full row-bit spreading across CA0r/CA0f (bank + 5 rising row bits + 8 falling row bits), which is a few LUT levels feeding the output register — well within the DFI clock budget. The formatter is not the controller's critical path; that is the arbiter upstream.

26.10 Verification Notes (cocotb test plan)

Scenario	What it proves
DDR2 ACT bank 3 row 0x1234 → `{ras,cas,we}={0,1,1}`, address=0x1234, bank=3	DDR2 ACT truth-table row
DDR2 RDA bank 3 col 0x40 → `{ras,cas,we}={1,0,1}`, address bit 10 set	DDR2 auto-precharge via A10
DDR2 PREA → `{ras,cas,we}={0,1,0}`, address bit 10	DDR2 all-bank precharge

Scenario	What it proves
set	via A10
DDR2 MRS MR0=0x533 → MR data on address (bit 10 present, not truncated)	MRS data on ROW field
LPDDR2 ACT bank 3 row 0x1234 → `w_lpddr2_ca` bit-exact vs `lpddr_ca.py` decode	LPDDR2 ACT CA packing
LPDDR2 WRA bank 3 col 0x40 → AP bit (CA0f) set	LPDDR2 auto-precharge via CA
LPDDR2 REFpb (CA2r=1, CA3r=0) vs REFab (CA2r=1, CA3r=1)	LPDDR2 refresh CA decode
LPDDR2 MRW MR63 → MA/OP packed on CA4r..CA9r / CA0f.. per Table 60	LPDDR2 MRW field packing
Multi-rank ACT rank 1 → `dfi_cs_n` bit 1 low, bit 0 high	Per-rank CS_n mask
rd_phase=1 → RD command lands on DFI phase 1, other phases NOP	Runtime phase placement
NOP / `!cmd_valid` → all-deselected (cs_n all-1), strobes all-1	Idle pattern
Reset → `dfi_signal_pack` drives cs_n=1, cke=0, wrdata_mask=all-1	Reset-safe NOP defaults

26.11 Open Questions / Future Work

- **ODT driving.** `dfi_odt_o` is held 0 in v1 (the decode leaves the ODT default at NOP). Multi-rank DDR2 needs the JEDEC cross-termination window (ODT-high on the non-accessed rank during a read, on the accessed rank during a write). The hooks exist (`mode_register.odt_o` decodes the DDR2 MR1 ODT bits); wiring the timed ODT window through the formatter is future work — see the absorbed ODT chapter.
- **`cmd_len_i` unused.** The burst-length input is reserved (tied into `unused_v1`); burst geometry is handled by the data-path serializers, so the formatter does not need it today.
- **Phase staggering / clk-disable.** `dfi_signal_pack` is a plain relatch; per-phase output staggering and `dfi_dram_clk_disable` driving (for power-down) are v2 items noted in the RTL header.

- **LPDDR2 BL16.** `mode_register.bl_o` is 4-bit and clips BL16 to BL8; only BL4/BL8 are wired end-to-end. Widening the burst path is a separate item.

---

27 DFI Layer and Host-Width Gearing (`pumice_dfi_layer` + `pumice_top_geared`)

Module: `pumice_dfi_layer.sv` (macro) / `pumice_top_geared.sv` (top wrapper) **Location:** `rtl/macro/` and `rtl/top/` **Category:** Macro / Top wrapper **Status:** Implemented (single-CDC DFI datapath; formal-IP host-width gearing wrapper)

> **Replaces the retired `gear_dfi_fub`.** The original chapter described a phase-packing FUB (`gear_dfi_fub` / `dfi_signal_pack`) that spread a single MC command and burst-of-`N` data beats into per-phase DFI slots. That phase-packing role is real but small — it lives inside `dfi_signal_pack` and the DFI-layer sub-FUBs. This chapter now covers the two live, distinct concepts that carry the word "gearing" in pumice, and it keeps them separate:
>
> 1. **The DFI layer** (`pumice_dfi_layer`) — the single controller/PHY clock-domain crossing plus the DFI-clock command/write/read datapath. This is where the internal `DW`-wide word is split into `DFI_RATE` DRAM beats.
> 2. **Host-width gearing** (`pumice_top_geared`) — an OPTIONAL top wrapper that lets a host SoC use any AXI data width, inserting the repo's formally verified AXI data-width converters between the host and the fixed-width core.
>
> These are orthogonal: (1) is about DRAM-beat phasing on the PHY side; (2) is about AXI beat width on the host side.

27.1 Concept 1 — The DFI Layer (`pumice_dfi_layer`)

27.1.1 Purpose

`pumice_dfi_layer` presents the controller-clock command/wrdata/rddata streams on one side and the DFI 2.1 pin bus on the other. It holds the **single** controller/PHY clock-domain crossing and the DFI-clock-side datapath. Its internal datapath unit is the DFI word (`dfi_wrdata` width = all `DFI_RATE` phases): one FIFO word equals one DFI cycle, which keeps the datapath bubble-free.

Two clock domains:

- **`ctl_clk`** — command from the scheduler, write data from the WR CAM, read data to the RD CAM, and the `init_start` / `init_complete` handshake.
- **`dfi_clk`** — the DFI command bus, `dfi_wrdata`/en/mask, `dfi_rddata`/en/valid, and `dfi_init_start`/complete.

27.1.2 Internal structure (verified against the RTL)

`pumice_dfi_layer` instantiates exactly four sub-FUBs:

Sub-FUB	Domain	Role
`pumice_dfi_cdc`	ctl↔dfi	The **single** clock crossing — async gaxi FIFOs only (cmd, wrdata, rddata, plus init-start/complete bit crossings)
`pumice_dfi_cmd_path`	dfi_clk	cmd FIFO → DFI command bus + `wr_fire`/`rd_fire` strobes; instantiates `dfi_cmd_formatter` (§2.14) and `dfi_signal_pack`
`pumice_dfi_wr_serializer`	dfi_clk	wrdata FIFO → `dfi_wrdata` at `t_phy_wrlat`
`pumice_dfi_rd_aligner`	dfi_clk	`dfi_rddata` → rddata FIFO at `t_rddata_en`

The CDC (`pumice_dfi_cdc`) is the only place the domains meet; everything downstream of it runs entirely in `dfi_clk`. The command path emits `wr_fire_o` / `rd_fire_o` strobes that time the serializer and aligner relative to the issued command.

27.1.3 Key parameters

Parameter	Default	Effect
`NUM_RANKS`	1	CS/rank fan-out
`NUM_BANKS`	8	Bank-field width

Parameter	Default	Effect
ROW_WIDTH	14	Row / DFI address width
COL_WIDTH	10	Column width
DFI_RATE	2	DRAM beats per DFI word (phase count)
DRAM_BEAT_WIDTH	64	One DRAM beat's data width
BL	8	DRAM beats per burst; BL_WORDS = BL / DFI_RATE DFI words/burst
CMD/WD/RD_FIFO_DEPTH	8/16/16	CDC FIFO depths
N_FLOP_CROSS	2	Synchronizer flop stages in the async FIFOs

Derived DFI geometry: DFI_DATA_WIDTH = DRAM_BEAT_WIDTH * DFI_RATE, DFI_STRB_WIDTH = DFI_DATA_WIDTH/8, DFI_EN_WIDTH = DFI_VALID_WIDTH = DFI_RATE. The FIFO payloads pack the command (CMD_DW), {last,strb,data} for writes (WD_DW), and {last,resp,data} for reads (RD_DW).

27.1.4 DFI-clock runtime knobs

Signal	Width	Description
memtype_i	enum	DDR2 / LPDDR2 selector for the command path
rd_phase_i	PHW	DFI sub-phase carrying the READ command (see §2.14)
wr_phase_i	PHW	DFI sub-phase carrying the WRITE command
t_phy_wrlat_i	[7:0]	PHY write latency — when the serializer launches wrdata (CSR reset 0; Nexys A7 tuple programs 1)
t_rddata_en_i	[7:0]	When the aligner strobes dfi_rddata_en (valid = en + PHY read_latency); board tuple 6

These are the PHY-integration knobs surfaced through the CSR (DFI_PHASE, DFI_TIMING); they carry the PHY-specific latencies rather than baking them into the datapath, which is what let on-board bring-up dial in t_rddata_en / phase placement without an RTL change.

27.2 Concept 2 — Host-Width Gearing (pumice_top_geared)

27.2.1 Purpose

The controller core is fixed at its natural width DW = DRAM_BEAT_WIDTH * DFI_RATE (default 128): one AXI beat == one DFI word == DFI_RATE DRAM beats, and one AXI burst (BL/DFI_RATE beats) == one DRAM burst (BL beats). pumice_top_geared wraps pumice_top with a **free** HOST_AXI_DATA_WIDTH, inserting the repo's formally verified data-width converters (axi4_dwidth_converter_wr / _rd) between a host-width AXI slave and the fixed-DW core.

This is the family-wide gearing path (DDR2/DDR3/DDR4/LPDDR2), reusing proven + formal IP rather than re-verifying a bespoke gearbox inside the freshly stabilized controller datapath.

> **The old coupling is gone.** Do not describe the retired "AXI_DATA_WIDTH == DRAM_BEAT_WIDTH" constraint. Internal geometry is DW = DRAM_BEAT_WIDTH * DFI_RATE; the DW→phase split happens inside the DFI layer / dfi_signal_pack, and the host AXI width is decoupled from it by this wrapper.

27.2.2 GEAR-1 bypass (bit-identical)

When HOST_AXI_DATA_WIDTH == DW, a generate selects the g_direct branch: the host AXI signals are wired straight through to the core with no converter and no added latency. Existing DW-width builds are therefore bit-identical whether they instantiate pumice_top directly or through pumice_top_geared.

27.2.3 Geared branch

When HOST_AXI_DATA_WIDTH != DW, the g_conv branch instantiates the two formal converters:

```
axi4_dwidth_converter_wr
#(.S_AXI_DATA_WIDTH(HOST_AXI_DATA_WIDTH), .M_AXI_DATA_WIDTH(DW), ...)
axi4_dwidth_converter_rd
#(.S_AXI_DATA_WIDTH(HOST_AXI_DATA_WIDTH), .M_AXI_DATA_WIDTH(DW), ...)
```

The host slave ports (s_axi_* at HOST_AXI_DATA_WIDTH) feed the converters; the converters' master ports (the internal c_* nets at DW) feed pumice_top. The CSR cpuif and the DFI pin bus pass straight through the wrapper unmodified.

Burst geometry. The core still requires one AXI burst == one DRAM burst at its DW side ((awlen+1) * DFI_RATE == BL). The host issues bursts at host width; the converter translates them to DW-width bursts of that geometry. Host burst sizing is the host's contract (the same spirit as the core's ragged-burst check). Verified for host widths in {64, 128, 256}. Full scope in docs/AXI_DRAM_GEARING_SCOPE.md.

27.2.4 Key parameters

Parameter	Default	Effect
HOST_AXI_DATA_WIDTH	128	Host AXI data width (free); == DW triggers the bypass
DW (derived)	DRAM_BEAT_WIDTH * DFI_RATE	Fixed core width
core geometry params	—	AXI_ID_WIDTH, AXI_ADDR_WIDTH, NUM_RANKS, NUM_BANKS, ROW_WIDTH, COL_WIDTH, DFI_RATE, DRAM_BEAT_WIDTH, BL, … mirror pumice_top

27.3 Narrow-Device (x16) Support — the Two Width Granularities

A distinction that is separate from both concepts above but which the DFI layer must respect: the **pumice DRAM beat** (DRAM_BEAT_WIDTH, one DFI phase's data) is not necessarily the **physical DRAM device word** (DRAM_DEVICE_WIDTH, the DQ width — e.g. 16 for an x16 device). When a beat is wider than the device (a 32-bit beat over an x16 device packs K = DRAM_BEAT_WIDTH / DRAM_DEVICE_WIDTH = 2 physical DDR words), three parts of the pipeline must reason in **device-word** units. Getting this wrong is invisible in a DFI-level behavioral sim but corrupts real silicon — the Nexys A7 x16 bring-up hit all three:

1. **Burst length (beats per command).** A JEDEC BL from MR0 (bl_o, BL4/BL8) is in physical device beats; the core scales it to pumice-beat units before feeding the burst-split and serializer, else an x16 BL4 is over-counted and the controller drives/captures an extra DFI cycle.
2. **Column address stride.** DRAM columns are device-word granular, so addr_mapper's BYTE_OFFSET_WIDTH must be log2(DRAM_DEVICE_WIDTH/8), not log2(DRAM_BEAT_WIDTH/8); using the beat width makes a split burst's chunk overwrite the previous chunk's tail (a ~50% read scramble on silicon).
3. **Read-data / valid alignment.** A PHY may present read data ahead of, or on a different phase than, rddata_valid. This is handled at runtime by the t_rddata_en knob and the DFI_PHASE CSR (below) rather than baked-in latencies.

Default DRAM_DEVICE_WIDTH = DRAM_BEAT_WIDTH (K=1) makes all of the above bit-identical to the wide-device behavior.

27.3.1 DFI_PHASE CSR (rd_phase / wr_phase)

dfi_cmd_formatter places the READ command on rd_phase and the WRITE command on wr_phase (defaults 0; all other commands on phase 0), runtime-settable via the DFI_PHASE CSR through pumice_dfi_layer's rd_phase_i / wr_phase_i. This matches a PHY that consumes a per-command rdphase/wrphase off the DFI bus. The LiteDRAM a7ddrphy instead takes the command on phase 0 and applies its rdphase internally, so on that PHY rd_phase stays 0 — the CSR exists for PHYs that do not.

27.4 Interface Summary

27.4.1 pumice_dfi_layer — controller side (ctl_clk)

Signal	Direction	Description
cmd_valid_i / cmd_ready_o / cmd_data_i[CMD_DW]	in/out/in	Command stream from the scheduler
wd_valid_i / wd_ready_o / wd_data_i[WD_DW]	in/out/in	Write data from the WR CAM drain ({last,strb,data})
rd_valid_o / rd_ready_i / rd_data_o[RD_DW]	out/in/out	Read data to the RD CAM ({last,resp,data})
init_start_i / init_complete_o	in/out	Init handshake, controller side

27.4.2 pumice_dfi_layer — PHY side (dfi_clk)

DFI command bus (dfi_address_o, dfi_bank_o, dfi_cas_n_o, dfi_ras_n_o, dfi_we_n_o, dfi_cs_n_o, dfi_odt_o), write data (dfi_wrdata_o, dfi_wrdata_en_o, dfi_wrdata_mask_o), read data (dfi_rddata_en_o, dfi_rddata_i, dfi_rddata_valid_i), and the DFI init handshake (dfi_init_start_o, dfi_init_complete_i), plus the runtime knobs above.

27.4.3 pumice_top_geared

Host AXI4 slave at HOST_AXI_DATA_WIDTH, PeakRDL cpuif passthrough (s_cpuif_*), init_done_o, and the DFI 2.1 pin bus straight through. Internally wires either directly (g_direct) or via the two converters (g_conv) to pumice_top at DW.

27.5 Verification Notes (cocotb test plan)

Scenario	What it proves
Command/wrdata/rddata cross the CDC bubble-free at DFI_RATE = 2	Single-CDC datapath, one FIFO word = one DFI cycle
Write burst: wr_fire → serializer launches dfi_wrdata at t_phy_wrlat	Write-latency alignment
Read burst: rd_fire → aligner strobes dfi_rddata_en at t_rddata_en	Read-capture alignment
rd_phase = 1: RD command lands on DFI phase 1	DFI_PHASE CSR routing
x16 device (K=2): BL4 uses one DFI cycle, columns stride by device words	Two-granularity handling
pumice_top_geared with HOST_AXI_DATA_WIDTH == DW: g_direct bypass	GEAR-1 bit-identical to bare pumice_top
Host 64 → DW 128 via axi4_dwidth_converter_wr/_rd: burst geometry preserved	Geared write/read path
Host 256 → DW 128: read data reassembled correctly at host width	Wide-host gearing
CSR cpuif and DFI pins identical whether geared or direct	Passthrough correctness

27.6 Open Questions / Future Work

- **Multi-command-per-cycle.** The command path places one issued command per DFI word today (multi-phase content passes through, but a single command occupies a word). Emitting multiple commands per DFI cycle is a scheduler-side feature; the bus widths are already in place.

- **`dfi_dram_clk_disable` / power-down.** `dfi_signal_pack` holds clk-disable at 0; wiring the power-state machine through the DFI layer is future work.
- **Higher gear ratios.** `DFI_RATE = 4` (and DDR3+ higher ratios) scale the DFI geometry linearly; validate the CDC FIFO depths and serializer/aligner timing when the family controller targets them.
- **Host width coverage.** Gearing is verified for host ∈ {64, 128, 256}; add 512 when a host SoC requires it. See `docs/AXI_DRAM_GEARING_SCOPE.md`.

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28 Transaction Queue (retired — replaced by the two command CAMs)

RETIRED — no standalone FUB

The unified transaction queue (`txn_queue_fub`) described by the SWAG was **never built as a separate block**. The scheduling-view function it was meant to provide is delivered directly by the two command CAMs:

- **Pending writes** live in `pumice_wr_data_cam` — `{bank, row, col}` key, free-running age, and the burst data in SRAM.
- **Pending reads** live in `pumice_rd_cmd_cam` — `{bank, row}` key, free-running age, per-entry return buffering.
- **Per-direction completion** is FIFO-based in the intakes (`pumice_wr_intake` / `pumice_rd_intake`): the write B FIFO and the read order + rd-data FIFOs.

This chapter is retained to document *where the queue's responsibilities went* and *why*. The SWAG's entry-format, four-state lifecycle, `cam_slot_idx` reverse pointer, and `row_hit_cached` broadcast are all superseded and should not be treated as the current architecture.

Status: Retired

28.1 What the CAMs do instead

The SWAG separated a narrow “scheduling view” (the queue) from a wide “metadata view” (the CAMs) to keep the scheduler’s per-cycle comparator fan-in small. The live design collapses the two: each CAM entry carries **both** the scheduling key and the metadata, and the scheduler reads the CAMs directly through purpose-built lookup ports rather than a wide parallel-entry snapshot.

28.1.1 Scheduler visibility (replaces the parallel `q_entries` bus)

Instead of a wide per-entry snapshot bus, each CAM exposes:

- **`N_SCHED_LU` lookup ports** (`N_LU = NUM_BANKS`, one per bank). The arbiter drives each port with `{bank j, that bank's open row}` and the CAM returns the oldest matching not-yet-committed entry — slot, column, id, and age. This is the row-hit query that used to be the `row_hit_cached` field.
- **An `oldest` port** — the oldest schedulable entry, used as the arbiter’s ACT fallback target.

The scheduler thus gets exactly the row-hit and fallback candidates it needs, computed inside the CAM combinationally, with no separate queue structure and no `row_hit_cached` coherency broadcast.

28.1.2 Entry lifecycle (replaces FREE/PENDING/ISSUED/COMPLETING)

The four-state queue lifecycle is replaced by per-entry flags in each CAM:

SWAG queue state	Live equivalent (write CAM)	Live equivalent (read CAM)
FREE	`!r_valid`	`!r_valid`
PENDING	`r_valid && r_fdone && !r_sched`	`r_valid && !r_issued`
ISSUED	`r_sched` (enqueued to drain FIFO)	`r_issued` (enqueued to issue FIFO)
COMPLETING	commit-drain streaming, then evict	data returning, then oldest-drain

The exclusion of already-committed/issued entries from the lookup/oldest ports (`r_sched` on the write side, `r_issued` on the read side) is what guarantees the arbiter never picks the same slot twice — the role the `pick_valid_mask` played in the SWAG.

28.1.3 Age (replaces the saturating age counter)

Both CAMs use a free-running `AGE_WIDTH` counter and wrap-safe relative age (`rel = age_ctr - entry_age`). Oldest = max `rel`. There is no saturation, no `AGE_MAX` parameter, and no `age_max_runtime` CSR clip; wrap-safety makes saturation unnecessary.

28.1.4 Backpressure (replaces `q_high_water` / `q_full`)

There is no queue occupancy watermark. Backpressure is structural: the write path stalls when the wr-data CAM has no free entry or its fill FIFO is full; the read path stalls when the rd-cmd CAM is full or the intake's order FIFO is full. The AXI channels back-pressure through the intake FIFOs.

28.2 Why the split disappeared

The queue existed to make a *single* wide scheduling snapshot cheap. Once the scheduler was rearchitected to query per-bank (one lookup per open row) rather than scan every entry, the wide snapshot was no longer needed — the CAM's own associative match is the scheduling view. Keeping a second copy of `{rank, bank, row}` in a queue, plus the `cam_slot_idx` reverse pointer and the `row_hit_cached` broadcast coherency machinery, would have been pure overhead.

See ch02/07 for the arbiter that consumes the CAM lookup ports and ch02/04 / ch02/05 for the CAM internals.

---

29 Bank Timer (`bank_timer` + `pumice_bank_timers`)

Module: `bank_timer.sv` (per-bank) / `pumice_bank_timers.sv` (aggregator) **Location:** `rtl/fub/` **Category:** FUB **Parent macro:** `pumice_mem_cmd_scheduler` **Status:** Implemented (FSM-free countdown-timer model)

> **Replaces the retired bank machine.** The original architecture placed a per-(rank, bank) seven-state FSM (`bank_machine_fub`) between the scheduler and the DFI encoder. That block **no longer exists**. Per-bank JEDEC timing is now enforced by `bank_timer.sv` — a set of preset/decrement countdown timers with a trivial row-open register and **no state machine at all**. `pumice_bank_timers.sv` stamps one `bank_timer` per (rank, bank) and fans the scheduler's command-event strobes to the addressed instance.

The retirement was deliberate: the old FSM double-registered readiness (state register + next-state combinational + accepts flop), so the scheduler saw readiness two cycles stale. That staleness was the root cause of the refresh-vs-ACT and column-vs-PRE hazards observed during bring-up. The FSM-free model collapses the readiness path to a **single register stage** (the timers themselves), so the arbiter sees the just-issued command's effect exactly one cycle later.

29.1 Purpose

`bank_timer` tracks one DRAM bank's JEDEC "safe" timing. For each JEDEC per-bank constraint it holds a down-counter that is preset on the triggering command and free-runs toward zero (saturating at 0). A constraint is "safe" when its counter reads 0. The per-command readiness outputs (`safe_act`, `safe_rd`, `safe_wr`, `safe_pre`) are a **combinational AND** of the relevant timers plus a one-bit row-open flag — no FSM, no multi-stage lag.

The scheduler (`pumice_cmd_arbiter`, the single command issuer in the design) drives one `set_*` strobe per issued command. Each timer reloads its config value on its trigger; the `safe_*` outputs therefore reflect the just-issued command one cycle later. The arbiter ANDs these per-bank `safe_*` signals with the channel-wide turnaround windows from `global_timers` (§2.10) to decide which command may fire.

`pumice_bank_timers` is a thin aggregator: it instantiates `NUM_RANKS × NUM_BANKS` copies of `bank_timer` and routes the arbiter's single command-event bus (`evt_*` + `evt_rank_i` / `evt_bank_i`) to the one addressed instance. All other instances see their `set_*` strobes deasserted that cycle and simply continue counting down.

29.2 Synthesis Parameters

29.2.1 `bank_timer`

Parameter	Default	Effect on this FUB
`ROW_WIDTH`	14	Width of `open_row_o` register and the `row_i` input
`TW`	8	Timer width (bits) for all five countdown timers

29.2.2 `pumice_bank_timers`

Parameter	Default	Effect
`NUM_RANK`	1	Rank dimension of the instance array; width of

Parameter	Default	Effect
S		evt_rank_i sel
NUM_BANKS	8	Bank dimension of the instance array; width of evt_bank_i sel
ROW_WIDTH	14	Passed to each bank_timer
RKW	clog2(NUM_RANKS)	Rank-select width (min 1)
BKW	clog2(NUM_BANKS)	Bank-select width

There is **no per-instance rank/bank identifier parameter**. Identity comes from the genvar position in pumice_bank_timers' nested generate loop; the arbiter selects an instance by comparing evt_rank_i / evt_bank_i against that position (w_sel). All five timers share the same TW-bit width; the JEDEC cycle counts are supplied at runtime on the t_*_i ports (CSR-backed), not fixed as parameters.

29.3 Instantiation Pattern

From pumice_bank_timers.sv — the nested generate that stamps the array:

```
for (genvar k = 0; k < NUM_RANKS; k++) begin : g_rank
    for (genvar b = 0; b < NUM_BANKS; b++) begin : g_bank
        // route the scheduler's command event to THIS bank only
        logic w_sel;
        assign w_sel = (evt_rank_i == RKW'(k)) && (evt_bank_i ==
BKW'(b));

        bank_timer #(.ROW_WIDTH(ROW_WIDTH)) u_bt (
            .clk(aclk), .rst_n(aresetn),
            .t_rcd_i(t_rcd_i), .t_rp_i(t_rp_i), .t_ras_i(t_ras_i),
            .t_rc_i(t_rc_i), .t_wr_i(t_wr_i), .t_rtp_i(t_rtp_i),
            .set_act_i(evt_act_i && w_sel),
            .set_rd_i (evt_rd_i  && w_sel),
            .set_wr_i (evt_wr_i  && w_sel),
            .set_pre_i(evt_pre_i && w_sel),
            .set_ap_i (evt_ap_i),
            .row_i(evt_row_i),
            .safe_act_o (bank_act_ready_o [k][b]),
            .safe_rd_o  (bank_rdwr_ready_o[k][b]),
            .safe_wr_o  (/* == safe_rd */),
            .safe_pre_o (bank_pre_ready_o [k][b]),
            .row_valid_o(bank_row_active_o[k][b]),
```

```
            .open_row_o (bank_open_row_o  [k][b]),
            .state_o    (bank_state_o     [k][b]),
            /* obs_* ... */
        );
    end
end
```

The per-(rank, bank) readiness outputs are 2-D-packed arrays (`logic [NUM_RANKS-1:0] [NUM_BANKS-1:0]`) consumed directly by `pumice_cmd_arbiter`. There is no aggregation logic beyond the packing — pure structural fan-out. Note that `safe_wr_o` is left unconnected in the aggregator because inside `bank_timer` it is identical to `safe_rd_o` (see the RTL: `assign safe_wr_o = safe_rd_o`); the arbiter uses the single `bank_rdwr_ready_o` bit for both RD and WR column commands.

29.4 The Timer Pool (per bank)

`bank_timer` holds five countdown registers (`r_rcd`, `r_ras`, `r_rc`, `r_rp`, `r_preblk`), each `TW` bits, plus a one-bit `r_row_valid`, the `r_open_row` register, and a single `r_ap_pending` auto-precharge bit. Every timer is a down-counter: reload on its trigger, else decrement while non-zero (saturate at 0).

Timer	JEDEC constraint	Reload trigger	Reload value	Gates
`r_rcd`	tRCD (ACT → RD/WR)	`set_act_i`	`t_rcd_i`	`safe_rd/wr`
`r_ras`	tRAS (ACT → PRE min)	`set_act_i`	`t_ras_i`	`safe_pre`
`r_rc`	tRC (ACT → ACT same bank)	`set_act_i`	`t_rc_i`	`safe_act`
`r_rp`	tRP (PRE → ACT)	`set_pre_i` OR auto-PRE fire	`t_rp_i`	`safe_act`
`r_preblk`	tRTP (RD) / tWR (WR)	`set_rd_i` / `set_wr_i`	`t_rtp_i` / `t_wr_i`	`safe_pre`

The `t_*_i` reload values are supplied by the parent (`pumice_bank_timers` forwards its own 8-bit `t_*_i` inputs, which are CSR-backed JEDEC timing fields). They are sampled live on each reload — the timer does not stage them.

Note: the retired FSM's separate `t_cl_cnt` / `t_cwl_cnt` (column-busy windows) are **gone**. Column-to-column pacing (tCCD) and the read-drives-DQ / write-recovery windows are enforced

globally in `global_timers` (tCCD/tWTR/tRTW) and, for the write-recovery-before-precharge case, folded into the `r_preblk` timer here. There is no per-bank “RD_BUSY / WR_BUSY” occupancy state anymore.

29.5 Row-Open Register and Auto-Precharge (no FSM)

Row state is a two-element register — `r_row_valid` + `r_open_row` — plus one `r_ap_pending` bit. There is no enumerated state; the priority in the RTL is **ACT** > **explicit PRE** > **auto-PRE fire** > **column**:

```
if (set_act_i) begin
    r_row_valid  <= 1'b1;
    r_open_row   <= row_i;
    r_ap_pending <= 1'b0;
end else if (set_pre_i) begin
    r_row_valid  <= 1'b0;
    r_ap_pending <= 1'b0;
end else if (w_ap_fire) begin
    r_row_valid  <= 1'b0;
    r_ap_pending <= 1'b0;
end else if (set_rd_i || set_wr_i) begin
    r_ap_pending <= set_ap_i;   // latch the RDA/WRA qualifier
end
```

Auto-precharge (RDA / WRA) is a single bit, not a state. `set_ap_i` qualifies a RD/WR issue; if set, `r_ap_pending` latches. The auto-precharge “fires” combinationally once the read/write recovery window and tRAS have both elapsed:

```
assign w_ap_fire = r_ap_pending && (r_preblk == '0) && (r_ras == '0);
```

When `w_ap_fire` asserts, the row closes internally (`r_row_valid <= 0`) and `r_rp` reloads with `t_rp_i` — exactly as an explicit PRE would. No PRE command is consumed from the scheduler and no extra state is entered. This is the same degeneracy the old FSM’s “auto-pre pending” trick achieved, but here it is one flop and one AND gate.

Open-page behavior. A bare RD/WR (with `set_ap_i` low) only reloads `r_preblk` and leaves `r_row_valid` set, so the row stays open. Successive column commands to the open row stream at the channel-wide tCCD rate enforced by `global_timers`; the bank timer imposes no additional per-column pacing.

29.6 Combinational `safe_*` Outputs

The readiness outputs are pure combinational functions of the timer/flag registers — one register stage behind the arbiter’s issue decision:

```
// ACT: bank precharged (row closed), tRP since PRE, tRC since last
ACT.
assign safe_act_o = !r_row_valid && (r_rp == '0) && (r_rc == '0);
// RD/WR: row open, tRCD met, not auto-precharging.
assign safe_rd_o  = r_row_valid && (r_rcd == '0) && !r_ap_pending;
assign safe_wr_o  = safe_rd_o;
// PRE: row open, tRAS + tRTP/tWR met, not auto-precharging.
assign safe_pre_o = r_row_valid && (r_ras == '0) && (r_preblk == '0)
&& !r_ap_pending;
```

The `!r_ap_pending` term on `safe_rd_o` / `safe_pre_o` is what makes an auto-precharging bank refuse further column commands and a redundant explicit PRE while its internal auto-PRE is still pending — the arbiter naturally skips the bank until the row closes and `safe_act_o` re-asserts.

Because these are combinational off single-stage registers, the arbiter's per-cycle pick sees the effect of the command it issued on cycle *T* on cycle *T+1*. There is no second flop between the timer and the arbiter, so a command issued this cycle cannot be double-issued next cycle — the class of hazard that motivated retiring the FSM.

29.7 Derived State (observability only)

A `bank_state_e` (`state_o`) is produced **purely for observability** — no downstream logic reads it. It is decoded combinationally from the row-valid flag and the timers:

```
if (r_row_valid) state_o = (r_rcd != '0) ? BANK_ACTIVATING :
BANK_ACTIVE;
else             state_o = (r_rp  != '0) ? BANK_PRECHARGING :
BANK_IDLE;
```

This gives waveform viewers and CSR telemetry the familiar IDLE / ACTIVATING / ACTIVE / PRECHARGING names without the design actually implementing them as a sequencer. It is a decode of the timers, not the source of truth.

The remaining observability outputs are single-bit non-zero flags on the timers and the auto-pre bit: `obs_rcd_nz_o`, `obs_preblk_nz_o`, `obs_ras_nz_o`, `obs_ap_pending_o` (per bank, fanned out to `obs_*` arrays in the aggregator).

29.8 Interface (`pumice_bank_timers`)

29.8.1 Clocks / Reset

Signal	Direction	Description
`aclk`	input	Controller clock

Signal	Direction	Description
aresetn	input	Active-low async reset

29.8.2 Timing config (CSR-backed, per-bank JEDEC cycle counts)

Signal	Width	Constraint
t_rcd_i	8	tRCD
t_rp_i	8	tRP
t_ras_i	8	tRAS
t_rc_i	8	tRC
t_wr_i	8	tWR (WR cmd → earliest PRE, incl WL+BL/2)
t_rtp_i	8	tRTP (RD cmd → earliest PRE)

29.8.3 Command-event strobes from the arbiter

Signal	Width	Description
evt_act_i	1	ACT issued this cycle
evt_rd_i	1	RD issued this cycle
evt_wr_i	1	WR issued this cycle
evt_pre_i	1	PRE issued this cycle
evt_ap_i	1	Auto-precharge qualifier (with RD/WR)
evt_rank_i	RKW	Rank of the issued command (selects instance)
evt_bank_i	BKW	Bank of the issued command (selects instance)
evt_row_i	ROW_WIDTH	Row operand (latched on ACT)

29.8.4 Per-bank readiness to the arbiter (combinational, single-stage)

Signal	Width	Description
bank_act_ready_o[NR][NB]	1 each	safe_act per bank
bank_rdwr_ready_o[NR][NB]	1 each	safe_rd (== safe_wr) per bank

Signal	Width	Description
bank_pre_ready_o[NR][NB]	1 each	safe_pre per bank
bank_row_active_o[NR][NB]	1 each	row_valid per bank
bank_open_row_o[NR][NB]	ROW_WIDTH each	Open row per bank
bank_state_o[NR][NB]	bank_state_e	Decoded state (observability)

29.8.5 Observability

obs_act_cnt_nz_o, obs_preblk_nz_o, obs_ras_nz_o, obs_ap_pending_o — each a [NR][NB] bit array, mirroring the per-bank obs_* outputs.

29.9 Timing Budget

The path from a command event to updated readiness is single-cycle:

```
cycle T:   arbiter asserts evt_act_i with evt_rank_i=r, evt_bank_i=b
cycle T:   w_sel picks instance (r,b); r_rcd/r_ras/r_rc load,
r_row_valid <- 1
cycle T+1: safe_act_o[r][b] = 0 (r_rc/row_valid now block ACT),
           safe_rd_o[r][b]  gated by r_rcd countdown
```

safe_*_o is combinational from the registers, so the arbiter consumes the post-event readiness on the very next cycle — no race allowing a second issue to the same bank while its constraint is still counting. This one-cycle discipline is the whole point of the FSM-free rework: readiness is exactly one register stage deep, versus the retired FSM's two (state + accepts).

29.10 Verification Notes (cocotb test plan)

Scenario	What it proves
ACT → wait tRCD → RD; RD blocked until r_rcd == 0	tRCD gate on safe_rd
ACT → RDA; row auto-closes once r_preblk and r_ras clear	w_ap_fire closes row, reloads r_rp; no scheduler PRE
ACT → WRA; r_preblk loads tWR; auto-PRE waits for tWR AND tRAS	Write-recovery folded into r_preblk
Bare RD (set_ap=0) keeps row open; second	Open-page: r_row_valid stays set

Scenario	What it proves
RD to same row allowed	
ACT → PRE (explicit) → ACT; second ACT blocked until `r_rp == 0`	tRP gate on `safe_act`
Two ACTs same bank; second blocked until `r_rc == 0`	tRC gate on `safe_act`
`safe_rd` deasserts while `r_ap_pending` set (mid auto-PRE)	Auto-precharging bank refuses new column commands
Aggregator: `evt_bank_i=b` only reloads instance b; others count down	`w_sel` routing correctness
`state_o` decode matches timers (ACTIVATING while `r_rcd != 0`)	Observability decode
Reset during any countdown → all timers and row flag clear	Reset behavior

29.11 Open Questions / Future Work

- **Timer width.** `TW = 8` covers the DDR2/LPDDR2 board targets (tRC ≈ tens of cycles). A DDR3/DDR4 family part at a higher CK:MC ratio could need wider timers; `TW` is already a parameter, so this is a re-elaboration, not a rework.
- **Per-bank vs shared tRC.** tRC is enforced per bank here. If a future family needs same-bank-group tRC differentiation (DDR4 bank groups), that logic would land in `global_timers`, not here — `bank_timer` stays bank-local by design.
- **Auto-PRE vs explicit PRE priority.** The RTL priority is ACT > explicit PRE > auto-PRE. If the arbiter ever issued an explicit PRE to a bank with a pending auto-PRE, the explicit PRE wins and `r_ap_pending` clears. The arbiter does not do this today (it treats an auto-precharging bank as un-schedulable via `safe_*`), but the RTL is safe if it ever did.

30 ODT Control (absorbed — no standalone block)

ABSORBED — No standalone FUB

There is **no `odt_ctrl` module** in the live RTL. The dedicated ODT-control FUB the original architecture planned was never built as a separate block. ODT responsibility is split between two modules that already exist:

- **`dfi_cmd_formatter.sv`** (see the command-formatter chapter) owns the `dfi_odt_o` output. ODT follows deterministically from the issued command, so it belongs on the same output stage as the ras/cas/we truth table rather than in a separate FSM.
- **`mode_register.sv`** decodes the DDR2 ODT rule bits from the mode registers into an `odt_o` field (see the mode-register chapter).

This chapter is retained only to explain where ODT lives and to record the current (minimal) state honestly.

Status: Absorbed / minimal — ODT is decoded but not yet actively driven

---

30.1 Where ODT Lives Today

Concern	Live location	State
`dfi_odt_o` bus (per-rank, per-phase)	`dfi_cmd_formatter.sv` output	Driven to 0 in v1 (decode leaves `w_p0_odt` at the NOP default for every op)
`dfi_odt_o` reset value	`dfi_signal_pack.sv`	0 (ODT off during reset / before init)
DDR2 ODT-rule decode from MRs	`mode_register.sv` `odt_o[1:0]`	Decoded from MR1 (`{w_mr1[6], w_mr1[2]}`); informational, not wired to the pin driver
LPDDR2 ODT	`mode_register.sv`	`odt_o = 0` (LPDDR2 typically point-to-point)

`mode_register` exposes `odt_o` as one of its live decoded outputs, but the RTL comment marks it "informational; not used" — the value is computed from the DDR2 MR1 ODT bits but is not currently consumed to time a termination window. `dfi_cmd_formatter` drives `dfi_odt_o` from

`w_p0_odt`, which the DDR2 decode never raises above its NOP default, so on the DDR2/LPDDR2 board targets (both effectively single-rank point-to-point) ODT stays off.

30.2 Why ODT Is Off on the Board Target

ODT is a JEDEC impedance-matching feature: a DDR2 device contains internal termination resistors that the controller selectively enables to terminate the DQ/DQS bus. The rule is asymmetric and only matters with more than one device on the bus:

- **During a read**, the accessed rank drives DQ; *other* ranks should terminate.
- **During a write**, the controller drives DQ; the *target* rank should terminate.

For a single-rank point-to-point system there is nothing else on the bus, so ODT serves no purpose and 0 is correct. The pumice board bring-up (Nexys A7, single x16 DDR2 device) is exactly this case — which is why the v1 controller ships with ODT decoded-but-not-driven and passes on silicon.

30.3 What a Full ODT Implementation Would Add

If a multi-rank DDR2 target is brought up, the timed ODT window would be added inside `dfi_cmd_formatter`'s output stage (not as a new block), consuming `mode_register.odt_o` plus CL/CWL and burst length to compute per-rank turn-on/ turn-off. The JEDEC cross-termination pattern (ODT-high on the non-accessed rank during a read, on the accessed rank during a write) would drive `w_p0_odt` per target rank instead of the current constant 0. Because the rule follows deterministically from the issued op and the MR-decoded termination value, it stays in the formatter's truth table — the reason the standalone block was never needed.

30.4 Verification Notes (cocotb test plan)

Scenario	What it proves
Single-rank DDR2: `dfi_odt_o` stays 0 for all traffic	Current behavior (point-to-point board target)
Reset: `dfi_signal_pack` drives `dfi_odt_o = 0`	Reset-safe ODT-off
`mode_register` MR1 ODT bits decode into `odt_o`	Decode present for future use
LPDDR2: `mode_register.odt_o = 0`	No ODT on LPDDR2 point-to-point

30.5 Open Questions / Future Work

- **Timed ODT window (multi-rank DDR2).** Not implemented. Would live in `dfi_cmd_formatter`'s output stage, driven by `mode_register.odt_o` + CL/CWL + BL. Add when a multi-rank DDR2 board is targeted.
- **Dynamic ODT (DDR3+).** DDR3 introduced Rtt_Wr vs Rtt_Nom; DDR2/LPDDR2 have only Rtt_Nom. A DDR3+ family controller would extend the decode.
- **ODT during refresh/precharge.** Some boards want ODT held during refresh to avoid a floating bus. Revisit only if signal-integrity characterization flags it on a multi-rank target.

---

31 AXI4 Slave Protocol

This chapter is the **wire-level contract** of the controller's host AXI4 slave face – what's supported, what's relaxed, what's omitted, and the timing / ordering / backpressure semantics the integrator can rely on.

The interface is implemented by `pumice_axi4_ifc` (section 2.1): a pair of AMBA burst splitters feed the dumb `pumice_wr_intake` / `pumice_rd_intake` blocks (each wrapping `axi4_slave_wr` / `axi4_slave_rd`), which push into the write / read CAMs.

31.1 Compliance Profile

Aspect	Support level
AXI4 (ARM IHI 0022) signal set	Full AW/W/B/AR/R
AXI4-Lite	Not supported (use the register cpuif for control)
AXI4-Stream	Not applicable
AXI5	Not supported

31.2 Data Width

The core's AXI data width is the **DFI word**: `DW = DRAM_BEAT_WIDTH * DFI_RATE` (128 by default). One AXI beat == one DFI word == `DFI_RATE` DRAM beats. If the SoC master runs a different width, wrap the core in `pumice_top_geared`, which inserts the formally-verified `axi4_dwidth_converter_wr/_rd` between a host-width AXI slave and the fixed-`DW` core (`HOST_AXI_DATA_WIDTH == DW` is a bit-identical generate bypass). See `docs/AXI_DRAM_GEARING_SCOPE.md`.

Hard width rule (compile-enforced). `HOST_AXI_DATA_WIDTH : DW` **MUST** be an exact power-of-two ratio (`AXI:DFI = G:1` or `1:G`, `G ∈ {1,2,4,8,…}`). An `initial assert … $fatal` in `pumice_top_geared` **fails elaboration / Vivado synthesis** on any other pairing — a bad width choice is a compile error, not a silent broken build. This mirrors LiteDRAM exactly (AXI frontend 1:1 with its native port; all width change via a power-of-two stride converter). Because the DFI word is the atomic memory-side transfer, an AXI beat must be a whole power-of-two number of DFI words. `HOST_AXI_DATA_WIDTH` and `DW` are the **only** compile-time width parameters; **gear ratio and burst length are runtime CSRs** (`gear_ratio`, `bl`), built for max and selected at runtime — a wrong value is bad config programming, never a parameter/synthesis mismatch.

31.3 Burst Splitting

Each host burst is split at DRAM-burst-byte boundaries by `axi_master_wr_splitter` / `axi_master_rd_splitter` (`ALIGN_MASK = BL*(DRAM_BEAT_WIDTH/8) - 1`), so every command handed to an intake is exactly one DRAM burst. This is how an arbitrary AXI `awlen`/`arlen` maps onto the fixed DRAM burst length.

31.4 Supported Features

- INCR burst type (mandatory)
- Arbitrary INCR burst length (split into DRAM-burst commands as above)
- ID-carried requests; per-ID in-order completion (AXI4 mandate)
- Cross-ID reordering at the DRAM layer, with AXI response ordering honored at the response side (per HAS 3.1)
- `awqos`/`arqos`/`awregion`/`arregion` carried through the splitters (observed, not yet used for scheduler priority)
- Single-bit `awuser`/`aruser`/`wuser` ports carried through the front-end (echoed on B/R `buser`/`ruser`)

31.5 Unsupported / Ignored

- AXI4 exclusive accesses (`awlock`, `arlock` observed but ignored; treated as normal accesses)
- AXI4 cache-coherent behavior (`awcache`/`arcache` observed, no behavior)

- FIXED / WRAP burst types are not a design target; the front-end is built for INCR traffic

31.6 Backpressure Semantics

The slave asserts backpressure via the standard AXI handshake. Backpressure originates in the intake FIFOs and CAM fill:

Channel	Backpressure reason
AW	AW-meta FIFO full in `pumice_wr_intake`, or `pumice_wr_data_cam` has no free slot
W	Write-data FIFO / write-data SRAM full in the write path
AR	`pumice_rd_cmd_cam` has no free slot
B	Standard `bready` handshake
R	Standard `rready` handshake; the drain mover in `pumice_rd_cmd_cam` stalls until `rready`

Total in-flight depth per direction is set by `NUM_ENTRIES` (CAM depth) with `N_SRAM_SLOTS` burst-data SRAM slots per CAM.

31.7 Response Ordering

1. **Per-ID in-order** – always enforced. Two accesses with the same ID complete in issue order.
2. **Cross-ID** – reordering is allowed at the DRAM layer (the scheduler picks by bank/row/age), but the CAMs drain and the intakes emit responses so that the AXI-visible ordering rules hold.

31.8 Read-Your-Write Forwarding (snarf)

`pumice_rd_intake` probes `pumice_wr_data_cam` before a read is scheduled. On a hit against an unscheduled write with the same id and same burst length, the read is streamed straight from the write CAM's SRAM (the **snarf mover**) with no DRAM round-trip. On a miss the read goes through the normal DRAM path.

31.9 Timing Contract

Path	Behavior
`awvalid` -> `awready`	<= 1 cycle when the AW-meta FIFO has

Path	Behavior
	space
awready accept -> CAM push	1 cycle
CAM push -> DRAM issue	scheduler / refresh dependent (typically 0 to ~256 cycles)
Last W beat -> B response	commit + write-recovery window + scheduler backlog
arvalid -> first R beat	DRAM access latency (typically 30-100 ns) + bus rounding, or ~0 on a snarf hit
Successive R beats (same burst)	1 per aclk cycle (sustained streaming)

The slave does not embed an AXI register slice. If a target needs one for closure, add an external axi_register_slice from the AMBA library, or use pumice_top_geared (whose dwidth converters register the boundary).

31.10 Error Responses

Condition	Response	Notes
Traffic before init_done_o	held / SLVERR	Poll STATUS.init_done (or the init_done_o pin) before issuing traffic
DFI read return flagged bad by the aligner	SLVERR	Propagated on the affected R beats

DECERR is never returned – the address space is monolithic.

31.11 Open Questions / Future Work

- **QoS-driven priority.** awqos/arqos are carried but not yet consumed by pumice_cmd_arbiter. A future revision would fold them into the pick.
- **Wider USER signals.** The front-end carries single-bit user today; wider sideband would need a USER_WIDTH parameter threaded through the intakes and CAMs.
- **AXI register-slice option.** Currently external / via the geared wrapper; a build-time embed could be added for very-high-frequency targets.

32 DFI v2.1 Master Protocol

This chapter is the wire-level contract of the controller's DFI 2.1 master face – what's driven, what's tied, what's sampled, what's deferred.

The interface is implemented by `pumice_dfi_layer` (section 2.4): `pumice_dfi_cmd_path` (with `dfi_cmd_formatter` + `dfi_signal_pack`) drives the command bus, `pumice_dfi_wr_serializer` drives write data, `pumice_dfi_rd_aligner` captures read data, and `pumice_dfi_cdc` is the single clock crossing to `aclk`.

32.1 DFI Spec Reference

Canonical: the DFI 2.1 specification (Denali cleartext copy at `/home/seang/github/cold_storage/MemorySpecs/`). The controller covers the subset of DFI 2.1 sub-interfaces needed for DDR2/LPDDR2 operation; higher-version sub-interfaces are not driven.

32.2 Sub-Interface Support Summary

Sub-Interface	Direction	Supported?	Notes
Command	Controller -> PHY	Yes	Per-phase; command placed on `wr_phase`/`rd_phase`
Write-Data	Controller -> PHY	Yes	`dfi_wrdata` presented at `t_phy_wrlat` after WR fire
Read-Data	PHY -> Controller	Yes	`dfi_rddata_en` at `t_rddata_en` after RD fire; captured on `dfi_rddata_valid`
Status (init)	Both	Yes	`dfi_init_start` / `dfi_init_complete` only
Update	Both	No	`ctrlupd_*` / `phyupd_*` not driven in v1
Training	Both	No	PHY-side (`a7ddrphy` self-trains at startup)
Frequency Change	Both	No	No use case

Sub-Interface	Direction	Supported?	Notes
Low-Power	Controller -> PHY	No	No PHY-side low-power coordination
Error / CRC	-	No	Not used in DDR2/LPDDR2 here

32.3 Clock and CDC

The DFI pin bus is driven in the **`dfi_clk`** domain. Unlike the classic "DFI is on the controller clock" model, this controller keeps the whole DFI datapath on a dedicated PHY clock and places the **single** controller-to-PHY CDC inside `pumice_dfi_layer` (`pumice_dfi_cdc`, async gaxi FIFOs only). The command, write-data, and read-data streams plus the init handshake cross there; the internal datapath unit is the whole DFI word, so one FIFO word is one `dfi_clk` cycle and the path is bubble-free at rate. `a7ddrphy` handles any further gearing to the DRAM clock.

32.4 Command Sub-Interface

32.4.1 Phase Topology

The command bus is `per-phase x DFI_RATE` wide. `dfi_address_o` is `ROW_WIDTH * DFI_RATE`, `dfi_bank_o` is `$clog2(NUM_BANKS) * DFI_RATE`, and each control strobe (`dfi_cas_n_o` / `dfi_ras_n_o` / `dfi_we_n_o`) is `1 * DFI_RATE`. `pumice_dfi_cmd_path` places the active JEDEC command on the `wr_phase` / `rd_phase` slot (programmed by the `DFI_PHASE` CSR) and drives NOP-equivalent idle on the other phases.

32.4.2 Chip-Select and ODT

`dfi_cs_n_o` and `dfi_odt_o` are each `NUM_RANKS * DFI_RATE` wide (rank inner, phase outer, as the `DFI_CS_BUS_W = NUM_RANKS * DFI_RATE` packing implies). The formatter drives the addressed rank's CS_n active for a single-rank command; init-time REF broadcasts. ODT is driven inside `dfi_cmd_formatter` / `mode_register` – there is no standalone `odt_ctrl` block.

32.4.3 DDR2 vs LPDDR2 Encoding

For `memtype_i = DDR2`, commands use the classic `ras/cas/we/cs` strobe encoding. For `memtype_i = LPDDR2`, `dfi_cmd_formatter` takes the memtype branch and emits bit-exact JESD209-2F CA-bus commands (Table 60): the 10-bit CA bus over 2 edges packed as a flat 20-bit word on `dfi_address`, with `ras/cas/we` idle. See `rtl/LPDDR2_CA_ENCODING.md`.

32.5 Write-Data Sub-Interface

`pumice_dfi_wr_serializer` presents `dfi_wrdata` / `dfi_wrdata_en` / `dfi_wrdata_mask` exactly `t_phy_wrlat` `dfi_clk` cycles after the `wr_fire` strobe from the command path. The

write burst arrives from the write CAM as whole DFI words `{last, strb, data}`; the serializer splits into the per-phase data / enable / mask.

`dfi_wrdata_en_o` is `DFI_RATE` wide (one enable per phase). A mask bit of 1 means "do not write this byte" per DFI convention; the mask payload carried from the CAM is aligned to that convention.

32.6 Read-Data Sub-Interface

`pumice_dfi_rd_aligner` drives `dfi_rddata_en_o` (`DFI_RATE` wide) exactly `t_rddata_en` `dfi_clk` cycles after the `rd_fire` strobe, then captures `dfi_rddata_i` when `dfi_rddata_valid_i` asserts, packing whole DFI words (`{last, resp, data}`) into the read FIFO for the crossing back to `aclk`.

32.6.1 Read-Data Error Handling

`dfi_rddata` carries no error bit in DFI 2.1. A missing / malformed read is surfaced by the aligner's `resp` field and propagated as `SLVERR` on the affected AXI R beats. On-silicon this path was hardened during board bring-up: `t_rddata_en` and the DFI read-data delay are runtime CSRs, and `rd_phase = 0` is correct for `a7ddrphy` (which handles read gearing internally).

32.7 Status / Init Sub-Interface

`dfi_init_start_o` and `dfi_init_complete_i` are the only status signals. The `init_sequencer` (in `pumice_mem_cmd_scheduler`) asserts `dfi_init_start` and walks the JEDEC MRS init, waiting on `dfi_init_complete` from the PHY plus the programmed init-timing CSRs. These cross the CDC in `pumice_dfi_cdc`.

32.8 Pin Tie-Offs

Pin	Tie-off (when applicable)
`dfi_ras_n_o` / `dfi_cas_n_o` / `dfi_we_n_o`	Idle in LPDDR2 (command rides the CA bus on `dfi_address`)
Update / frequency / low-power buses	Not present / not driven

32.9 Open Questions / Future Work

- **DFI error / update.** `ctrlupd_*` / `phyupd_*` and a `dfi_error` input are not wired in v1; a later revision could add quiet-point update handshaking and route read errors to an IRQ.

- **Per-rank CS width form.** The controller produces the flat `NUM_RANKS * DFI_RATE` vector form; a PHY expecting a different packing needs a thin wrapper at the boundary.
- **DFI 3+ migration.** When the DDR3-LPDDR3 family controller arrives, the sub-interface set grows substantially. Swap `pumice_dfi_layer` and add a dedicated DFI section in that MAS; keep this one DDR2/LPDDR2-scoped.

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33 Register Access Interface (PeakRDL cpuif)

Per HAS §2.4 §3 for the per-signal port list and HAS §4.3 for the architectural view. This chapter is the **wire-level contract** for the controller's register access interface.

The register block is a PeakRDL-generated module (`pumice_csr`, from `rtl/macro/pumice_csr.rdl`) instantiated directly in `pumice_top`. It exposes a **PeakRDL passthrough CPU interface** (`s_cpuif_*`), not a hand-written APB slave. Any APB (or AXI-Lite) bridge is **external/optional** — the SoC attaches whatever register transport it uses and drives the cpuif. All register storage lives in the `aclk` (MC) domain; there is no separate `apb_pclk` domain and no on-die CDC inside the register block.

33.1 The cpuif Contract

`pumice_top` presents the PeakRDL passthrough interface on `aclk`/`aresetn`:

Direction	Signal	Meaning
in	`s_cpuif_req`	Request strobe
in	`s_cpuif_req_is_wr`	1 = write, 0 = read
in	`s_cpuif_addr[11:0]`	Byte address (12-bit, 4 KB region)
in	`s_cpuif_wr_data[31:0]`	Write data

Direction	Signal	Meaning
in	`s_cpuif_wr_biten[31:0]`	Per-bit write-enable (bit strobe)
out	`s_cpuif_req_stall_wr`	Back-pressure a write request
out	`s_cpuif_req_stall_rd`	Back-pressure a read request
out	`s_cpuif_rd_ack`	Read response valid
out	`s_cpuif_rd_err`	Read decode error (unmapped address)
out	`s_cpuif_rd_data[31:0]`	Read data
out	`s_cpuif_wr_ack`	Write response valid
out	`s_cpuif_wr_err`	Write decode error (unmapped address)

The interface is single-request/single-response with a stall handshake. Registers are 32-bit, naturally aligned on 4-byte boundaries. Sub-word writes use `s_cpuif_wr_biten` (per-bit granularity); a full-word write drives all 32 biten bits.

33.2 Attaching an APB / AXI-Lite Bridge

Because the interface is a generic passthrough cpuif, the register transport is an integration choice made **outside** this controller:

- An APB3/APB4 slave bridge translates `psel`/`penable`/`pwrite`/`paddr`/`pwdata`/`prdata`/`pslverr` to the cpuif and back. `pslverr` maps from `s_cpuif_rd_err`/`s_cpuif_wr_err`.
- An AXI-Lite slave bridge is equally valid; the cpuif does not assume APB.
- If the register bus is in a different clock domain than `aclk`, the CDC lives in that external bridge — not in `pumice_csr`.

No `pumice_top` parameter selects the transport; the bridge is a separate module in the SoC's register fabric.

33.3 Address Space

The register block decodes a 4 KB region (12-bit address). Layout, matching `rtl/macro/pumice_csr.rdl`:

Address range	Region
0x000 – 0x008	Control / Status / Status-history

Address range	Region
0x010 – 0x01C	Timing parameters (RC/RCD/RP/RAS, RFC/REFI, RRD/FAW/WTR/CCD, CL/CWL/WR)
0x020 – 0x02C	Mode Register values (MR0..MR3)
0x030 – 0x038	LPDDR2-specific (PASR bank/segment masks, temp derate)
0x040 – 0x05C	Scheduler / Page / Refresh / Addr-map / Init tuning + init timing
0x054, 0x060, 0x064	tRTP/tRTW, DFI command phase, PHY/DFI data timing
0x080 – 0x09C	Per-bank row-hit observation (NUM_BANKS = 8)
0x0C0 – 0x0DC	Per-bank refresh-latency observation
0x100 – 0x138	System observation / telemetry
0x1C0 – 0x1E0	Packed observation-word harvest (9 words)
0xFF0 – 0xFF4	Module identification + build hash

The complete register/field table is in §4.2; it is generated from the RDL and must match it exactly.

33.4 Behavior

33.4.1 Access Cycles

Each request drives `s_cpuif_req` with `s_cpuif_req_is_wr`, `s_cpuif_addr`, and (for writes) `s_cpuif_wr_data`/`s_cpuif_wr_biten`. The block responds with `s_cpuif_rd_ack`/`s_cpuif_wr_ack` (and optionally asserts a stall while it is busy). Reads return `s_cpuif_rd_data`; unmapped accesses assert `s_cpuif_rd_err`/`s_cpuif_wr_err`.

33.4.2 Sub-Word Accesses

`s_cpuif_wr_biten` carries a per-bit write mask, so byte- or field-level writes are supported natively. A bridge that lacks byte strobes simply drives all biten bits on every write.

33.4.3 Error Conditions

The block asserts a decode error (`s_cpuif_rd_err` / `s_cpuif_wr_err`) for:

Condition	Notes
Access to an unmapped address (gaps / RSVD)	`rd_err`/`wr_err`; read data is 0
Write to a read-only register/field	Write is dropped; field unchanged

Writes to reserved (`sw = r`) fields inside an otherwise-writable register are ignored per the RDL field access; the register keeps its value. There is no scheme/rule sub-decode error (the old `ADDR_MAP_TUNING` scheme decode is retired — address mapping is now the plain `ADDR_MAP.bank_lsb` field, see §4.2/§4.4).

33.5 Config-Drive Model

`pumice_top` wires the register block's `hwif_out.*` outputs **by name** directly into `pumice_core` (see the top-level instantiation in `rtl/top/pumice_top.sv`). Configuration is therefore live combinational drive from the register storage into the datapath — software writes a field and the corresponding core input tracks it. There is no staging/commit or quiet-point protocol in this architecture (see §4.3). Status/observation readback (`hwif_in.*`) is currently tied off in `pumice_top`.

33.6 Open Questions / Future Work

- **Observation readback.** `hwif_in` is presently tied to 0; wiring the live status/telemetry counters back into the read path is a follow-up.
- **Bridge selection.** The passthrough cpuif is transport-agnostic. A packaged APB and AXI-Lite bridge pair (with the CDC option) would simplify SoC integration.
- **Quiet-point commit.** The old two-cell staging/commit model was removed with the config-drive refactor. If a future field must not change mid-transaction, a per-field commit gate can be reintroduced.

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34 Register Map

This MAS chapter is the **authoritative field-level register map**, transcribed directly from `rtl/macro/pumice_csr.rdl` and mirrored by the generated `dv/tbclasses/pumice_regmap.py`. If this table and the RDL ever disagree, the RDL wins. Offsets are byte addresses in the 4 KB (12-bit) region.

34.1 Source of Truth

The register map is a SystemRDL source, `rtl/macro/pumice_csr.rdl`. It is compiled by `bin/peakrdl_generate.py` into:

Artifact	Consumer
`regs/generated/rtl/pumice_csr.sv` + `pumice_csr_pkg.sv`	The PeakRDL passthrough register block instantiated in `pumice_top` (`hwif_in`/`hwif_out` structs)
`dv/tbclasses/pumice_regmap.py`	The DV `RegisterMap` by-name access model (offset/field/default)

`pumice_top` drives `hwif_out.*` fields **by name** straight into `pumice_core` — there is no hand-written register file and no APB slave inside the controller (see §4.1). DV programs every register by name via `pumice_regmap.py`; hardcoded offsets are forbidden.

34.2 Register Summary

Offset	Register	Purpose
0x000	`CTRL`	Init / power / soft-reset request bits
0x004	`STATUS`	Init / power / version status (RO, hw-written)
0x008	`STATUS_HISTORY`	Last 8 power-state transitions (RO)
0x010	`TIMINGS_RC_RCD_RP_RAS`	tRC / tRCD / tRP / tRAS
0x014	`TIMINGS_RFC_REFI`	tRFC / tREFI
0x018	`TIMINGS_RRD_FAW_WTR_CCD`	tRRD / tFAW / tWTR / tCCD
0x01C	`TIMINGS_CL_CWL_WR`	CL / CWL / tWR / tRFCpb
0x020	`MR0`	Mode Register 0 value
0x024	`MR1`	Mode Register 1 value

Offset	Register	Purpose
0x028	MR2	Mode Register 2 value
0x02C	MR3	Mode Register 3 value
0x030	PASR_BANK_MASK_RANK0	LPDDR2 PASR per-bank mask (rank 0)
0x034	PASR_SEG_MASK_RANK0	LPDDR2 PASR segment mask (rank 0)
0x038	TEMP_DERATE_RANK0	LPDDR2 MR4 temperature class (rank 0, RO)
0x040	SCHED_TUNING	Scheduler runtime knobs
0x044	PAGE_PRED_TUNING	HAPPY-mode predictor knobs
0x048	REFRESH_TUNING	Refresh policy + page-policy override + ZQCS interval
0x04C	ADDR_MAP	Address-map: bank_lsb + XOR-hash (replaces ADDR_MAP_TUNING)
0x050	INIT_TUNING	ZQ retries + per-step init timeout
0x054	TIMINGS_RTP_RTW	tRTP / tRTW
0x058	INIT_TIMING0	Init waits: tINIT / tDLLK
0x05C	INIT_TIMING1	Init waits: tMRD / tRP / tRFC
0x060	DFI_PHASE	DFI READ/WRITE command sub-phase placement
0x064	PHY_TIMING	t_phy_wrlat / t_rddata_en / memtype / refresh_burst
0x080..0x	OBS_ROW_HIT[8]	Per-bank row-hit count (RO, read-clear)

Offset	Register	Purpose
09C		
0x0C0..0x0DC	OBS_REF_LATENCY[8]	Per-bank refresh-blocking cycles (RO)
0x100..0x138	OBS_*	System observation / telemetry (RO)
0x1C0..0x1E0	OBS_WORDS[9]	Packed obs_* harvest words (RO)
0xFF0	ID	Module ID (version / memtype / n_phases / 0xD2)
0xFF4	BUILD	Build hash

34.3 Field-Level Detail

All registers are 32-bit; unlisted bits are reserved (RSVD, `sw = r`). “Default” is the RDL reset value.

34.3.1 CTRL @ 0x000 (rw)

Bits	Field	Default	Notes
0	init_start	0	Write 1 to start init (swmod)
1	init_force_restart	0	Write 1 to force re-init mid-sequence (swmod)
4	pwr_req_low_power	0	Request power-down
5	pwr_req_dpd	0	Request DPD (LPDDR2 only)
6	pwr_req_active	0	Request return to ACTIVE
7	pwr_req_self_refresh	0	Request self-refresh
31	soft_reset	0	Write 1 to assert internal soft reset (self-clearing, swmod)

34.3.2 STATUS @ 0x004 (RO, hw-written)

Bits	Field	Notes
0	init_done	Init complete
1	init_error	Init error
7:4	power_state	Current power-state FSM state (encoded)
8	pasr_active	LPDDR2: PASR mask is non-zero
23:16	init_step_dbg	Current init step number (bring-up)
31	version_match	Build matches expected version

34.3.3 STATUS_HISTORY @ 0x008 (RO)

Bits	Field	Notes
31:0	history	8 x 4-bit power-state history; most recent in [3:0]

34.3.4 TIMINGS_RC_RCD_RP_RAS @ 0x010 (rw)

Bits	Field	Default (dec)	Notes
7:0	tRC	60	MC cycles
15:8	tRCD	15	
23:16	tRP	15	
31:24	tRAS	40	

34.3.5 TIMINGS_RFC_REFI @ 0x014 (rw)

Bits	Field	Default (dec)	Notes
15:0	tRFC	16	or tRFCab; mission-mode REF recovery (arbiter down-counter)
31:16	tREFI	1950	

34.3.6 TIMINGS_RRD_FAW_WTR_CCD @ 0x018 (rw)

Bits	Field	Default (dec)
7:0	tRRD	6

Bits	Field	Default (dec)
15:8	tFAW	35
23:16	tWTR	4
31:24	tCCD	4

34.3.7 TIMINGS_CL_CWL_WR @ 0x01C (rw)

Bits	Field	Default (dec)	Notes
7:0	CL	6	CAS latency
15:8	CWL	4	CAS write latency
23:16	tWR	15	Write recovery
31:24	tRFCpb	70	LPDDR2 per-bank tRFC

34.3.8 MR0..MR3 @ 0x020 / 0x024 / 0x028 / 0x02C (rw)

Bits	Field	Default	Notes
15:0	VAL	0	Mode-register value loaded during init

34.3.9 PASR_BANK_MASK_RANK0 @ 0x030 (rw)

Bits	Field	Default	Notes
7:0	pasr_banks	0	LPDDR2 MR16; bit N=1 masks bank N

34.3.10 PASR_SEG_MASK_RANK0 @ 0x034 (rw)

Bits	Field	Default	Notes
7:0	pasr_segs	0	LPDDR2 segment mask

34.3.11 TEMP_DERATE_RANK0 @ 0x038 (RO, hw-written)

Bits	Field	Notes
1:0	temp_class	LPDDR2 MR4: 00 nominal, 01 2x refresh, 10 4x refresh

34.3.12 SCHED_TUNING @ 0x040 (rw)

Bits	Field	Default	Access	Notes
3:0	lookahead_active	0	rw	Active lookahead window (0 disables)
4	force_inorder	0	rw	1 = force first-ready FIFO
5	happy_enable	1	rw	HAPPY predictor active (if synthesized)
15:8	age_max_runtime	0	rw	Runtime AGE_MAX override (0 = build default)
23:16	txn_queue_high_water	0	rw	Backpressure threshold
27:24	lookahead_max_obs	0	RO	Echo of build-time LOOKAHEAD_DEPTH_MAX

34.3.13 PAGE_PRED_TUNING @ 0x044 (rw)

Bits	Field	Default (dec)	Notes
15:0	warmup_cycles	1024	
23:16	hysteresis	2	

34.3.14 REFRESH_TUNING @ 0x048 (rw)

Bits	Field	Default	Notes
1:0	refpb_policy_or	0	00 build-time, 01 RR, 10 OLDEST_FIRST, 11 DARP
3:2	page_policy_or	0	00 build-time, 01 OPEN, 10 CLOSE, 11 HAPPY_HYBRID
7:4	refresh_defer_active	1	Active refresh deferral count
31:16	zqcs_freq_hz	1	Periodic ZQCS interval in Hz (0 disables)

34.3.15 ADDR_MAP @ 0x04C (rw) — replaces the retired ADDR_MAP_TUNING

Bits	Field	Default	Notes
4:0	bank_lsb	0x0A	Bank-field LSB in the byte-offset-stripped word

Bits	Field	Default	Notes
		(=COL_WIDTH, ROW_MAJOR)	address; RTL clamps to [0, COL_WIDTH]
8	hash_en	0	Enable bank XOR-hash: `bank ^= fold(row) ^ hash_seed`
23:16	hash_seed	0	XOR-hash seed (`seed[BW-1:0]`)

There is no scheme selector, `scheme_or`, or `synth_mask_obs`. ROW_MAJOR / BANK_INTERLEAVE / XOR_HASH are just settings of this one register (see §4.4 and `rtl/fub/addr_mapper.sv`).

34.3.16 INIT_TUNING @ 0x050 (rw)

Bits	Field	Default (dec)	Notes
3:0	zq_retries	3	ZQ retries (1..8)
15:8	init_timeout_ms	10	Init timeout ms

34.3.17 TIMINGS_RTP_RTW @ 0x054 (rw)

Bits	Field	Default (dec)	Notes
7:0	tRTP	4	Read to precharge
15:8	tRTW	6	Read to write

34.3.18 INIT_TIMING0 @ 0x058 (rw)

Bits	Field	Default (dec)	Notes
15:0	t_init_wait	512	CKE/tINIT settle
31:16	t_dll_wait	256	DLL lock (tDLLK)

34.3.19 INIT_TIMING1 @ 0x05C (rw)

Bits	Field	Default (dec)	Notes
7:0	t_mrd_wait	8	post mode-reg (tMRD)
15:8	t_rp_wait	8	post precharge (tRP)
23:16	t_rfc_wait	16	post refresh (tRFC)

34.3.20 DFI_PHASE @ 0x060 (rw)

Bits	Field	Default	Notes
2:0	rd_phase	0	READ command DFI sub-phase
6:4	wr_phase	0	WRITE command DFI sub-phase

Sliced to clog2(DFI_RATE) bits downstream; upper bits ignored when DFI_RATE is small. On the Nexys A7 a7ddrphy, rd_phase=0 (the PHY handles rdphase internally).

34.3.21 PHY_TIMING @ 0x064 (rw)

Bits	Field	Default (dec)	Notes
7:0	t_phy_wrlat	0	WR cmd -> dfi_wrdata_en (Nexys A7 bring-up tuple programs 1)
15:8	t_rddata_en	6	RD cmd -> dfi_rddata_en window
16	memtype	0	0 = DDR2, 1 = LPDDR2
23:20	refresh_burst	1	REFs drained per request (1..8)

34.3.22 Observation registers (RO)

Offset	Register / array	Field	Notes
0x080..0x09C	OBS_ROW_HIT[8]	VAL[31:0]	Per-bank row-hit count; read-clear (onread = rclr)
0x0C0..0x0DC	OBS_REF_LATENCY[8]	VAL[31:0]	Per-bank refresh-blocking cycles

Offset	Register / array	Field	Notes
0x100	OBS_TXN_QUEUE_DEPTH_MAX	VAL	Max queue depth observed
0x104	OBS_TXN_QUEUE_DEPTH_AVG	VAL	Time-averaged queue depth
0x108	OBS_REFRESH_PENDING_MAX	VAL	Max refresh_pending observed
0x10C..0x118	OBS_REFRESH_DEFER_HIST_0..3	VAL	Refresh-deferral histogram bins
0x120	OBS_PAGE_PRED_ACCURACY	VAL	HAPPY rolling prediction accuracy (%)
0x130	OBS_AXI_R_LATENCY_AVG	VAL	Avg AXI read latency (cycles)
0x134	OBS_AXI_R_LATENCY_P99	VAL	99th-pct AXI read latency
0x138	OBS_AXI_W_LATENCY_AVG	VAL	Avg AXI write latency
0x1C0..0x1E0	OBS_WORDS[9]	VAL	Packed obs_* harvest words

34.3.23 ID @ 0xFF0 (RO) — reset 0xD2020001

Bits	Field	Value	Notes
7:0	version	0x01	Build version
15:8	memtype	0x00	0 = DDR2, 1 = LPDDR2
23:16	n_phases	0x02	Gear ratio (1/2/4)
31:24	module_id	0xD2	Fixed 0xD2

34.3.24 BUILD @ 0xFF4 (RO)

Bits	Field	Default	Notes
31:0	VAL	0	Build hash word

34.4 Multi-Rank / Multi-Bank Registers

The RDL declares per-bank observation arrays with `NUM_BANKS = 8` for rank 0 (`OBS_ROW_HIT[8]` at 0x080, `OBS_REF_LATENCY[8]` at 0x0C0). PASR / temperature registers are declared per rank as

*_RANK0 for the default single-rank build. The generated pumice_regmap.py flattens arrays into indexed register names (e.g., OBS_ROW_HIT0_ROW_HIT, OBS_REF_LATENCY7_REF_LAT, OBS_WORDS8_WORD). Multi-rank builds add the corresponding *_RANK{N} registers; there is no separate capability vector register in this RDL — software reads memtype/n_phases from ID (0xFF0).

34.5 Reset Values

Reset defaults come straight from the RDL = <value> initializers (echoed in the default/offset entries of pumice_regmap.py). They are chosen so a "do nothing" bring-up sees a sane DDR2 baseline; DV programs the workload-specific values by name before triggering init.

34.6 Open Questions / Future Work

- **Observation readback wiring.** The RO/telemetry registers are declared and generated, but pumice_top currently ties hwif_in to 0 (see §4.1). Connecting the live counters is a follow-up.
- **Multi-rank register generation.** The single-rank build declares only *_RANK0. A NUM_RANKS-driven RDL loop for the PASR/temp/observation windows is the natural extension.
- **RDL <-> regmap drift check.** pumice_regmap.py is generated from the RDL; a CI diff would catch manual edits.

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35 Runtime CSR Fields (Config-Drive Model)

Per HAS §5.1 for the build-time-vs-runtime principle and §4.2 for the field-level register map. This MAS chapter is the **implementation protocol** for how runtime CSR fields reach the live datapath.

In the rearchitected controller, pumice_top wires the register block's hwif_out.* fields **by name** directly into pumice_core (see rtl/top/pumice_top.sv). There is **no** staging/commit two-cell architecture, **no** CTRL.config_apply, **no** STATUS.config_settled, and **no** quiet-point drain FSM. A CSR write updates the field's storage and the corresponding core input tracks it combinationally.

35.1 What “Runtime” Means Here

Most of the controller’s behavior is programmed at runtime by CSR fields, not fixed by parameters. The following are all live CSR-driven core inputs (see the `u_core` port map in `rtl/top/pumice_top.sv`):

Domain	CSR field(s)	Core input
Memory type	`PHY_TIMING.memtype`	`memtype_i` (0=DDR2, 1=LPDDR2)
Page policy	`REFRESH_TUNING.page_policy_or`	`page_policy_i`
Address mapping	`ADDR_MAP.bank_lsb` / `.hash_en` / `.hash_seed`	`bank_lsb_i` / `hash_en_i` / `hash_seed_i`
Core JEDEC timing	`TIMINGS_RC_RCD_RP_RAS`, `TIMINGS_CL_CWL_WR.tWR`, `TIMINGS_RTP_RTW`, `TIMINGS_RRD_FAW_WTR_CCD`, `TIMINGS_RFC_REFI.tREFI`	`t_rcd_i`/`t_rp_i`/ `t_ras_i`/`t_rc_i`/ `t_wr_i`/`t_rtp_i`/ `t_rtw_i`/`t_faw_i`/ `t_rrd_i`/`t_wtr_i`/ `t_ccd_i`/`t_refi_i`
Refresh burst	`PHY_TIMING.refresh_burst`	`refresh_burst_i`
Init waits	`INIT_TIMING0` / `INIT_TIMING1`	`t_init_wait_i`/ `t_dll_wait_i`/ `t_mrd_wait_i`/ `t_rp_wait_i`/ `t_rfc_wait_i`
DFI phase	`DFI_PHASE.rd_phase` / `.wr_phase`	`rd_phase_i`/`wr_phase_i` (sliced to `clog2(DFI_RATE)`)
PHY data timing	`PHY_TIMING.t_phy_wrlat` / `.t_rddata_en`	`t_phy_wrlat_i`/ `t_rddata_en_i`

CL/CWL/BL are **not** driven from the timing CSRs at the core boundary — they are decoded from the mode-register shadow inside the scheduler layer (`mode_register.sv`) as the init sequencer programs MR0..MR3 (see §5.1). The `TIMINGS_CL_CWL_WR.CL`/`.CWL` fields are informational in this build.

35.2 Commit Semantics

Because the fields drive the core directly, a write takes effect as soon as it lands in the register storage. There is no explicit apply step. Two practical timing classes:

Class	Fields	When it takes effect
Sampled at an event boundary	JEDEC timings, refresh interval/burst, page policy, scheduler knobs	On the next counter reload / arbiter decision that consumes the value
Sampled continuously	`memtype`, `bank_lsb`/`hash_*`, DFI phase, PHY data timing	Combinationally; affects the next command that uses the mapper/formatter

35.3 Safe-Change Guidance

The controller does not enforce a quiet point, so **software** is responsible for not changing a field mid-transaction when that would corrupt in-flight state:

- **Timings, page policy, scheduler knobs, refresh interval**: safe to change during light traffic; the new value is picked up at the next event boundary. For a clean swap, quiesce AXI traffic first (stop issuing, let outstanding responses drain) then write.
- **`memtype`, address-map (`bank_lsb`/`hash_*`)**: these define how addresses decode and how commands are encoded. Change them **only before init** (or with the datapath fully idle). Changing address mapping under traffic re-decodes in-flight addresses inconsistently.
- **DFI phase / PHY data timing**: board bring-up knobs; set once during bring-up and leave fixed. `rd_phase`/`t_rddata_en`/`t_phy_wrlat` are matched to the attached PHY (the Nexys A7 a7ddrphy bring-up tuple, board-validated 2026-07-21: `rd_phase=0`, `t_rddata_en=6`, `t_phy_wrlat=1`, plus harness `DFI_TUNING.rddata_delay=7`).

35.4 Recommended Programming Order

For a clean bring-up the register writes happen while init is still gated (before `CTRL.init_start`):

```
// 1. Select memory type and address mapping (must be set before init)
csr_write(PHY_TIMING, memtype_field | t_rddata_en_field |
t_phy_wrlat_field | refresh_burst_field);
```

```
csr_write(ADDR_MAP,   bank_lsb_field | hash_en_field |
hash_seed_field);

// 2. Program JEDEC timings for the attached part
csr_write(TIMINGS_RC_RCD_RP_RAS, ...);
csr_write(TIMINGS_RFC_REFI,      ...);
csr_write(TIMINGS_RRD_FAW_WTR_CCD, ...);
csr_write(TIMINGS_CL_CWL_WR,     ...);
csr_write(TIMINGS_RTP_RTW,       ...);
csr_write(INIT_TIMING0, ...);  csr_write(INIT_TIMING1, ...);

// 3. Program MR0..MR3 and (LPDDR2) PASR
csr_write(MR0, ...); csr_write(MR1, ...); csr_write(MR2, ...);
csr_write(MR3, ...);

// 4. Trigger init — the values above are already live at the core
boundary
csr_write(CTRL, CTRL_INIT_START);
```

There is no apply/poll handshake between the writes — each write is already visible to the core.

35.5 Open Questions / Future Work

- **Mid-traffic re-tuning guard.** A future revision could add an optional per-field commit gate (re-introducing a lightweight quiet point) for fields that software wants to change under load. Not present in this build.
- **CL/CWL CSR coupling.** The `TIMINGS_CL_CWL_WR.CL/.CWL` fields are currently informational; the live decode comes from the MR shadow. Deciding whether the CSR or the MR shadow is authoritative is a cleanup item.
- **Observation readback.** `STATUS/OBS_*` are declared but `hwif_in` is tied off in `pumice_top` (see §4.1); wiring them back is a follow-up.

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36 Family Config Knobs (DDR2 / LPDDR2)

Per HAS §5.4 for the family-wide config philosophy. This MAS chapter is the **implementation-level** detail for **this controller** specifically: which

CSR fields select DDR2-vs-LPDDR2 behavior and which fields are present-but-inert on this generation.

The controller is a two-family design (DDR2 + LPDDR2). The single most important family selector is `PHY_TIMING.memtype` (0 = DDR2, 1 = LPDDR2), which `pumice_top` maps to the core's `memtype_i` and which the init sequencer and DFI command formatter branch on. There is **no** separate family capability vector register; `ID` @ 0xFF0 echoes the build's memtype and phase count.

36.1 The Family Selector: `PHY_TIMING.memtype`

memtype	Family	What it changes
0	DDR2	`init_sequencer` runs the DDR2 MRS/precharge/refresh chain; `dfi_cmd_formatter` drives the DDR2 RAS/CAS/WE command bus
1	LPDDR2	`init_sequencer` runs the LPDDR2 MRW chain (MR63/MR10/MR1/MR2/MR3); `dfi_cmd_formatter` packs the JESD209-2F CA-bus word onto `dfi_address`

`memtype` is set before init and left fixed (see §4.3). DDR2 is the board target; LPDDR2 is the family-reuse path. Both pass the full sim suite.

36.2 Field-by-Field Applicability

Fields below are the real CSR fields in `rtl/macro/pumice_csr.rdl` (see §4.2). "DDR2 / LPDDR2" notes whether the field is meaningful on each family in this build.

36.2.1 SCHED_TUNING (0x040)

Field	Applicability
`lookahead_active`	Both; scheduler lookahead window (0 disables)
`force_inorder`	Both; forces first-ready FIFO ordering
`happy_enable`	Both, only meaningful when the HAPPY predictor is synthesized/selected
`age_max_runtime`	Both; anti-starvation AGE_MAX override
`txn_queue_high_water`	Both; backpressure threshold

Field	Applicability
lookahead_max_obs	RO echo of build-time LOOKAHEAD_DEPTH_MAX

36.2.2 REFRESH_TUNING (0x048)

Field	Applicability
page_policy_or	Both; drives page_policy_i (00 build-time, 01 OPEN, 10 CLOSE, 11 HAPPY_HYBRID)
refresh_defer_active	Both; refresh deferral / batching count
zqcs_freq_hz	Both; periodic ZQCS interval (DDR2 has no ZQCL, but ZQCS calibration short is JEDEC)
refpb_policy_or	**LPDDR2-relevant** (per-bank refresh); DDR2 uses all-bank REFab only

36.2.3 ADDR_MAP (0x04C) — family-agnostic

Field	Applicability
bank_lsb	Both; the single address-map knob. = COL_WIDTH -> ROW_MAJOR; smaller -> interleave. No scheme selector exists.
hash_en	Both; enables the bank XOR-hash (the old XOR_HASH “scheme”)
hash_seed	Both; XOR-hash seed

DDR2/LPDDR2 have no bank groups, so there is no bank-group field. The retired ADDR_MAP_TUNING register (with scheme_or / synth_mask_obs / xor_seed_runtime) does not exist; all address mapping is the ADDR_MAP register above.

36.2.4 INIT_TUNING (0x050) / INIT_TIMING0/1 (0x058/0x05C)

Field	Applicability
zq_retries, init_timeout_ms	Both
t_init_wait, t_dll_wait	Both; on LPDDR2 the sequencer reuses t_init_wait for tINIT4 and t_dll_wait for tZQINIT (see §5.1)
t_mrd_wait, t_rp_wait, t_rfc_wait	DDR2 uses all three; LPDDR2 reuses t_mrd_wait for post-MRW (t_rp_wait/t_rfc_wait are inert on the LPDDR2 chain)

36.2.5 PHY_TIMING (0x064)

Field	Applicability
memtype	The family selector (above)
t_phy_wrlat	Both; WR cmd -> dfi_wrdata_en (0 = a7ddrphy pre-pull)
t_rddata_en	Both; RD cmd -> dfi_rddata_en window
refresh_burst	Both; REFs drained per request

36.2.6 LPDDR2-only registers

Register	Applicability
PASR_BANK_MASK_RANK0 (0x030)	LPDDR2 partial-array self-refresh (MR16); inert on DDR2
PASR_SEG_MASK_RANK0 (0x034)	LPDDR2 PASR segment mask; inert on DDR2
TEMP_DERATE_RANK0 (0x038)	LPDDR2 MR4 temperature class (RO); inert on DDR2
TIMINGS_CL_CWL_WR.tRFCpb	LPDDR2 per-bank tRFC; unused by DDR2
CTRL.pwr_req_dpd	LPDDR2 Deep Power Down request; inert on DDR2

36.3 Feature Discovery

There is no 0xFF8 capability vector in this RDL. Software identifies the build via the ID register at 0xFF0:

Bits	Field	Meaning
7:0	version	Build version (0x01)
15:8	memtype	Build memtype echo (0 DDR2 / 1 LPDDR2)
23:16	n_phases	Gear ratio (1 / 2 / 4)
31:24	module_id	Fixed 0xD2

Note the ID.memtype field is a build-time echo (default reflects the DDR2 build); the **runtime** family select is PHY_TIMING.memtype, which software programs before init.

36.4 Not Present in This Generation

The following belong to higher DDR/LPDDR generations and have **no** field in this controller's CSR map: fine-granularity refresh (FGR), write-leveling, MPR, CA training, command-bus training (CBT), bank groups, and inline ECC. They were speculative entries in an earlier family registry and are omitted here rather than tied off.

36.5 Open Questions / Future Work

- **Multi-rank family fields.** PASR/temperature registers are declared for rank 0 only; a `NUM_RANKS` loop in the RDL would add `*_RANK{N}`.
- **Capability register.** If a family capability vector proves useful for a generic SoC driver, it can be re-added as an RO register; for now `ID` carries the essentials.
- **DPD power-state path.** `CTRL.pwr_req_dpd` is defined; the LPDDR2 DPD entry/exit datapath is a follow-up (see §5.2).

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37 Initialization Sequence

Per HAS §6.2 for the architectural sequence and §2.12 for the FUB-level detail. This chapter is the **software-side cookbook** — what bring-up firmware actually does to get from power-on to "DRAM ready for AXI traffic." The sequence is driven by `rtl/fub/init_sequencer.sv`.

37.1 Pre-Reset Setup

Before `aresetn` (the MC-domain reset) deasserts, the SoC's PMU should:

1. Apply DRAM power rails (VDD, VDDQ; LPDDR2 also VDD1/VDD2/VDDCA)
2. Wait for power rails to be stable (typically 10–100 µs per JEDEC)
3. Bring the register transport out of reset (the PeakRDL cpuif is R/W-able; the init sequencer sits in `S_RESET`/`S_DFI_INIT`)

4. Program the family selector and address map: `PHY_TIMING.memtype`, `ADDR_MAP.bank_lsb/hash_*` (these must be set before init — see §4.3)
5. Load timing values (`TIMINGS_*`, `INIT_TIMING0/1`, `MR0..MR3`, PASR if LPDDR2)
6. Release the datapath reset

Because config is driven by name into the core (see §4.3), the values are already live at the core boundary when init runs — there is no staging/commit step.

37.2 Init Trigger

```
void start_dram_init(void) {
    // Family + address map first (must be set before init)
    csr_write(PHY_TIMING, MEMTYPE(DDR2) | T_RDDATA_EN(6) |
T_PHY_WRLAT(1) | REFRESH_BURST(1));
    csr_write(ADDR_MAP,   BANK_LSB(COL_WIDTH));   // ROW_MAJOR; add
HASH_EN|HASH_SEED if wanted

    // Load timings (example values for DDR2-800)
    csr_write(TIMINGS_RC_RCD_RP_RAS,   TIMING_PACK(60, 15, 15, 40));
// tRC, tRCD, tRP, tRAS
    csr_write(TIMINGS_RFC_REFI,        REFI_PACK(16, 1950));
// tRFC, tREFI
    csr_write(TIMINGS_RRD_FAW_WTR_CCD, RRD_PACK(6, 35, 4, 4));
// tRRD, tFAW, tWTR, tCCD
    csr_write(TIMINGS_CL_CWL_WR,       CL_PACK(6, 4, 15, 70));
// CL, CWL, tWR, tRFCpb
    csr_write(TIMINGS_RTP_RTW,         RTP_PACK(4, 6));
// tRTP, tRTW

    // JEDEC init-sequence waits (or leave RDL defaults
512/256/8/8/16)
    csr_write(INIT_TIMING0, INIT_T0_PACK(512, 256));
// tINIT, tDLLK
    csr_write(INIT_TIMING1, INIT_T1_PACK(8, 8, 16));
// tMRD, tRP, tRFC

    // MR values: for DDR2 the sequencer uses its own JEDEC-correct MR
words
    // (0x0533/0x0433/0x0380/0x0000 at the MR0.VAL reset default); the
MR0..MR3 CSRs are the software-visible
    // shadow. For LPDDR2 the sequencer drives the MRW OP values
internally.
    csr_write(MR0, mr0_value); csr_write(MR1, mr1_value);
    csr_write(MR2, mr2_value); csr_write(MR3, mr3_value);
```

```
    // (LPDDR2) PASR if needed
    csr_write(PASR_BANK_MASK_RANK0, pasr_bank_mask);

    // Optional: ZQ retry count / timeout
    csr_write(INIT_TUNING, INIT_TUNING_PACK(3, 10));  // 3 retries,
10ms timeout

    // Trigger init
    csr_write(CTRL, CTRL_INIT_START);

    // Poll for completion
    while (1) {
        uint32_t s = csr_read(STATUS);
        if (s & STATUS_INIT_DONE)  break;             // DRAM ready
        if (s & STATUS_INIT_ERROR) {                  //
STATUS.init_step_dbg shows where
            log_error("DRAM init failed at step %d",
STATUS_INIT_STEP_DBG(s));
            return -EIO;
        }
    }
}
```

Note (this build): `STATUS` readback (`hwif_in`) is currently tied off in `pumice_top` (see §4.1); `init_done` is also exposed directly as the top-level `init_done_o` port. Poll whichever is wired on the platform.

37.3 Init Sequence Details

The step-by-step JEDEC sequence is a hardware FSM in `init_sequencer.sv` (`r_state`); software does not interleave commands during init. The sequencer both **issues** the commands to the DRAM (through the scheduler to `dfi_cmd_formatter`) and updates the mode-register shadow so the live CL/CWL/BL decode tracks what was programmed. The `memtype_i` input selects the DDR2 or LPDDR2 chain.

37.3.1 DDR2 chain (mirrors LiteDRAM's proven Nexys A7 sequence)

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i` (PHY DLL-lock / IO training), then wait tINIT (`t_init_wait`, CKE settle)
2. **Precharge All** (wait tRP)
3. **EMRS(2)** = 0, **EMRS(3)** = 0, **EMRS(1)** = 0 — JEDEC MRS order EMR2 -> EMR3 -> EMR1, each followed by tMRD
4. **MRS(0)** + **DLL reset** (`MR0.VAL | 0x100`, reset default 0x0533: BL8/CL3/tWR3/DLL_RESET); wait tDLLK (DLL lock)
5. **Precharge All** (wait tRP)

6. **Auto Refresh x2** (each followed by tRFC)
7. **MRS(0)** without DLL-reset (`MR0.VAL`, reset default 0x0433 — clears the reset bit); wait tMRD
8. **EMRS(1)** + **OCD Default** (0x0380) -> **EMRS(1)** + **OCD Exit** (0x0000)
9. `init_done_o = 1`

37.3.2 LPDDR2 chain (JESD209-2F power-up + MR configuration)

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i`; tINIT settle
2. **MRW(MR63)** = Reset (wait tINIT4, reuses `t_init_wait`)
3. **MRW(MR10)** = 0xFF ZQ Init Calibration (wait tZQINIT, reuses `t_dll_wait`)
4. **MRW(MR1)** = BL8/nWR3 (0x23), **MRW(MR2)** = RL3/WL1 (0x01), **MRW(MR3)** = DS 40ohm (0x02) — each followed by tMRW (reuses `t_mrd_wait`)
5. `init_done_o = 1`

The MR index (MA, up to MR63) and data (OP) are packed as `{MA[5:0], OP[7:0]}` into the row request field; `dfi_cmd_formatter` unpacks it for the LPDDR2 CA-bus MRW word (a 3-bit bank port cannot reach MR10/MR63). Only MR1/MR2/MR3 update the CL/CWL/BL decode shadow; MR63/MR10 are issued but not shadowed.

The init waits are the `INIT_TIMING0/1` CSR fields (`t_init_wait`, `t_dll_wait`, `t_mrd_wait`, `t_rp_wait`, `t_rfc_wait`); RDL defaults (512/256/8/8/16 MC cycles) comfortably cover the JEDEC minimums at the board frequency. For fast sim, program small values before triggering init.

37.4 Multi-Rank Init Differences

The current `init_sequencer.sv` targets a single rank (the board build is `NUM_RANKS = 1`). For a `NUM_RANKS > 1` build the intended extension is:

- MRS/MRW steps iterate over ranks (one rank per step iteration)
- ZQ calibration is per-rank (each rank has its own reference)
- Each rank's PASR mask is programmed via its own `PASR_*_RANK{N}` register (see §4.2 open items)

Software programs the MR values once; per-rank iteration is absorbed by the sequencer. Per-rank register generation is a follow-up (the RDL declares only `*_RANK0` today).

37.5 ZQ Retry

The `INIT_TUNING.zq_retries` field (default 3) is exposed for the retry count; DDR2 uses OCD calibration during init (no ZQCL) and LPDDR2 uses the MR10 ZQ Init step. Raise the retry budget if signal-integrity issues cause occasional failures.

37.6 Post-Init Power-State

After `init_done`, the controller is in ACTIVE for all ranks with CKE high. The refresh manager starts its tREFI counter and begins normal refresh scheduling. The scheduler is unblocked and any AXI bursts that accumulated during init begin issuing.

37.7 Software Wait Times

Phase	Wait (real silicon, 200 MHz MC clock)
Power-up settling	200 µs
MRS loads + per-rank iteration	~10 cycles × NR × 4 MRs ≈ 160 cycles
ZQ calibration	~1 µs
Initial refresh batches	~10 µs
Total init duration	~250 µs (single-rank), ~280 µs (4-rank)

The init duration scales weakly with rank count — the per-rank serialization is dwarfed by the fixed settling delays.

37.8 Reset Recovery

If init fails (`STATUS.init_error = 1`):

```
void recover_init_failure(void) {
    uint8_t step = STATUS_INIT_STEP_DBG(csr_read(STATUS));
    log_error("Init failed at step %d", step);

    // Inspect step-specific telemetry (per step table)
    // ...

    // Force re-init
    csr_write(CTRL, CTRL_INIT_FORCE_RESTART);

    // Wait a few cycles for FSM to settle
    for (int i = 0; i < 10; i++) asm volatile("nop");

    // Re-init
    start_dram_init();
}
```

A hard fault (e.g., persistent ZQ calibration failure) requires SoC-level intervention — typically a power-cycle and re-bring-up.

37.9 Open Questions / Future Work

- **Init telemetry capture.** When init fails, telemetry beyond `init_step_dbg` would help (DFI signal trace, per-rank fault detection). The current debug surface is sparse; expand if bring-up calls it out.
- **Init via JTAG.** Some platforms init via a JTAG side-band rather than CPU+APB. The CSR map supports this through any APB-attached transport; no controller changes needed.
- **Partial init for warm reset.** Coming out of self-refresh doesn't require full init. The current init engine doesn't have a "warm reset" path. Add when DDR3+ needs it for frequency change.

---

38 Refresh and Power-State Programming

Per HAS §3.4 / §3.5 for architecture and §2.11 for FUB detail. This chapter is the **software-side** view of refresh and power configuration. Register writes are the PeakRDL cpuif (`csr_write`/`csr_read`); there is no apply/commit handshake — fields drive the core live (see §4.3).

38.1 Tuning Refresh Behavior

```
void configure_refresh(refresh_config_t* cfg) {
    // Base tREFI / tRFC (also set during init; retunable at runtime)
    csr_write(TIMINGS_RFC_REFI, REFI_PACK(cfg->trfc, cfg->trefi));

    // REFRESH_TUNING packs deferral count, page/refpb policy, ZQCS
interval.
    uint32_t v = REFRESH_DEFER_ACTIVE(cfg->defer_active);   // 1..8

    // REFpb policy override (LPDDR2 only): 01 RR, 10 OLDEST_FIRST, 11
DARP
    if (cfg->refpb_policy == DARP)              v |=
REFPB_POLICY_OR(3);
    else if (cfg->refpb_policy == OLDEST)       v |=
REFPB_POLICY_OR(2);
```

```
    else if (cfg->refpb_policy == ROUND_ROBIN) v |=
REFPB_POLICY_OR(1);

    // Periodic ZQCS interval in Hz (0 = disable)
    v |= ZQCS_FREQ_HZ(cfg->zqcs_freq_hz);

    csr_write(REFRESH_TUNING, v);   // live on the next refresh event
boundary
}
```

`PHY_TIMING.refresh_burst` (1..8) additionally controls how many REFs are drained per refresh request.

38.1.1 When to Tune

Workload	Recommended config
Streaming (DMA, video)	`defer_active` = 8, ZQCS = 1 Hz
Low-latency bursty (CPU access)	`defer_active` = 1 (no batching), ZQCS = 0.1 Hz
Power-sensitive (sleep-heavy)	`defer_active` = 4, ZQCS = 0.1 Hz
Real-time / safety-critical	`defer_active` = 1, deterministic refresh latency

38.2 LPDDR2 PASR (Partial Array Self-Refresh)

PASR masks DRAM regions that hold no data, reducing refresh power.

```
void set_pasr_rank0(uint8_t bank_mask, uint8_t seg_mask) {
    csr_write(PASR_BANK_MASK_RANK0, bank_mask);
    csr_write(PASR_SEG_MASK_RANK0,  seg_mask);
    // Propagated to DRAM via MR16/MR17 at the next self-refresh
entry.
}
```

The single-rank build exposes `PASR_BANK_MASK_RANK0` / `PASR_SEG_MASK_RANK0`; multi-rank `*_RANK{N}` registers are a follow-up (see §4.2). The PASR mask is propagated lazily — typically at the next self-refresh entry — to avoid an extra bus-blocking MR write during normal operation.

38.3 Temperature Compensation (LPDDR2)

LPDDR2 devices expose temperature classification in MR4. The SoC reads this via PHY-side mechanisms (out of scope for the controller; the PHY signals temperature change via interrupt or polled register).

Software updates the controller's tREFI scaling:

`TEMP_DERATE_RANK0.temp_class` (0 = nominal, 1 = 2x refresh, 2 = 4x refresh) is a **read-only, hardware-written** field in this build — the controller captures the LPDDR2 MR4 class rather than software programming it. Software reads it to inform its own tREFI scaling via `TIMINGS_RFC_REFI`:

```
uint8_t temp = csr_read(TEMP_DERATE_RANK0) & 0x3;
// Scale tREFI down for 2x/4x refresh as temp rises
csr_write(TIMINGS_RFC_REFI, REFI_PACK(trfc, trefi >> temp));  // live
next reload
```

38.4 Self-Refresh Entry

For periods of inactivity, software (or auto-detection) can put the DRAM into self-refresh:

The power-state request bits live in `CTRL`: `pwr_req_low_power`, `pwr_req_self_refresh`, `pwr_req_active`, `pwr_req_dpd`. `STATUS.power_state[7:4]` reports the current encoded state (RO, hw-written).

```
void enter_self_refresh(void) {
    csr_write(CTRL, CTRL_PWR_REQ_SELF_REFRESH);
    while (STATUS_POWER_STATE(csr_read(STATUS)) !=
POWER_STATE_SELF_REFRESH);
}

void exit_self_refresh(void) {
    csr_write(CTRL, CTRL_PWR_REQ_ACTIVE);
    while (STATUS_POWER_STATE(csr_read(STATUS)) !=
POWER_STATE_ACTIVE);
}
```

Note: any AXI traffic to a rank will auto-wake it from SR. Explicit exit is only needed when the application wants to be ready before issuing.

There is no `POWER_TUNING` register in this build — idle-threshold auto-APD/SRF is not a CSR knob here. Power-state entry is software-requested via `CTRL`. (An auto-low-power threshold register is a possible follow-up; see below.)

38.5 DPD (LPDDR2 Deep Power Down)

DPD is the deepest power state on LPDDR2 — DRAM is fully off; software must full-init to exit. The request bit is `CTRL.pwr_req_dpd` (inert on DDR2). Check the build memtype via `ID` (0xFF0) rather than a capability vector (there is no `0xFF8`):

```
void enter_dpd(void) {
    if (((csr_read(ID) >> 8) & 0xFF) != MEMTYPE_LPDDR2) {
        log_error("DPD is LPDDR2-only");
        return;
    }
```

```
    csr_write(CTRL, CTRL_PWR_REQ_DPD);
    while (STATUS_POWER_STATE(csr_read(STATUS)) != POWER_STATE_DPD);
}

void exit_dpd(void) {
    start_dram_init();   // full re-init
}
```

DPD is rare in practice (DRAM fully unused for hours). Note the runtime family select is `PHY_TIMING.memtype`; `ID.memtype` is the build-time echo.

38.6 Open Questions / Future Work

- **Auto power-down thresholds.** This build has no `POWER_TUNING` register — APD/SRF is request-driven. Adding idle-threshold CSRs for automatic low-power entry is a candidate feature.
- **Per-rank power control.** `CTRL.pwr_req_*` is channel-wide. Per-rank request registers are a multi-rank follow-up.
- **Observation readback.** `STATUS.power_state`, `STATUS_HISTORY`, and the `OBS_REFRESH_DEFER_HIST_*` telemetry are declared but `hwif_in` is tied off in `pumice_top` today (see §4.1); wiring them enables telemetry-driven auto-tuning.

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39 Multi-Rank Programming

Per HAS §2.1 (multi-rank as differentiator), HAS §3.3 / §3.4 / §3.6 (architecture). FUB-level detail in MAS §2 (`pumice_bank_timers` per (rank,bank), `refresh_ctrl` round-robin, `dfi_cmd_formatter` CS_n / ODT).

Note: the default board build is single-rank (`NUM_RANKS = 1`). This chapter documents the intended multi-rank programming model; the single-rank RDL declares only `*_RANK0` registers, and per-rank register generation is a follow-up (see §4.2).

39.1 Discovery

There is no capability vector register in this build. Software knows `NUM_RANKS` from the build parameter (it is a synthesis-time geometry parameter, see §6.1); the `ID` register (0xFF0) reports memtype and phase count but not rank count. Treat the build parameter as authoritative.

39.2 Per-Rank Mode Register Loads

During init, the sequencer (intended multi-rank extension) iterates the MRS/MRW loads across ranks. Software writes MR0..MR3 once:

```
csr_write(MR0, mr0_value);   // applied to all ranks during init
csr_write(MR1, mr1_value);
csr_write(MR2, mr2_value);
csr_write(MR3, mr3_value);
```

For DDR2 the sequencer uses its own JEDEC-correct MR words; the `MR*` CSRs are the software-visible shadow (see §5.1). Per-rank MR variation (rare) is a future feature.

39.3 Per-Rank PASR (LPDDR2)

Already covered in §5.2. In the single-rank build:

```
csr_write(PASR_BANK_MASK_RANK0, mask);
```

A multi-rank build adds `PASR_BANK_MASK_RANK{N}` registers (each rank has independent MR16/MR17).

39.4 ODT

ODT is handled inside `dfi_cmd_formatter` / `mode_register` (there is no standalone ODT control block and no `RANK_TUNING.odt_rule_or` CSR). `dfi_odt_o` is driven per command. Board-specific ODT rule selection is not a runtime CSR knob in this build; it is baked into the formatter's per-command ODT logic. For boards with non-standard impedance budgets, adjust the formatter's ODT policy at build time.

39.5 Per-Rank Disable

There is no `RANK_TUNING` rank-enable CSR in this build. Rank enable/disable is a multi-rank feature to be added alongside per-rank register generation. In the single-rank build there is one rank and it is always active.

39.6 Address Mapping for Multi-Rank

The rank field always sits in the high bits of the (byte-offset-stripped) word address, above the row — its position is invariant. The single `ADDR_MAP.bank_lsb` knob only slides the **bank** field within

the column region (see §4.4 and `rtl/fub/addr_mapper.sv`); the rank field is unaffected by `bank_lsb`. The `hash_en` XOR-hash folds row bits into the bank index only — it does not touch the rank field.

There is no rank-interleave mode. Software-managed rank interleaving (the OS striping allocations across rank-aligned regions) is the workaround if consecutive-line rank striping is desired.

39.7 Refresh Behavior with Multi-Rank

Per HAS §3.4 and MAS §2 (`refresh_ctrl`, intended multi-rank extension):

- REFab dispatches per-rank in round-robin (not all-rank simultaneously)
- REFpb (LPDDR2) selects (rank, bank) tuples via the `REFRESH_TUNING.refpb_policy_or` policy
- Per-rank PASR masks are honored independently

A multi-rank system distributes refresh across the timeline so any single rank only blocks for tRFC per refresh — non-target ranks keep operating.

39.8 Power State Per-Rank

- Channel-wide CSR `CTRL.pwr_req_*` applies to all ranks (there is no per-rank power request CSR)
- `STATUS.power_state[7:4]` reports the encoded state
- Per-rank auto-low-power is not a CSR feature in this build (there is no `POWER_TUNING`; see §5.2)

39.9 Per-Rank / Per-Bank Observation

The RDL declares per-bank observation arrays for rank 0: `OBS_ROW_HIT[8]` (0x080..0x09C) and `OBS_REF_LATENCY[8]` (0x0C0..0x0DC). The generated regmap flattens them to indexed names:

```
uint32_t get_row_hit(uint8_t bank) {
    return csr_read(OBS_ROW_HIT0_ROW_HIT + bank * 4);   // read-clear
}
```

A multi-rank build adds the per-rank arrays. (Note: `hwif_in` observation readback is tied off in `pumice_top` today — see §4.1.)

39.10 Multi-Rank Bring-Up Checklist

Step	Why
Verify the build's `NUM_RANKS` matches the	Build / board mismatch

Step	Why
expected DIMM rank count	
Program `PASR_*_RANK{N}` if using LPDDR2	PASR for LPDDR2 bring-up
Verify per-rank ZQ calibration succeeds during init	Each rank's drive impedance
Sweep `OBS_REF_LATENCY[bank]` across rank workload mix	Per-rank refresh fairness
Stress-test rank-switching (read alternating ranks)	tRTRS / tCS timing

39.11 Open Questions / Future Work

- **Per-rank register generation.** The RDL declares only `*_RANK0`; a `NUM_RANKS` loop is needed for PASR/temp/observation and rank-enable.
- **Per-rank MR override.** Some DIMMs have rank-specific tuning (rare).
- **Per-rank power control.** `CTRL.pwr_req_*` is channel-wide; per-rank request registers are a follow-up.
- **Runtime ODT rule select.** ODT is currently baked into `dfi_cmd_formatter`; a runtime CSR knob could be reintroduced if board diversity needs it.

40 Error Handling

The controller's error-reporting paths and the recommended software response model. Register accesses are the PeakRDL cpuif (`csr_write`/`csr_read`); config fields drive the core live with no apply/commit step (see §4.3).

40.1 Error Categories

Category	Detection	Reporting
Init failure	`init_sequencer` timeout / ZQ retries exhausted	`STATUS.init_error = 1`; `STATUS.init_step_dbg` shows where
DFI rddata-valid timeout	RD issued, no rddata in the expected window	`SLVERR` on the AXI R (bring-up: check DFI phase / `t_rddata_en`)
AXI burst boundary violation	Per §3.1 (one AXI burst == one DRAM burst)	`SLVERR` on B/R
ZQ calibration failure (silicon)	DFI signal-integrity, caught at PHY layer	Not this controller
Multi-bit DRAM error	Out of scope (no inline ECC)	Would need ECC sideband

Note: `STATUS` is currently a declared RO register whose `hwif_in` is tied off in `pumice_top` (see §4.1). `init_done` is exposed directly as the top-level `init_done_o` port. There is no dedicated IRQ port list wired in this build; software polls `STATUS`/`init_done_o`. A queue-overflow / refresh-miss IRQ is a possible follow-up.

40.2 Software Response Patterns

40.2.1 Init Failure

```
void handle_init_error(void) {
    uint32_t s = csr_read(STATUS);
    uint8_t step = STATUS_INIT_STEP_DBG(s);

    log_error("DRAM init failed at step %d", step);

    // init_sequencer step encodings (r_state), for reference:
    //   S_DFI_INIT  -> PHY init did not complete (check PHY config)
    //   S_EMR*/S_MR0* / S_L_* -> MRS/MRW issue phase (verify MR
values, CA-bus)
    //   S_REF1/S_REF2 -> refresh phase
    // Force re-init (config CSRs are preserved)
    csr_write(CTRL, CTRL_INIT_FORCE_RESTART);
}
```

40.2.2 DFI Read-Data Timeout (board bring-up)

A read that returns no data almost always means the DFI read window is misaligned for the attached PHY. The knobs are `DFI_PHASE.rd_phase`, `PHY_TIMING.t_rddata_en`, and `PHY_TIMING.t_phy_wrlat`:

```
// Nexys A7 a7ddrphy known-good: rd_phase=0, t_rddata_en=6,
t_phy_wrlat=1 (+ harness rddata_delay=7)
csr_write(DFI_PHASE,   RD_PHASE(0) | WR_PHASE(0));
csr_write(PHY_TIMING,  MEMTYPE(DDR2) | T_RDDATA_EN(6) | T_PHY_WRLAT(1)
| REFRESH_BURST(1));
```

Set these during bring-up before triggering init; they take effect immediately (config-drive, no commit step).

40.2.3 Refresh Pressure

Watch `OBS_REFRESH_PENDING_MAX` (0x108). If it approaches the deferral budget, reduce batching:

```
uint32_t v = csr_read(REFRESH_TUNING);
v = (v & ~REFRESH_DEFER_ACTIVE_MASK) | REFRESH_DEFER_ACTIVE(1);  // no
batching
csr_write(REFRESH_TUNING, v);   // live on the next refresh event
boundary
```

40.3 STATUS_HISTORY for Bring-Up

The `STATUS_HISTORY` register (0x008) captures the last 8 power-state transitions (4 bits each, most recent in [3:0]). Useful when the controller oscillates between power states:

```
void dump_state_history(void) {
    uint32_t hist = csr_read(STATUS_HISTORY);
    for (int i = 0; i < 8; i++)
        log_info("State[-%d]: %s", i, power_state_name((hist >> (i*4))
& 0xF));
}
```

(As with `STATUS`, this readback depends on `hwif_in` being wired — a follow-up per §4.1.)

40.4 Diagnostic Sequence for Suspected Bug

```
1. Read STATUS.init_done (or the init_done_o port) — must be 1
2. Read STATUS.power_state — should be ACTIVE
3. Read STATUS_HISTORY — look for oscillation
4. Read OBS_TXN_QUEUE_DEPTH_MAX / OBS_REFRESH_PENDING_MAX — check for
backlog
5. Read OBS_AXI_R_LATENCY_P99 — tail latency telemetry
```

```
6. Read OBS_ROW_HIT[bank] — per-bank traffic distribution
7. Read OBS_REF_LATENCY[bank] — per-bank refresh fairness
```

This is the bring-up team's first-pass diagnostic flow.

40.5 Soft Reset vs. Re-Init

`CTRL.soft_reset` (bit 31, self-clearing) requests an internal soft reset that re-clears datapath state without disturbing the CSR contents. Use it when the controller is in an inconsistent state but the software's config is correct:

```
void soft_reset(void) {
    csr_write(CTRL, CTRL_SOFT_RESET);
    for (int i = 0; i < 16; i++) asm volatile("nop");  // self-
clearing
    csr_write(CTRL, CTRL_INIT_START);                  // CSRs
preserved
}
```

For hard cases (silicon bug, persistent failure), the SoC PMU drives the asynchronous reset and software re-brings-up from scratch.

40.6 Open Questions / Future Work

- **IRQ port.** This build polls `STATUS/init_done_o`. A latched error/IRQ output (init error, refresh miss, rddata timeout) is a candidate feature.
- **Observation readback wiring.** The `STATUS/OBS_*` diagnostics assume `hwif_in` is connected; it is tied off today (§4.1).
- **Per-rank error reporting.** When multi-rank lands, localizing a fault to a rank would help board bring-up.

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41 Build-Time Configuration Reference

Per HAS §5.2 for the parameter table and §5.1 for the build-vs-runtime philosophy. This chapter is the **build-script-author's** view: the synthesis-time parameters and how the data-width geometry works. Most behavior is runtime CSR-driven (see §4/§5), so the build parameter set is deliberately small — mostly memory geometry.

41.1 Parameters

The synthesis-time parameters are the ports of `pumice_top` (`rtl/top/pumice_top.sv`). They are geometry, not policy — memtype / timings / page policy / address map are all runtime CSR fields.

```
pumice_top #(
    .AXI_ID_WIDTH    (8),
    .AXI_ADDR_WIDTH  (32),
    .NUM_RANKS       (1),      // board build is single-rank
    .NUM_BANKS       (8),
    .ROW_WIDTH       (14),
    .COL_WIDTH       (10),
    .DFI_RATE        (2),      // DFI phases per MC clock (gear ratio)
    .DRAM_BEAT_WIDTH (64),     // bits per DRAM beat (device data bus)
    .BL              (8),      // DRAM burst length
    .NUM_ENTRIES     (8),      // CAM depth
    .N_SRAM_SLOTS    (8)
    // DW is DERIVED: DW = DRAM_BEAT_WIDTH * DFI_RATE
) u_pumice ( ... );
```

There is **no** `MEMTYPE`, `PAGE_POLICY`, `SCHEDULER_MODE`, `ODT_RULE_*`, `ADDR_MAP_SCHEMES_SYNTH`, or `DFI_ADDR_WIDTH` parameter — those are runtime CSR fields or derived from geometry. `memtype` is `PHY_TIMING.memtype`; page policy is `REFRESH_TUNING.page_policy_or`; address mapping is `ADDR_MAP.bank_lsb`.

41.2 Data-Width Geometry (DW = DRAM_BEAT_WIDTH x DFI_RATE)

The controller's internal / AXI-facing data width is **derived**, not a free parameter:

```
DW = DRAM_BEAT_WIDTH * DFI_RATE          // e.g. 64 * 2 = 128
(default)
```

- **One AXI beat == one DFI word == DFI_RATE DRAM beats.**
- **One AXI burst (BL / DFI_RATE beats) == one DRAM burst (BL beats).**
- The DW -> per-phase split happens in `dfi_signal_pack` / the DFI layer (`DFI_DATA_WIDTH = DW`, `DFI_EN_WIDTH = DFI_RATE`).

The old "AXI_DATA_WIDTH == DRAM_BEAT_WIDTH" coupling is **gone** — do not describe it. The host AXI data width the core presents is exactly `DW`.

41.2.1 Host-Width Freedom (`pumice_top_geared`)

To let a host SoC use a convenient fixed AXI width regardless of the DRAM point, wrap `pumice_top` in `pumice_top_geared` (`rtl/top/pumice_top_geared.sv`), which adds a free `HOST_AXI_DATA_WIDTH` parameter:

- It inserts the repo's **formally-verified** `axi4_dwidth_converter_wr` / `_rd` between a host-width AXI slave and the fixed-`DW` core.
- `HOST_AXI_DATA_WIDTH == DW` is a **generate bypass** — bit-identical to bare `pumice_top`, no converter, no added latency.
- Verified host widths: 64, 128, 256.
- Burst geometry contract is unchanged at the core's DW side; the converter re-sizes host bursts to DW-width bursts.

Doc: `docs/AXI_DRAM_GEARING_SCOPE.md`.

41.3 Parameter Sanity

The natural elaboration-time constraints:

```
generate
    if (NUM_RANKS != 1 && NUM_RANKS != 2 && NUM_RANKS != 4)
        $error("NUM_RANKS must be 1, 2, or 4");
    if (NUM_BANKS != 4 && NUM_BANKS != 8)
        $error("NUM_BANKS must be 4 or 8");
    if (DFI_RATE != 1 && DFI_RATE != 2 && DFI_RATE != 4)
        $error("DFI_RATE must be 1, 2, or 4 (gear ratio)");
    // BL must be an integer multiple of DFI_RATE so a DRAM burst is a
whole
    // number of AXI beats.
endgenerate
```

`DFI_RATE` must equal the attached PHY's phase count (e.g., the on-board 300 MT/s point is `DFI_RATE = 4` at sys = 37.5 MHz / CK = 150 MHz; regenerating for a different phase count alone produces a broken hybrid — see project notes).

41.4 Filelists

FUB filelists live in `rtl/filelists/`. The live module hierarchy (top -> core -> `pumice_axi4_ifc` / `pumice_mem_cmd_scheduler` / `pumice_dfi_layer`) is fixed; the DDR2 vs LPDDR2 command paths coexist in `dfi_cmd_formatter` and branch on `memtype`, so a single build supports both families with dead-code elimination handling the unused branch. `page_predictor.sv` / `powerdown_ctrl.sv` are optional/not in the default top build (confirm via the filelists).

41.5 Open Questions / Future Work

- **Multi-rank build.** `NUM_RANKS > 1` requires the per-rank register generation and per-rank init iteration described in §4.2 / §5.3.
- **Curated small-area filelist.** Excluding the optional predictor/power-down FUBs shrinks the smallest build.

- **Geared-top width matrix.** Extending the verified `HOST_AXI_DATA_WIDTH` set beyond {64,128,256} as needed.

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42 Runtime Configuration Reference

Per §4.2 for the full CSR map and §4.3 for the config-drive model (fields drive the core live; no apply/commit step). This chapter is the **driver author's** cookbook for runtime tunables: what to tune, when, and what to watch for. Register access is the PeakRDL cpuif (`csr_write`/`csr_read`).

42.1 Bring-Up Configuration Order

When bring-up software first comes up, recommended order (address map and memtype are set **before** init — see §5.1):

1. **Family + address map (pre-init)** — `PHY_TIMING.memtype`; `ADDR_MAP.bank_lsb` / `.hash_en` / `.hash_seed`.
2. **Set page policy** — `REFRESH_TUNING.page_policy_or` (00 build-time, 01 OPEN, 10 CLOSE, 11 HAPPY_HYBRID).
3. **Set lookahead depth** — `SCHED_TUNING.lookahead_active` (0 disables).
4. **Set refresh deferral** — `REFRESH_TUNING.refresh_defer_active`; `PHY_TIMING.refresh_burst`.
5. **Set ZQCS frequency** — `REFRESH_TUNING.zqcs_freq_hz` (1 Hz default; 0 to disable).

Each write is live immediately at the core boundary; there is no `config_apply` and no quiet-point drain. Quiesce AXI traffic before changing a field that would corrupt in-flight state (see §4.3).

42.2 Address-Map Tuning (single knob)

Address mapping is `ADDR_MAP.bank_lsb` alone (plus the optional XOR-hash). There is no scheme selector, `scheme_or`, or `xor_seed_runtime`:

```
// ROW_MAJOR: bank field above the whole column
csr_write(ADDR_MAP, BANK_LSB(COL_WIDTH));

// Max BANK_INTERLEAVE: bank field just above the burst's low column
bits
csr_write(ADDR_MAP, BANK_LSB(log2_cols_per_burst));

// XOR_HASH folded on top of any placement
csr_write(ADDR_MAP, BANK_LSB(COL_WIDTH) | HASH_EN | HASH_SEED(seed));
```

RTL clamps `bank_lsb` to `[0, COL_WIDTH]`; keep `log2(BL/DFI_RATE) <= bank_lsb <= COL_WIDTH` so a DRAM burst stays inside one bank (see §4.4 and `rtl/fub/addr_mapper.sv`). Change address mapping only before init or with the datapath idle.

42.3 Characterization Sweep Order

Sweep order	Knob	Why
1	`ADDR_MAP.bank_lsb`	Largest impact on row-hit / bank parallelism
2	`ADDR_MAP.hash_en` / `.hash_seed`	Defeat power-of-two-stride hot-banking
3	`REFRESH_TUNING.page_policy_or`	OPEN vs CLOSE vs HAPPY for the workload mix
4	`SCHED_TUNING.lookahead_active`	Issue rate vs lookahead depth
5	`SCHED_TUNING.happy_enable`	A/B test the predictor (HAPPY only)
6	`REFRESH_TUNING.refresh_defer_active`	Refresh latency vs sustained BW
7	`SCHED_TUNING.age_max_runtime`	Anti-starvation tuning
8	`SCHED_TUNING.txn_queue_high_water`	Backpressure timing

42.4 Telemetry to Watch

Observation registers per §4.2 (RO; note `hwif_in` readback is tied off in `pumice_top` today — see §4.1):

Telemetry register	What it tells you	Tune action
`OBS_AXI_R_LATENCY_AVG` / `_P99`	AXI read latency (avg / tail)	scheduler / lookahead / page policy / age_max

Telemetry register	What it tells you	Tune action
OBS_AXI_W_LATENCY_AVG	AXI write latency	write-path / CWL alignment
OBS_ROW_HIT[bank]	Per-bank row-hit rate (read-clear)	address mapping (bank_lsb/hash), page policy
OBS_REF_LATENCY[bank]	Per-bank refresh blocking	refresh deferral / refpb policy
OBS_TXN_QUEUE_DEPTH_MAX/AVG	Queue pressure	txn_queue_high_water
OBS_REFRESH_PENDING_MAX	Proximity to refresh-deadline miss	lower refresh_defer_active
OBS_REFRESH_DEFER_HIST_0..3	Refresh batch histogram	validate refresh_defer_active
OBS_PAGE_PRED_ACCURACY	HAPPY prediction accuracy	warmup_cycles / hysteresis
OBS_WORDS[9]	Packed obs_* harvest	FUB-internal diagnostics

42.5 Workload-Specific Recipes

42.5.1 Streaming (DMA, video, audio capture)

```
// Maximize row-hit, batch refresh, interleave banks
csr_write(SCHED_TUNING,   LOOKAHEAD_ACTIVE(4) | HAPPY_ENABLE);
csr_write(REFRESH_TUNING, REFRESH_DEFER_ACTIVE(8) | PAGE_POLICY_OR(1
/*OPEN*/) | ZQCS_FREQ_HZ(1));
csr_write(ADDR_MAP,       BANK_LSB(log2_cols_per_burst));   // bank-
interleave (pre-init)
```

42.5.2 Low-Latency Bursty (CPU)

```
csr_write(SCHED_TUNING,   LOOKAHEAD_ACTIVE(2) | HAPPY_ENABLE);
csr_write(REFRESH_TUNING, REFRESH_DEFER_ACTIVE(1) | PAGE_POLICY_OR(3
/*HAPPY*/));
csr_write(ADDR_MAP,       BANK_LSB(COL_WIDTH) | HASH_EN |
HASH_SEED(seed));  // pre-init
```

42.5.3 Real-Time / Safety-Critical

```
csr_write(SCHED_TUNING,   FORCE_INORDER | LOOKAHEAD_ACTIVE(0));
csr_write(REFRESH_TUNING, REFRESH_DEFER_ACTIVE(1) | PAGE_POLICY_OR(2
/*CLOSE*/));
```

42.6 Telemetry-Driven Auto-Tuning Loop

For SoCs with firmware capable of background loops (once observation readback is wired):

```
void periodic_autotune(void) {
    static uint8_t defer = 8;
    uint32_t pending_max = csr_read(OBS_REFRESH_PENDING_MAX);

    if (pending_max > DEFER_BUDGET * 7 / 8)      { if (defer > 1)
defer--; }
    else if (pending_max < DEFER_BUDGET / 4)     { if (defer < 8)
defer++; }

    uint32_t v = csr_read(REFRESH_TUNING);
    v = (v & ~REFRESH_DEFER_ACTIVE_MASK) |
REFRESH_DEFER_ACTIVE(defer);
    csr_write(REFRESH_TUNING, v);   // live on the next refresh event
boundary
}
```

Run every ~100 ms. Writes are cheap (no commit drain) and workload-dependent tuning is automatic.

42.7 Open Questions / Future Work

- **Observation readback.** The telemetry recipes assume `hwif_in` is wired; it is tied off today (§4.1).
- **Profile-select CSR.** A single "workload profile" field that switches all knobs at once would simplify firmware; adds CSR area. Punt.
- **QoS priority.** `awqos`/`arqos` are on the AXI port but not yet consumed by the scheduler; a v2 hook.